Thermodynamic Bounds and an Optimal Tempo for Goal-Holding Agents

David Charlot, Open Interface Engineering

Research paper · OpenIE research.openie.dev
https://research.openie.dev/papers/agency/ · PDF https://research.openie.dev/pdfs/agency.pdf

Measurement. Research paper, not a journal final. Theorems, experiments and positions are cited from their sources. Estimates are never measured_j. No board power is claimed. No fabricated citations.

> Agency first, kinematics second. The proposed Universal Law of Agency is a definition plus a gate, and it is required before anything counts as an agent. The kinematic bounds say how fast agency moves per second and per joule. Intelligence is the calculus of agency: a derivative read in hindsight off a completed run, ι = dX/dJ. Energy to run is the only true metric of computer intelligence. Estimates are never measured_j.

The proposed Universal Law of Agency, stated as a definition plus a gate.

Agency first. Kinematics second. Intelligence is the calculus of agency, read in joules.

---

1. Thesis

The proposal has three layers, and the order is fixed.

1. **Agency comes first, and its definition is required.** This track proposes it as the Universal Law of Agency. In form it is a definition plus a gate: four clauses say what counts as an agent, and the fourth clause refuses any act that breaks the joule budget. Nothing is an agent until the definition is met. Nothing downstream is defined without it.
2. **The kinematic bounds of agency follow from the definition.** They say how fast agency can move, per second and per joule, and what it must pay to move.
3. **Intelligence is the calculus of agency.** Intelligence is a hindsight characteristic. It is read off a completed run, the way velocity is read off a recorded path. It is therefore a derivative of agency: $\iota = dX/dJ$, where X is confirmed agency and J is joules.

The thesis: Energy to run is the only true metric of computer intelligence. All other factors collapse to zero.

The thesis has an exact form. Correctness is a gate. Joules are the ranking. For a task q with completeness predicate $C(z)$, the joules to correct completion of system S are

$$J_S^*(q) = \min_{\pi} \mathbb{E}[J_\pi] \quad \text{subject to} \quad \Pr[C(z) = 1] \geq 1 - \delta$$

The minimum runs over the policies π that S can execute, and δ is the failure rate the task allows. A deadline, when the task has one, is written into $C(z)$: an answer after its deadline fails the predicate. So correctness, deadlines and task size appear in the formulas of this track, and each appears as a constraint. None appears as a score. Among systems that pass the gate, joules alone order them. Parameters, tokens, benchmark scores and prices carry zero weight in that order. That is what “collapse to zero” means. Correctness becomes the constraint, and energy becomes the only metric. That is the thesis. Energy-to-solution in high-performance computing and the accuracy-gated measurements of MLPerf Power have the same form [260]. This track puts an eligibility definition in front of the gate and reads the joules as a metered density. Section 2.3 states the ranking. Proposition 8 proves that it is a total order on systems that survives bounded meter error, and that integrated ι is not.

The order matters. A derivative needs a path. A path needs something that moves. The definition says what moves. The kinematics say how it can move. Intelligence is read off the path afterward.

This track tests that order against the literature. It sorts every source into three bins.

- **Proven:** theorems and derivations. The result follows from stated assumptions.
- **Demonstrated:** experiments, measurements and simulations. The result was observed.
- **Opinion:** position papers, frameworks, definitions and philosophy. The result is argued.

No source is excluded for its method. Each source is placed by what it gives the definition, the kinematics or the calculus.

Terms

- **SI:** the International System of Units.
- **Joule (J):** the SI unit of energy. One watt for one second.
- **k_B:** the Boltzmann constant, 1.380649×10^{-23} J/K, exact by SI definition.
- **Entropy production (Σ):** the total entropy a process creates, in units of k_B. It is zero only for a reversible process.
- **Mutual information (I):** the number of bits one variable carries about another.
- **Dynamical activity (A):** the rate of jumps between states in a stochastic process. It measures how busy the dynamics are.
- **TUR (thermodynamic uncertainty relation):** a theorem that ties the precision of a current to its entropy production.
- **NMR (nuclear magnetic resonance):** a method that controls and reads molecular spins.
- **Instability rate (Λ):** how fast an uncontrolled world drifts from the goal, in nats per second. For a linear plant run at f steps per second, $\Lambda = f \sum_{|\lambda_i|>1} \ln |\lambda_i|$.
- **Reset error (ε):** the probability that a reset record is not in its most likely state.
- **Counting observable:** a quantity of a trajectory that counts selected jumps. It never decreases.
- **Radon-Nikodym derivative:** the density of one measure with respect to another. It exists when the first measure gives zero mass to every set that the second gives zero mass to.

- **Mixture of Limits:** the OpenIE philosophy. Use the cheapest gear that is sufficient. There exist floors past which additional tokens, parameters, or joules do not purchase verifiable progress. Spend past the floor is waste. That floor is the plateau (threshold convergence). Mixture of Limits is not one fixed cascade. A fixed cascade order is one instance of it. The σ-law selector is another.
- **Notational Intelligence:** the OpenIE receipt study. Every act carries its prediction, its outcome and its joules.

---

2. Formal framework

Definitions fix the terms. Propositions are proven from cited theorems, with their assumptions stated.

2.1 Layer 1. The definition and its gate: eligibility

For a system S, an act a and an episode budget B in joules, eligibility $E(S, a) \in \{0, 1\}$ equals 1 only when all four clauses hold.

1. **Prediction first.** A result template $R^*(a)$ is fixed at time t_0, before the act, and it discriminates: $D(R^*(a)) \geq \delta$.
2. **Coupling to the world.** $I(\mathrm{do}(a); W') > 0$. Here W' is the world state downstream of the actuator, and $\mathrm{do}(a)$ is Pearl's intervention operator [1]. The coupling must survive changes to the system's own inputs. Editing your own sensors does not count [2].
3. **Consequence borne.** The comparison $m = M(R, R^*(a)) \in \{1, \mu, 0\}$ (match, partial, mismatch) changes the system: $s_{t+1} = f(s_t, m)$.
4. **Physical floor.** $J(a) \leq B$. If no act meets clauses 1 to 4, the act is refusal, with a receipt.

Physics already prices clauses 1 to 3. The second law with feedback [3, 4] and the measurement and erasure bound [5] give, for one bath at temperature T:

$$W_{\mathrm{ext}} \leq -\Delta F + k_B T\, I, \qquad W_{\mathrm{meas}} + W_{\mathrm{erase}} \geq k_B T\, I$$

Here I is the mutual information the act's comparison acquires, in nats. One bit is $\ln 2$ nats, so each bit costs $k_B T \ln 2$. An acceptor that compares and then resets pays at least $k_B T \ln 2$ per bit over a closed cycle [6, 7]. The definition is physical from its first clause.

Definition 1. Coupling

Clause 2 is measured by causal information flow. Let act a be drawn from the policy $p(a)$ and set by intervention, and let W' be the world downstream of the actuator. Then

$$I(\mathrm{do}(a); W') = \sum_a p(a) \sum_w p(w \mid \mathrm{do}(a)) \log_2 \frac{p(w \mid \mathrm{do}(a))}{\sum_{a'} p(a')\, p(w \mid \mathrm{do}(a'))}$$

This is the information flow of Ay and Polani from the act node to W' [8]. It equals the mutual information of the interventional joint distribution $p(a)\,p(w \mid \mathrm{do}(a))$. Its maximum over policies is the capacity of the channel from acts to the world:

$$0 \;\le\; I(\mathrm{do}(a); W') \;\le\; \max_{p(a)} I(\mathrm{do}(a); W') \;\le\; \log_2 |\mathcal{A}|$$

Here $\mathcal{A}$ is the set of available acts. Klyubin, Polani and Nehaniv defined empowerment as this capacity, with the agent's own future sensors in place of W' [9]. Clause 2 takes the world side, so editing a sensor adds nothing [2].

Realized, not capacity. The quantity credited to an act is the realized flow under the policy the agent ran: $I(\mathrm{do}(a); W')$ at that $p(a)$. The capacity is an upper bound and is never credited. A system is not credited for coupling it could have made and did not.

Estimation. The flow is estimated from receipts. The agent's randomized policy assigns acts, so the receipt's joint counts of (a, w) are interventional samples. The plug-in mutual information of those counts is biased upward by about $(|\mathcal{A}| - 1)(|\mathcal{W}| - 1)/(2n)$ nats for n samples, with $\mathcal{W}$ the set of distinguishable world outcomes. That Miller-Madow term is subtracted, and a bootstrap interval is reported [315]. Where the policy is deterministic in a context, acts are not randomized there, and the flow is not identified from that run. The receipt then records the capacity bound in its own column, labeled as a bound, and X for that context is reported as an upper bound. The run of Section 8.11 credited the capacity bound $b = n_{\text{empty}} \log_2 9$. Its X and its ι are upper bounds on realized coupling and realized intelligence. Its J^* ranking does not use X, so it stands.

Why this measure:

- **It is interventional.** Pearl's do-calculus gives the operator [1]. Kenton, Kumar, Farquhar, Richens, MacDermott and Everitt define agents as systems that would adapt their policy if their acts influenced the world in a different way, and they discover agents from interventional data [10]. A receipt is interventional by construction. The agent sets its acts, and the outcome is read from the world.
- **Observational measures fail.** James, Barnett and Crutchfield showed by example that transfer entropy does not measure information flow [11]. Lizier and Prokopenko separated information transfer from causal effect and assigned causal effect to causal information flow [12].
- **It is priced.** Under intervention the flow is a mutual information. Mutual information is the quantity that the feedback second law and the measurement and erasure bound price in joules [3, 5].
- **It needs no causal graph.** Janzing, Balduzzi, Grosse-Wentrup and Schölkopf derived a measure of causal influence from postulates, given the causal graph and the joint distribution [13]. A receipt carries acts and outcomes, not the agent's full graph. The flow is computed from them directly.
- **Credit goes to the act that caused it.** When several acts touch one outcome, the bits credited to act k follow the account of actual causation for single transitions given by Albantakis, Marshall, Hoel and Tononi [14].

Definition 2. Agent boundary

Let the physical system be a finite set V of components, each behind a metered wall. A candidate agent is a subset $S \subseteq V$, or a coarse-graining of one, that holds the record of every comparison m_k (clause 3), with W' outside S. For a completed run, let X_S be the confirmed coupled bits credited with S as the agent, and let J_S be the joules metered at the wall of S.

The agent is a candidate that maximizes X_S/J_S.

V is finite, so there are finitely many candidates. A maximizer exists whenever some candidate has $J_S > 0$. The receipt names the maximizer and records any tie.

Act-closed metering. Clause 4 charges an act's joules to the agent that acts. So a candidate credited with the coupling of an act is charged every joule that realizes it: the joules of the actuator that executes the act and of the link that carries the command. J_S in Definition 2 is read under this rule. Takahashi and Hayashi show why the rule is needed. With an open boundary, information gain and in-boundary dissipation decouple, and bits per joule can be made as large as one likes [239].

Lemma 1. The maximizing boundary contains the actuator

Setting. A component $\alpha \in V$, the actuator, is a cut between the rest of V and the world: every causal path from a component of $V \setminus \{\alpha\}$ to W' passes through the output a of α. A candidate S with $\alpha \notin S$ acts on W' only through a command c that it sends to α. Walls are disjoint, as in Proposition 7, so $J_{S\cup\{\alpha\}} = J_S + J_\alpha + J_0$, with link joules $J_0 \geq 0$.

Assumptions.

- (B1) Cut. For a candidate without α, $c \to a \to W'$ is a Markov chain under intervention on c.
- (B2) Act-closed metering. A candidate without α that is credited coupling through α is charged $J_S + J_\alpha + J_0$.
- (B3) A tie between candidates goes to the larger by inclusion.

Statement. For every candidate S with $\alpha \notin S$ and positive charged joules, $X_S/(J_S + J_\alpha + J_0) \leq X_{S\cup\{\alpha\}}/J_{S\cup\{\alpha\}}$. So some maximizer of Definition 2 contains α, and under (B3) the named maximizer does. The inequality is strict whenever the actuator adds coupling, $X_S < X_{S\cup\{\alpha\}}$.

Proof.

1. Fix an act. Both candidates hold the same comparison records, because S holds every record (Definition 2) and $S \cup \{\alpha\}$ contains S. So E_k and m_k are the same for both. By (B1) and the data-processing inequality, applied to the interventional joint distribution, $I(\mathrm{do}(c); W') \leq I(\mathrm{do}(a); W')$. Summing over acts, $X_S \leq X_{S\cup\{\alpha\}}$.
2. By (B2) and the composition of walls (Proposition 7, part 1), the joules charged to S equal $J_{S\cup\{\alpha\}}$.
3. Step 1 over the common denominator of step 2 gives the inequality. It is strict when $X_S < X_{S\cup\{\alpha\}}$.

4. V is finite, so a maximizer S^* exists. If $\alpha \notin S^*$, step 3 makes $S^* \cup \{\alpha\}$ a maximizer too, and (B3) names it. ■

Reading. Without (B2), Definition 2 is degenerate. A small subset that holds the records and leaves the actuator's draw outside its wall reports the same coupled bits on fewer joules. Act-closed metering removes that, at no cost in practice, because the actuator's joules are already on the receipt under clause 4. Components off the causal path are still removed by the ratio: dropping one leaves X unchanged and lowers J. The search does not run over all $2^{|V|}$ subsets. It runs over candidates that hold the records and contain the actuator. On a host those are few: processor, accelerator, memory, actuator. One meter suffices when its domains attribute power to those components (Section 8.1).

Why this boundary:

- **The ratio, not X_S alone.** Maximizing X_S alone puts no price on components that add joules and no confirmed bits. The ratio removes them.
- **Not the Markov blanket.** Aguilera, Millidge, Tschantz and Buckley derived that the Markov blanket condition holds only in a narrow range of parameters, even for weakly coupled linear stochastic systems [15]. Biehl, Pollock and Kanai proved by counterexample that the definitions of the Markov blanket in use are not equivalent [16]. Bruineberg, Dołęga, Dewhurst and Baltieri separate the blanket as a tool of inference from the blanket as a physical boundary [17].
- **No observer.** Orseau, McGregor McGill and Legg define agent and device by a Bayesian comparison of two descriptions of the same behavior, so the label depends on the observer's priors [18]. X_S and J_S are read off receipts and meters.
- **A boundary found by a measure.** IIT 4.0 (integrated information theory) takes the system to be the set of elements with maximal integrated cause-effect power [19]. Krakauer, Bertschinger, Olbrich, Flack and Ay identify individuals as aggregates that propagate information from their past into their future [20]. Definition 2 keeps the form and uses the definition's own currency: confirmed coupled bits per joule.
- **Coarse-grainings count.** Hoel, Albantakis and Tononi showed that a macro description can carry more causal power than the micro one [21].
- **Every candidate has an interface.** Rosas, Mediano, Biehl, Chandaria and Polani proved that every bipartite stochastic process has a causal blanket, and gave an algorithm that finds it from data with no steady-state or Markov assumption [22].

2.2 Layer 2. Kinematics

Cumulative joules are $J(t) = \int_0^t P(t')\,dt'$, with P the metered power.

Agency displacement counts bits of confirmed, coupled change:

$$X(t) = \sum_{k:\, t_k \le t} E_k\, m_k\, I(\mathrm{do}(a_k); W'_k)$$

Per act, $I(\mathrm{do}(a_k); W'_k)$ is the realized flow of the policy that chose a_k, estimated over the run's acts in the same context (Definition 1).

Velocity per second is $v_t = dX/dt$. Velocity per joule is $v_J = dX/dJ$. Acceleration per joule is $\alpha_J = d^2X/dJ^2$. The budget is $J(T) \le B$.

The kinematic bounds are the limits on these quantities.

K1. Landauer floor, closed cycle. Each confirmed bit is recorded and later erased to reset the acceptor. So

$$v_J \le \frac{1}{k_B T \ln 2}.$$

At $T = 300$ K, $k_B T \ln 2 \approx 2.87 \times 10^{-21}$ J, computed from the SI constants. That is about 3.5×10^{20} bits per joule. Proposition 1 proves K1 act by act, for the mean joules of each act, under stated assumptions on the record.

K2. Classical speed limit. Shiraishi, Funo and Saito proved this bound for a Markov jump process with local detailed balance [23]:

$$\tau \;\ge\; \frac{L\big(p(0), p(\tau)\big)^2}{2\,\Sigma_{\mathrm{tot}}\,\langle A \rangle_\tau}$$

- $L(p, p') = \sum_i |p_i - p'_i|$ is the full L_1 distance between two probability distributions. There is no factor of 1/2. With total variation distance $d_{TV} = L/2$, the same bound reads $\tau \ge 2d_{TV}^2/(\Sigma_{\mathrm{tot}}\langle A \rangle_\tau)$.
- $\Sigma_{\mathrm{tot}} = \int_0^\tau \dot{\Sigma}\, dt$ is total entropy production, system plus baths, in units of k_B.
- $A(t) = \sum_{i \ne j} W_{ij}(t)\, p_j(t)$ is the dynamical activity, with W_{ij} the jump rate from state j to state i. $\langle A \rangle_\tau = \frac{1}{\tau}\int_0^\tau A(t)\, dt$ is its time average.

Rearranged, $\Sigma_{\mathrm{tot}} \ge L^2/(2\tau\langle A \rangle_\tau)$. Moving a distribution far, fast, with few jumps, costs entropy.

Joule form. For one bath, $k_B T\, \Sigma_{\mathrm{tot}} = W - \Delta F_{\mathrm{neq}}$, with F_{neq} the nonequilibrium free energy [24]. So the dissipated work obeys

$$W_{\mathrm{diss}} \;\ge\; \frac{k_B T\, L^2}{2\,\tau\,\langle A \rangle_\tau}.$$

Stationary currents. The same paper gives $\tau \ge c^* L^2/(2\,\Sigma_{HS}\,\langle A \rangle_\tau)$, with $c^* = 0.896\ldots$ and Σ_{HS} the Hatano-Sasa, or excess, entropy production [23].

K3. Precision costs dissipation. For a time-antisymmetric current Y in a steady state, $\mathrm{Var}(Y)/\langle Y \rangle^2 \ge 2/\Sigma_{\mathrm{tot}}$ [25, 26]. Agents use feedback, and feedback beats this bound in experiment [27]. The bound for agents is the feedback form, in which mutual information enters the bound [28]. Proposition 3 states it for confirmed agency.

K4. Finite-time cost. Erasing a bit in finite time τ costs more than $k_B T \ln 2$. The excess grows as τ shrinks [29]. Near equilibrium, the excess work along a protocol is bounded below by the squared thermodynamic length over the duration [30]. Proposition 1 gives the floor for a whole act.

K5. Quantum ceiling on rate. A system with mean energy $\langle E \rangle$ above its ground state makes at most $2\langle E \rangle / (\pi\hbar)$ orthogonal transitions per second [31]. That is about 6.04×10^{33} transitions per second per joule, computed from the constants. A figure of about 3×10^{33} matches $2E/h$, not $2E/(\pi\hbar)$.

Proposition 1. Joule floor of one act in finite time

Setting. Act k writes its comparison to a record M_k with N states, then resets the record. The record is a Markov jump process with local detailed balance, in contact with one bath at temperature T. During the write, the compared outcome R_k is held fixed, and the record's jump rates may depend on it. During the reset, the record's rates depend on nothing else, so the record evolves alone. The reset lasts τ_k. $\langle A \rangle_k$ is the record's time-averaged dynamical activity during the reset. Joules and works are means over the ensemble of act k.

Assumptions.

- (R1) The record's states have equal energy. The record shares no interaction energy with R_k at the start or the end of the write. At the start of the act the record is in its ready distribution q^0, independent of R_k.
- (R2) The credited bits do not exceed the information the write leaves in the record: $b_k \le I(M_k; R_k)$, in bits, at the end of the write. The record is unchanged between the write and the reset.
- (R3) The reset returns the record to q^0. The reset error is $\varepsilon = 1 - \max_s q^0_s$.
- (R4) The work of the write and of the reset passes through the meter: $J_k \ge W_k^{\mathrm{write}} + W_k^{\mathrm{reset}}$.

Statement.

$$J_k \;\ge\; b_k \, k_B T \ln 2 \;+\; \frac{2\, k_B T \left[\left(\phi_N^{-1}(b_k) - \varepsilon \right)_+ \right]^2}{\tau_k \, \langle A \rangle_k}$$

Here $(x)_+ = \max(x, 0)$, $h_2(x) = -x \log_2 x - (1-x)\log_2(1-x)$ is the binary entropy, and $\phi_N(x) = h_2(x) + x \log_2(N-1)$. On $[0, 1 - 1/N]$, ϕ_N increases from 0 to $\log_2 N$, so its inverse is defined on $[0, \log_2 N]$. For a one-bit record that confirms one bit, $\phi_2^{-1}(1) = 1/2$ and

$$J_k \;\ge\; k_B T \ln 2 \;+\; \frac{2\, k_B T \left[(1/2 - \varepsilon)_+ \right]^2}{\tau_k \, \langle A \rangle_k}$$

As $\varepsilon \to 0$ this tends to $k_B T \ln 2 + k_B T / (2\tau_k \langle A \rangle_k)$.

Proof.

1. For an isothermal Markov jump process with local detailed balance, $W = \Delta F_{\mathrm{neq}} + k_B T\, \Sigma_{\mathrm{tot}}$ with $\Sigma_{\mathrm{tot}} \geq 0$. Here $F_{\mathrm{neq}} = \langle E \rangle - k_B T\, S$, with S the Shannon entropy in nats [24].
2. The write. Apply step 1 to the pair (M_k, R_k) with R_k fixed. By (R1) the mean energy does not change and the pair starts independent. The entropy of R_k does not change. So the joint entropy changes by $H(q') - I - H(q^0)$, where q' is the record's distribution after the write and $I = I(M_k; R_k)$, all in nats. So $W_k^{\mathrm{write}} \geq k_B T\, [\, I - H(q') + H(q^0)\,]$.
3. The reset. The record evolves alone, so step 1 applies to the record. Its energy does not change. By (R2) and (R3) its entropy goes from $H(q')$ to $H(q^0)$. So $W_k^{\mathrm{reset}} = k_B T\, [\, H(q') - H(q^0)\,] + k_B T\, \Sigma^{\mathrm{reset}}$.
4. Steps 2 and 3 and (R4) give $J_k \geq k_B T\, I + k_B T\, \Sigma^{\mathrm{reset}}$. The record's entropies cancel over the cycle. In bits, $k_B T\, I = k_B T \ln 2\; I(M_k; R_k) \geq b_k\, k_B T \ln 2$ by (R2). This is the measurement and erasure bound of Sagawa and Ueda [5], here for any ready distribution.
5. The classical speed limit gives $\Sigma^{\mathrm{reset}} \geq L^2 / (2 \tau_k \langle A \rangle_k)$, with $L = \sum_s |q'_s - q^0_s|$ [23]. Take s with $q^0_s = \max q^0 = 1 - \varepsilon$. The other states carry the opposite change in total, so $L \geq 2(q^0_s - q'_s) \geq 2\, [(1 - q'_{\max}) - \varepsilon]$.
6. Fano's inequality, with no side information and the most likely state as the guess, gives $H(q') \leq h_2(1 - q'_{\max}) + (1 - q'_{\max}) \log_2(N - 1) = \phi_N(1 - q'_{\max})$, in bits. Since $1 - q'_{\max} \leq 1 - 1/N$ and ϕ_N increases there, $1 - q'_{\max} \geq \phi_N^{-1}(H(q'))$. By (R2), $H(q') \geq I(M_k; R_k) \geq b_k$, and ϕ_N^{-1} increases. So $L \geq 2\big(\phi_N^{-1}(b_k) - \varepsilon\big)_+$.
7. Steps 4 to 6 give the statement. ■

Reading. The floor is a sum of two prices. The information price is $k_B T \ln 2$ per confirmed bit, paid over the closed cycle whatever the ready state. The speed price belongs to the reset. A reset with finite rates never reaches a single state in finite time, so $\varepsilon > 0$ at every finite τ_k, and the speed price carries that error. K1 holds act by act at every τ_k. Tighter erasure bounds for particular dynamics are also lower bounds on the same reset (Section 4.1). Per act, the velocity per joule is

$$v_{J,k} = \frac{b_k}{J_k} \;\leq\; \frac{b_k}{b_k\, k_B T \ln 2 + 2 k_B T \left[\big(\phi_N^{-1}(b_k) - \varepsilon\big)_+\right]^2 / (\tau_k \langle A \rangle_k)} \;\leq\; \frac{1}{k_B T \ln 2}$$

The last inequality is strict whenever $\phi_N^{-1}(b_k) > \varepsilon$ at finite $\tau_k \langle A \rangle_k$.

Proposition 2. Standing joules to hold a goal

Setting. The world is a linear plant $x_{t+1} = A x_t + B u_t + w_t$, with $x_t \in \mathbb{R}^n$. At step t the agent writes a finite-valued record c_t of the plant. Its memory s_t is a finite-valued function of (s_{t-1}, c_t), with s_0 fixed, and its act u_t is a function of s_t. Keeping every record, $s_t = (s_{t-1}, c_t)$, is one such memory. The goal is to hold the plant: $\sup_t \mathbb{E}\|x_t\|^2 < \infty$.

Assumptions.

- (C1) The noise w_t is independent of (x_t, s_t).
- (C2) x_0 has a density with finite differential entropy and a finite second moment.

- (C3) Closed-cycle accounting per step. J_t is the mean joules attributed to the record c_t: its write and its later erasure, whenever they are paid. $J_t \geq k_B T\, I(x_t; c_t \mid s_{t-1})$, with the information in nats. This is Proposition 1, steps 1 to 4, at each value of the memory s_{t-1}, with a fresh record in its ready state and x_t held fixed during the write [5]. A memory of bounded size erases each record within a bounded number of steps, so the attributed joules fall inside the run up to a boundary term.

Statement. This is a joule-priced corollary of the data-rate theorem of Tatikonda and Mitter [169] and of Nair and Evans [170]. The closed-loop inequality of Touchette and Lloyd [32, 33] carries the rate to the record, and Landauer's erasure price [6] converts each bit to $k_B T \ln 2$ joules. It is new only as that composition. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A. Then

$$\liminf_{N \to \infty} \frac{1}{N} \sum_{t=1}^{N} J_t \;\geq\; k_B T \ln 2 \sum_{i:\, |\lambda_i| > 1} \log_2 |\lambda_i|$$

Proof.

1. Choose real coordinates in which $A = \mathrm{diag}(A_u, A_s)$, with A_u carrying exactly the eigenvalues of modulus above 1. Let y_t be the unstable block of x_t. Then $y_{t+1} = A_u y_t + v_t + e_t$, where v_t is a function of s_t and e_t is independent of (y_t, s_t) by (C1). A linear change of coordinates shifts differential entropies by a constant, so finiteness is unchanged.
2. Let $D_t = h(y_t \mid s_t)$, a conditional differential entropy in nats. Adding independent noise does not lower a conditional entropy. Shifting by a function of s_t does not change it. The linear map A_u adds $\ln |\det A_u|$. So $h(y_{t+1} \mid s_t) \geq D_t + \ln |\det A_u|$.
3. s_{t+1} is a function of (s_t, c_{t+1}), and conditioning on less never lowers entropy. So $D_{t+1} \geq h(y_{t+1} \mid s_t, c_{t+1}) = h(y_{t+1} \mid s_t) - I(y_{t+1}; c_{t+1} \mid s_t)$. Since y is a function of x, $I(y_{t+1}; c_{t+1} \mid s_t) \leq I(x_{t+1}; c_{t+1} \mid s_t)$. This is the closed-loop inequality of Touchette and Lloyd [32, 33].
4. Steps 2 and 3 give $I(x_{t+1}; c_{t+1} \mid s_t) \geq \ln |\det A_u| - (D_{t+1} - D_t)$. Summing over N steps, $\sum_{t=1}^{N} I(x_t; c_t \mid s_{t-1}) \geq N \ln |\det A_u| - (D_N - D_0)$.
5. Each record is finite-valued, so each information term is finite. $D_0 = h(y_0)$ is finite: the second moment bounds it above, and $h(x_0) \leq h(y_0) + h(x_0^{\mathrm{s}})$, with x_0^{s} the stable block, bounds it below, by (C2). $D_N \leq h(y_N)$, and a Gaussian has the largest entropy for a given covariance, so the bounded second moment bounds D_N above. Dividing by N, the average information per step is at least $\ln |\det A_u| = \sum_{|\lambda_i| > 1} \ln |\lambda_i|$ in the limit.
6. (C3) prices each nat at $k_B T$, which is $k_B T \ln 2$ per bit. ■

Reading. An agent that holds a goal against an unstable world pays a standing power of at least $f\, k_B T \ln 2 \sum_{|\lambda_i| > 1} \log_2 |\lambda_i|$ at f control steps per second. One unstable mode with $|\lambda| = 2$ needs one bit per step, which is $k_B T \ln 2 \approx 2.87 \times 10^{-21}$ J per step at 300 K. The record's bits per step bound its information from above, so the proof also gives a non-strict, averaged form of the data-rate theorem (Section 4.1). The composition is short. It is the data-rate theorem of Tatikonda and Mitter and of Nair and Evans [169, 170], priced at $k_B T \ln 2$ per bit through Touchette and Lloyd [32, 33]. What it adds is the price in joules, which is the floor Prediction 1 needs.

Proposition 3. Precision of confirmed agency

Setting. The agent, its world and its feedback form one time-homogeneous, irreducible Markov jump process on finitely many states, in its stationary state. Each confirmed act is one jump of a marked kind. $N(t)$ counts confirmed acts in $[0, t]$. $\langle K \rangle$ is the mean number of jumps of all kinds in $[0, t]$, and $\langle k \rangle = \langle K \rangle / t$ is the mean jump rate.

Statement.

1. In the long-time limit, $\mathrm{Var}(N)/\langle N \rangle^2 \geq 1/\langle K \rangle$. If every confirmed act credits the same b bits, then $X = bN$ and the same bound holds for X.
2. Let T_n be the time at which the n-th confirmed act closes. For large n, $\mathrm{Var}(T_n)/\langle T_n \rangle^2 \geq 1/(\langle T_n \rangle \langle k \rangle)$.
3. Let X instead be odd under the map to the backward experiment of a fluctuation relation that includes an information term, $P_B(-X, -\sigma_I)/P(X, \sigma_I) = e^{-\sigma_I}$. Then

$$\frac{\mathrm{Var}(X) + \mathrm{Var}_B(X)}{\big(\langle X \rangle + \langle X \rangle_B\big)^2} \;\geq\; \frac{1}{\exp\big[(\langle \sigma_I \rangle + \langle \sigma_I \rangle_B)/2\big] - 1}$$

When the backward experiment equals the forward one, this reads $\mathrm{Var}(X)/\langle X \rangle^2 \geq 2/\big(e^{\langle \sigma_I \rangle} - 1\big)$.

Proof. N is a counting observable: it counts a subset of jumps and never decreases. Garrahan proved parts 1 and 2 for counting observables in stationary Markov jump processes [34]. Part 1 is his bound on the variance of a counting observable by the mean activity. Part 2 is his first-passage bound, stated for counts whose jump weights are 0 or 1, which holds here because N counts one marked kind of jump. Scaling by b leaves a relative variance unchanged. Part 3 is the uncertainty relation that Potts and Samuelsson derived from any fluctuation relation, measurement and feedback included [28], applied with X as the observable. ■

Reading. Part 1 needs no time-reversal property and no model of the feedback. The precision of confirmed agency is bought with activity. Part 3 is the dissipation price, and it holds under feedback because it rests on the feedback fluctuation relation. The standard TUR does not bind feedback engines in experiment [27]. That is why the agent form carries the information term.

Proposition 4. Quantum agents keep the floor

Setting. The comparison m_k comes from a quantum measurement and is written to a classical register, the record of clause 3. The register's reset meets the setting of Proposition 1 and (R2) and (R3), with $I(M_k; R_k)$ the quantum-classical mutual information of the measurement. The reset's work passes through the meter: $J_k \geq W_k^{\mathrm{reset}}$.

Statement.

$$J_k \;\geq\; k_B T \ln 2 \left[\, b_k - \phi_N(\varepsilon) \,\right] \;+\; \frac{2\, k_B T \left[\big(\phi_N^{-1}(b_k) - \varepsilon\big)_+\right]^2}{\tau_k \, \langle A \rangle_k}$$

As $\varepsilon \to 0$ this is the floor of Proposition 1. When the write is metered as well, the measurement and erasure bound of Sagawa and Ueda gives $J_k \geq b_k \, k_B T \ln 2$ at every ε, under the additional assumption they state for quantum measurement [5].

Proof. Step 3 of Proposition 1 uses only the register and its reset dynamics. It gives $W_k^{\mathrm{reset}} = k_B T \ln 2 \, [H(q') - H(q^0)] + k_B T \, \Sigma^{\mathrm{reset}}$, with entropies in bits. The quantum-classical mutual information is at most the entropy of the outcomes [5], so $H(q') \geq b_k$. Fano's inequality gives $H(q^0) \leq \phi_N(\varepsilon)$, since $\varepsilon \leq 1 - 1/N$. Steps 5 and 6 of Proposition 1 bound Σ^{reset}. ■

Reading. Quantum resources change what an agent can do per act. They do not lower the price of its record. The record costs $k_B T \ln 2$ per bit, less a residue $\phi_N(\varepsilon)$ that vanishes as the reset becomes exact. The measurement that writes the record carries its own price, given in Section 4.1.

2.3 Layer 3. Calculus: intelligence as a derivative, read in hindsight

A run produces a receipt: the ordered record $\{(J_k, X_k)\}_{k=0}^{n}$ of metered joules and confirmed displacement. Each m_k is known only after act k closes. So the path $X(J)$ exists only once the run is complete.

- **Intelligence, marginal:** $\iota(J) = \dfrac{dX}{dJ}$, estimated from the receipt as $\iota_k = \Delta X_k / \Delta J_k$.
- **Run integral:** $X(B) = \int_0^B \iota(J) \, dJ$.
- **Joules to close a requirement:** $J^*(X_{\mathrm{req}}) = \int_0^{X_{\mathrm{req}}} \frac{dX}{\iota}$.
- **Optimum:** $\pi^* = \arg\min_\pi J_\pi$ subject to $X_\pi \geq X_{\mathrm{req}}$, $E_k = 1$ for every k, and $J_\pi \leq B$. If no policy is feasible, refuse with a receipt.
- **Ranking:** $S_1 \succ_q S_2 \iff J^*_{S_1}(q) < J^*_{S_2}(q)$.

X_{req} is set by the task's completeness predicate $C(z)$. It is a gate, not a score. Parameters, tokens and benchmarks enter only through J^*. All other factors collapse to zero.

Hindsight is literal here. Velocity is read off a recorded path. Intelligence is read off a recorded run. The receipt is the path.

Proposition 5. Intelligence exists as a bounded density on the joule axis

Setting. A run on $[0, T]$ has metered power $P(t) \geq 0$, integrable, and joule measure $dJ = P \, dt$. Act k occupies $[s_k, e_k]$ and credits $b_k = E_k \, m_k \, I(\mathrm{do}(a_k); W_k') \geq 0$ bits. Acts may overlap. Each act has an attributed power $P_k \geq 0$, zero outside $[s_k, e_k]$, with $\sum_k P_k \leq P$. Its joules are $J_k = \int P_k \, dt > 0$. For acts that do not overlap, $P_k = P$ on the act's interval.

Crediting rule. Each act's bits are spread over its own joules:

$$dX = \sum_k \frac{b_k}{J_k} \, P_k \, dt$$

Assumption (K1 per act). $b_k\, k_B T \ln 2 \le J_k$ for every k. Proposition 1 gives it under (R1) to (R4) for the mean joules of each act, so the statement holds for the mean run. On a single run it holds whenever each act draws more than $b_k\, k_B T \ln 2$. Every confirmed act in the run of Section 8.11 drew more than 3×10^{14} times that.

Statement.

1. For every measurable set $A \subseteq [0, T]$ of times, $0 \le X(A) \le J(A)/(k_B T \ln 2)$. So dX is absolutely continuous with respect to dJ.
2. The Radon-Nikodym derivative $\iota = dX/dJ$ exists, is unique up to sets of zero joules, and satisfies

$$0 \;\le\; \iota \;\le\; \frac{1}{k_B T \ln 2}$$

Where $P > 0$, $\iota = \sum_k (b_k/J_k)(P_k/P)$. For acts that do not overlap, $\iota = b_k/J_k$ on act k, which is the receipt's ι_k.

3. Confirmed agency is a function of cumulative joules: $X(t) = g(J(t))$, with g nondecreasing and Lipschitz with constant $1/(k_B T \ln 2)$. So g is differentiable at almost every joule, and $g'(J(t)) = \iota(t)$ for almost every joule.
4. If each act's bits are instead credited at the instant it closes, then $dX = \sum_k b_k\, \delta_{e_k}$ is singular with respect to dJ whenever some $b_k > 0$. Then g is a step function, $g' = 0$ at almost every joule, and all agency sits on a set of zero joules.

Proof.

1. $X(A) = \sum_k (b_k/J_k) \int_A P_k\, dt \le \frac{1}{k_B T \ln 2} \int_A \sum_k P_k\, dt \le \frac{J(A)}{k_B T \ln 2}$, by K1 per act and then $\sum_k P_k \le P$. If $J(A) = 0$, then $X(A) = 0$.
2. Both measures are finite, so the Radon-Nikodym theorem gives ι. If $\iota > 1/(k_B T \ln 2)$ on a set B with $J(B) > 0$, then $X(B) > J(B)/(k_B T \ln 2)$, against step 1. On $\{P > 0\}$, $dX = \sum_k (b_k/J_k)(P_k/P)\, dJ$, and $P_k = 0$ where $P = 0$. That gives the explicit form.
3. If $J(t_1) = J(t_2)$ with $t_1 < t_2$, step 1 gives $X((t_1, t_2]) = 0$, so g is well defined. For $t_1 < t_2$, $g(J(t_2)) - g(J(t_1)) = X((t_1, t_2]) \le (J(t_2) - J(t_1))/(k_B T \ln 2)$. J is continuous, so g is Lipschitz on $[0, J(T)]$. A Lipschitz function is absolutely continuous and differentiable almost everywhere. A change of variables gives $\int_0^t g'(J(s))\, dJ(s) = g(J(t)) = X([0, t])$. So $g' \circ J$ is also a density of X with respect to J, and it equals ι by uniqueness.
4. $X(\{e_k\}) = b_k > 0$ while $J(\{e_k\}) = 0$, because J is continuous. So dX is not absolutely continuous. It sits on finitely many points of zero joules, so it is singular. A function constant between finitely many jumps has zero derivative almost everywhere. ■

Reading. Hindsight intelligence is a well-defined density on the joule axis once each act's bits are spread over the joules that act used. The crediting rule is what makes dX/dJ exist. Point crediting puts all agency on a set of zero joules. Overlapping acts are handled by any attribution that never assigns more joules than the meter read. Each b_k is known only after act k closes, so ι is read off the completed run.

Proposition 5 holds by construction. The crediting rule makes X absolutely continuous with respect to J, so the density exists by definition. Its content is the choice of rule: part 4 shows that the other natural rule, point crediting, leaves no density at all.

Proposition 6. Sampled intelligence converges

Setting. The meter is read at cumulative joules $0 = j_0 < j_1 < \cdots < j_M = J(T)$, with mesh $\delta = \max_i (j_{i+1} - j_i)$. On $[j_i, j_{i+1})$ the sampled intelligence is

$$\hat{\iota} = \frac{g(j_{i+1}) - g(j_i)}{j_{i+1} - j_i}$$

Time samples refine joule samples, because J is continuous.

Statement. Under the setting of Proposition 5, $\hat{\iota}(j) \to g'(j)$ at almost every joule j as $\delta \to 0$, and $0 \le \hat{\iota} \le 1/(k_B T \ln 2)$ at every mesh. If the meter's reading of each sampled interval has relative error at most $\epsilon < 1$, the reported ratio lies in $[\hat{\iota}/(1+\epsilon),\ \hat{\iota}/(1-\epsilon)]$.

Proof. $\hat{\iota}(j)$ is the average of g' over a sampled interval $I \ni j$ of length $h \le \delta$. Since $I \subseteq [j - h, j + h]$,

$$\left|\hat{\iota}(j) - g'(j)\right| \ \le\ \frac{1}{h} \int_{j-h}^{j+h} \left|g'(u) - g'(j)\right| du$$

The right side tends to 0 at every Lebesgue point of g', and almost every j is a Lebesgue point by Lebesgue's differentiation theorem. The bounds on $\hat{\iota}$ are the monotonicity and the Lipschitz constant of g. A reading $\Delta\hat{J} \in [(1-\epsilon)\Delta J, (1+\epsilon)\Delta J]$ divides the same ΔX. ■

Proposition 7. Composition of joules and coupled bits

Setting. A group G of members $i = 1, \ldots, n$ acts on one world W'. Each member sits behind its own metered wall. The walls are disjoint and inside the group's wall, and every member draws its energy from outside the group. Members choose acts $a_1, \ldots, a_n$ by independent policies. Member i's coupling is computed with the other members acting under their own policies.

Statement.

1. $J_G = \sum_i J_i + J_0$, where $J_0 \ge 0$ is the energy that enters the group's wall and no member's wall.
2. $I(\mathrm{do}(a_1, \ldots, a_n); W') \ge \sum_i I(\mathrm{do}(a_i); W')$, with equality if and only if a_i is independent of $(a_1, \ldots, a_{i-1})$ given W' for every i.
3. The inequality can be strict when every member alone is uncoupled. With a_1, a_2 independent uniform bits and $W' = a_1 \oplus a_2$, each member's coupling is 0 bits and the group's is 1 bit.
4. Under K1 per act at the group's wall, Proposition 5 applies to G: ι_G exists and $0 \le \iota_G \le 1/(k_B T \ln 2)$. By Definition 2, G is the agent when X_G/J_G exceeds X_S/J_S for every other candidate, each member included.

Proof.

1. Every joule that enters a member's wall first enters the group's wall, and metered energy adds over disjoint walls.
2. Under intervention the acts have the product distribution $\prod_i p(a_i)$, and each coupling is a mutual information of the interventional joint distribution (Definition 1). Member i's coupling is $I(a_i; W')$ in that joint, by marginalizing the other acts. By the chain rule, $I(a_{1:n}; W') = \sum_i I(a_i; W' \mid a_{<i})$. Independence gives $I(a_i; a_{<i}) = 0$, so $I(a_i; W' \mid a_{<i}) = I(a_i; W', a_{<i}) = I(a_i; W') + I(a_i; a_{<i} \mid W') \geq I(a_i; W')$. Equality holds for every term if and only if $I(a_i; a_{<i} \mid W') = 0$ for every i.
3. Given a_1 alone, W' is still a uniform bit, so $I(a_1; W') = 0$, and likewise for a_2. Given both acts, W' is fixed, so $I(a_1, a_2; W') = H(W') = 1$ bit.
4. Proposition 5 needs only the run, the meter and K1 per act. Apply it at the group's wall. ■

Reading. Joules add, with the group's links on top. Coupled bits from independent members add at least, and some exist only at the group level. A group raises ι_G when it gains more coupled bits than its members hold alone, or spends fewer joules acting together than its members spend acting apart. Section 4.1 gives the theorems that price both terms.

Prediction 1. The optimal act time of an agent that holds a goal

Setting. A closed-cycle acceptor runs one act after another. Each act writes a record with N states and credits b bits when confirmed. The comparison, made before the reset, confirms with probability p. Each reset meets Proposition 1, with duration τ, mean activity $\langle A \rangle$ and reset error ε. Throughout, the agent holds a goal against a world with instability rate Λ, through a separate register. Every joule is metered at the agent's wall (Definition 2).

Derivation. Each act lasts at least τ. Proposition 2 prices the holding at $k_B T\, \Lambda$ per second, in the long run. Proposition 1 prices the act. The two registers are distinct, so their metered joules add. Per act, in the mean,

$$J(\tau) \;\geq\; b\, k_B T \ln 2 \;+\; \frac{\kappa}{\tau} \;+\; k_B T\, \Lambda\, \tau, \qquad \kappa = \frac{2 k_B T \left[\left(\phi_N^{-1}(b) - \varepsilon\right)_+\right]^2}{\langle A \rangle}$$

By Proposition 5, the run's intelligence is confirmed bits per joule, $\iota(\tau) = p\, b / J(\tau)$. The right side of the bound on J is least at $\tau^* = \sqrt{\kappa / (k_B T \Lambda)}$, where it equals $b\, k_B T \ln 2 + 2\sqrt{\kappa\, k_B T \Lambda}$. For a one-bit record and an exact reset, this gives two numbers:

$$\tau^* = \frac{1}{\sqrt{2\, \langle A \rangle\, \Lambda}}, \qquad \iota(\tau) \;\leq\; \frac{p}{k_B T \left[\ln 2 + \sqrt{2\Lambda / \langle A \rangle}\right]}$$

Prediction 1.

1. **Ceiling.** No acceptor that holds a goal at instability rate Λ while it acts reaches ι above $p / \left(k_B T [\ln 2 + \sqrt{2\Lambda / \langle A \rangle}]\right)$ for one-bit acts, at any act time. The general ceiling is $p\, b / \left(b\, k_B T \ln 2 + 2\sqrt{\kappa\, k_B T \Lambda}\right)$. It depends only on T, p, b, N, ε and the ratio $\Lambda / \langle A \rangle$.

2. **Optimal act time.** Let an acceptor's measured joules per act follow $J(\tau) = J_0 + a/\tau + s\,\tau$. Measure a and J_0 from a sweep of τ with no goal held, and the standing power s from the goal held with no acts. Then the measured $\iota(\tau)$ peaks at $\tau^* = \sqrt{a/s}$, at the value $p\,b/(J_0 + 2\sqrt{as})$, with $a \geq \kappa$ and $s \geq k_B T \Lambda$. The error band is

$$\frac{\delta\tau^*}{\tau^*} = \frac{1}{2}\sqrt{\left(\frac{\delta a}{a}\right)^2 + \left(\frac{\delta s}{s}\right)^2}$$

3. **Scaling.** Across a sweep of the instability rate, $\ln \tau^*$ falls against $\ln s$ with slope $-1/2$.

Falsifier. The measured maximizer of $\iota(\tau)$ lies outside $\tau^* \pm 2\,\delta\tau^*$, or the fitted slope of $\ln \tau^*$ against $\ln s$ over at least one decade of s lies outside $-1/2$ by more than two standard errors. Either defeats the claim that act joules and standing joules add on one meter, with ι read as their density. A measured ι above the ceiling defeats the accounting of that run (F12, Section 8.7).

What is new. The finite-time Landauer bound prices the memory alone and falls toward $k_B T \ln 2$ as τ grows [29]. There, the slowest reset is the cheapest. The data-rate theorem prices holding alone, per second (Section 4.1). Neither gives an interior optimum. The definition meters the whole agent, counts confirmed bits only, and reads ι as a density on the joule axis. Then holding a goal makes slowness expensive, and the best act time is the geometric mean of the jump time $1/\langle A \rangle$ and the drift time $1/\Lambda$, up to the factor $1/\sqrt{2}$. The controller's own costs enter the joules in the same way in the work of Ehrich, Still and Sivak (Section 5.1).

Where to test it. A colloidal feedback trap imposes virtual potentials in software and computes work from trajectories at the $k_B T$ scale (Section 3.2). One coordinate runs a one-bit act cycle in a virtual double well with reset time τ. The other is held at the top of a virtual inverted well of rate Λ. A single-electron box with a charge detector runs the same act cycle with every jump counted, so $\langle A \rangle$ is measured (Section 3.2). Section 8.8 gives the protocol.

Proposition 8. The gated ranking

Setting. Fix a task q, its predicate $C(z)$ and its allowed failure rate δ. For each system S, $J^*_S(q) \in (0, \infty]$ is the joules to correct completion of Section 1, with $J^*_S(q) = \infty$ when no policy of S passes the gate. Define $S_1 \preceq_q S_2$ when $J^*_{S_1}(q) \leq J^*_{S_2}(q)$.

Statement.

1. $\preceq_q$ is a total preorder on systems: reflexive, transitive and total. The ranking $\succ_q$ of Section 2.3 is its strict part.
2. Every system that passes the gate ranks above every system that fails it.
3. Let every reading of a policy's mean joules carry relative error at most $\epsilon < 1$. If $(1+\epsilon)\,J^*_{S_1}(q) < (1-\epsilon)\,J^*_{S_2}(q)$, then S_1 ranks above S_2 under every admissible reading.
4. Integrated ι does not induce this order. There are systems with $\iota_{S_1} > \iota_{S_2}$ and $S_2 \succ_q S_1$.

Proof.

1. $\preceq_q$ is the pullback of the total order $\leq$ on $(0, \infty]$ by the map $S \mapsto J^*_S(q)$. A pullback of a total order is a total preorder. Its strict part is $J^*_{S_1} < J^*_{S_2}$, which is the ranking of Section 2.3.
2. Every finite value is below ∞.
3. Each policy's reading lies in $[(1-\epsilon)J_\pi, (1+\epsilon)J_\pi]$, so the minimum of the readings over the feasible policies lies in $[(1-\epsilon)J^*, (1+\epsilon)J^*]$. Then $\hat{J}^*_{S_1} \leq (1+\epsilon)J^*_{S_1} < (1-\epsilon)J^*_{S_2} \leq \hat{J}^*_{S_2}$.
4. Let S_1 make one attempt per task at 0.1 J that passes with probability 1/2 and never retries. Let S_2 make one attempt at 1 J that always passes. Each pass credits b bits. Then $\iota_{S_1} = (b/2)/0.1 = 5b$ bits per joule and $\iota_{S_2} = b$ bits per joule. For any $\delta < 1/2$, S_1 never meets the gate, so $J^*_{S_1} = \infty$, while $J^*_{S_2} = 1$ J. ■

Reading. Part 4 is the mechanism of R-F6 in Section 8.11: cheap failures raise integrated ι and leave J^* unchanged or infinite. That is why ι is read as a density inside a run and J^* is the ranking across systems. Part 3 closes P6 up to a validated ϵ for each meter.

Proposition 9. When cheapest first is optimal

Setting. A task has gears $g_1, \ldots, g_n$. An attempt with gear g_i costs $c_i > 0$ joules, paid in full, and passes $C(z)$ with probability p_i. A policy tries gears in some order, stops at the first pass, and refuses after the last. Two structures are treated. Independent: passes of different gears are independent events. Nested: if g_i passes then every later gear passes, with pass probabilities $q_1 \leq \cdots \leq q_n$.

Statement.

1. Independent gears. The expected joules of an order are least when gears are tried in decreasing order of p_i/c_i. Cheapest first is optimal exactly when the order of rising cost is also an order of falling p_i/c_i.
2. Nested gears, two of them, with the heavy gear sufficient ($q_H = 1$). Trying the light gear first lowers the expected joules if and only if $q_L\, c_H > c_L$.
3. Budget. Under a hard budget B with $c_H \leq B < c_L + c_H$, light first closes the task with probability q_L. Heavy alone closes it with probability 1.

Proof.

1. The expected joules of an order σ are $\sum_k c_{\sigma(k)} \prod_{j<k}(1 - p_{\sigma(j)})$. Swap two adjacent gears i then j that follow a prefix with survival probability P. Their contribution changes from $P[c_i + (1-p_i)c_j]$ to $P[c_j + (1-p_j)c_i]$, and nothing else changes. So i first is no worse if and only if $p_i c_j \geq p_j c_i$, that is $p_i/c_i \geq p_j/c_j$. Any order not sorted by this ratio has an adjacent pair out of order, and swapping it does not raise the expected joules. Finitely many swaps sort the order. This is the ratio rule of Simon and Kadane for search with all-or-none solutions [318].
2. Light first costs $c_L + (1 - q_L)\, c_H$, because the heavy gear runs only when the light one fails and then passes. Heavy alone costs c_H. The first is smaller if and only if $c_L < q_L\, c_H$.
3. A failed light attempt leaves $B - c_L < c_H$, so the heavy gear cannot run. ■

Reading. The fixed cascade, cheapest sufficient first in a fixed order, is optimal when cheap gears pass often enough to pay for their failures: $q_L c_H > c_L$ at each step of a nested cascade, or rising cost matched by falling p/c among independent gears. When the condition fails, going straight to the heavier gear wins, and under a binding budget it can be the only way to close. That is the mechanism of the selector result in Section 8.11. The σ-law selector estimates the ratio with calibrated pass probabilities. It is a value-of-computation rule with joules as the cost, in the sense of Russell and Wefald and of Horvitz (Section 5.1). It stops when the next act no longer pays for its joules. Both rules stop at the floor past which more joules buy no verifiable progress, so both are Mixture of Limits instances. P7 keeps one open part: a regret bound when pass probabilities are estimated, and the nested case with more than two gears, which is a dynamic program over increasing subsequences of gears.

2.4 Where the studies sit

- **Mixture of Limits** is the optimum operator. The cheapest sufficient gear is the argmin over joules. It is a philosophy, not one fixed order: spend stops at the plateau, where more is waste.
- **Notational Intelligence** is the receipt. It carries R^*, R, m and J for every act.

Both are developed in the companion paper, Metered Commit Gates. Product placement is not part of this physics paper. It is kept in the Metabolic Intelligence white paper.

Shared thesis, stated once for every paper. Energy to run is the only true metric of computer intelligence. All other factors collapse to zero. The gate form defines what that means. Among runs that pass the task's completeness predicate $C(z)$ within the stated budget and deadline, systems are ranked by joules to completion, J^*. Correctness, deadlines and task size are the gate, not terms in the score. Parameters, tokens, benchmark scores and prices carry zero weight in the ranking (Agency, Section 1). A price trend is a separate claim, labeled as one. Epoch AI reports that the cost of a given level of AI performance has fallen about 47 percent per quarter since 2023, about 13 times per year (Emberson and Roodman, *The plunging price of thought*, 22 September 2026). If that trend continues, energy becomes the dominant marginal cost of inference. That is a cited trend with a condition, not a measured collapse. Spell Check is Global is an existence proof that some computer intelligence commoditizes to bundled zero price. Estimates are never `measured_j`.

---

3. Layer 1. The proposed Universal Law of Agency: a definition plus a gate

The definition has four clauses: prediction fixed before the act, coupling to the world through intervention, consequence borne by the system, and a physical floor in joules with refusal on a receipt.

3.1 Proven

Information has a price. Landauer showed that a logically irreversible operation dissipates at least $k_B T \ln 2$ per bit [6]. This is the floor under clause 4. Bennett built a reversible Turing machine and showed that only erasure must cost [35]. The unavoidable cost of agency sits at the reset of the acceptor, not at the thinking. Bennett then resolved Maxwell's demon by the cost of erasing the demon's memory [7]. The demon is a minimal agent. Its accounts close only when its memory is reset. That is clause 3 plus clause 4.

Feedback is priced by information. Sagawa and Ueda proved that with feedback, extractable work is bounded by $-\Delta F + k_B T I$ [3]. The value of acting on a measurement is capped by the information the measurement carries. They proved that measurement work plus erasure work is at least $k_B T I$ [5]. The full cycle of predict, compare and reset is priced in joules. They derived an exact Jarzynski equality under feedback [4] and a fluctuation theorem with information exchange between subsystems [36]. Coupling is a correlation that enters the second law. Hartle, Wolpert, Stier, Kempes and Manzano priced feedback with noisy measurements, limited memory and a limited repertoire of protocols. Against random action as the baseline, the benefit of feedback can exceed the information gained by orders of magnitude [37]. The bits bound the work against free energy. The value against a baseline is read off the meter.

Entropy converts to joules. Esposito and Van den Broeck derived $k_B T \Sigma = W - \Delta F_{\mathrm{neq}}$ for an isothermal driven system [24]. Every kinematic bound in Layer 2 becomes a joule bound through this identity. Horodecki and Oppenheim proved work costs for single shots at small scale [38]. Agency acts are single shots. The standard reviews state the settled account: information is a thermodynamic resource with a price [39, 40].

Control needs coupling. Conant and Ashby proved that a good regulator must be a model of the system it regulates [41]. A system that regulates well already carries a prediction. Touchette and Lloyd proved that the entropy a controller removes is bounded by its mutual information with the system [32, 33]. Coupling is the currency of control. Closed loop control, acting on a comparison, is what the definition requires. Pearl's do-calculus gives the intervention operator and the conditions under which its effects are identifiable [1]. Clause 2 is written in this operator. Everitt and colleagues classified how an agent can tamper with its own reward or inputs [2]. Coupling must be to the world, not to the agent's own sensor. Richens and Everitt proved that any agent that meets a regret bound under a large set of distributional shifts has learned an approximate causal model of the process that generates its data [42]. Richens, Abel, Bellot and Everitt proved that any agent that generalizes to multi-step goal-directed tasks has learned a predictive model of its environment, and that the model can be extracted from the policy [43]. A capable agent carries a prediction. That is clause 1, proven for modern agents.

Information flows inside the agent. Horowitz and Esposito showed that information flow between subsystems enters each subsystem's second law [44]. Coupling becomes a rate. Ito and Sagawa proved a second law for arbitrary causal networks with transfer entropy [45]. Prediction, act and outcome are nodes on such a network. Hartich, Barato and Seifert linked transfer entropy, learning rate and dissipation [46]. Allahverdyan, Janzing and Mahler bounded

the efficiency of information flow between coupled systems [47]. Esposito and Schaller showed that feedback that changes only barriers still enters the entropy balance [48]. A purely informational act is in the joule ledger.

Autonomous demons. Mandal and Jarzynski solved a demon that lifts a mass by writing bits to a tape [49]. It has no outside operator, and memory pays for its work. Boyd, Mandal and Crutchfield bounded the work of such ratchets by the entropy rates of their input and output tapes [50]. They showed that modular designs pay dissipation above Landauer [51]. How an agent is built changes its floor. Wolpert and Kolchinsky computed the extra entropy production a gate pays inside a circuit, set by the circuit's topology [52]. Wiring is part of the floor.

Prediction quality has a joule price. Kolchinsky and Wolpert proved that a process tuned for one input distribution dissipates extra on any other, by a relative entropy term [53]. Still, Sivak, Bell and Crooks showed that memory that does not predict costs dissipation [54]. Keeping only what predicts is the cheap design. Kolchinsky and Wolpert defined semantic information as information that is causally necessary for a system to keep itself out of equilibrium, with a thermodynamic value [55]. It is the closest existing account with joules. It stops before kinematics and before an intelligence operator. Ortega and Braun derived bounded-rational decisions from a free-energy principle with an information cost [56]. Kolchinsky and Wolpert bounded the heat of universal computation by program complexity [57]. Kolchinsky and Wolpert showed that constraints on the available protocols limit extractable work and the value of information [58]. Manzano and colleagues derived fluctuation relations and second-law-like inequalities for computations that halt at random times [59]. A run that stops when its predicate passes is such a computation.

Coupling and boundary rest on theorems. Ay and Polani defined causal information flow through interventions [8]. Transfer entropy does not measure that flow, shown by example [11]. The Markov blanket condition holds only in a narrow range of parameters, by derivation [15], and its definitions in use differ, by counterexample [16]. Every bipartite stochastic process has a causal blanket, by theorem [22]. Definitions 1 and 2 stand on these results.

3.2 Demonstrated

Landauer's floor, measured. A colloidal bead in an optical double well approached $k_B T \ln 2$ of mean heat as erasure slowed [60]. A feedback trap measured the same approach with high precision and extra cost at short cycle times [61]. The floor rises when the act is fast. That is a Layer 2 effect seen in a Layer 1 experiment. An adiabatic electronic circuit showed the Landauer cost for irreversible operation and dissipation below $k_B T$ for reversible operation [62]. The floor binds the reset, not the logic. Nanomagnetic memory bits showed the floor in memory hardware [63]. Erasure with near-zero work appears when the free-energy accounting is complete [64]. The floor is on total entropy, so the receipt must carry the full account. The entropy that enters the bound was measured to have the Gibbs-Shannon form [65]. A single trapped ion showed the quantum floor [66]. The floor holds on every substrate tested. An underdamped micromechanical oscillator reached the floor at high speed [67]. A quantum dot erased bits at minimal dissipation for a given duration, with protocols designed from thermodynamic length [68].

Feedback agents turn information into work. A colloid on a spiral staircase potential gained free energy above the work done, paid by information, and confirmed the generalized Jarzynski equality [69]. This is the first direct observation of an act whose value is set by a measurement made before it. A single-electron box extracted work near $k_B T \ln 2$ per bit [70] and measured the fluctuation theorem with mutual information [71]. An autonomous on-chip demon cooled a system with information while its own heating was measured [72]. The agent's cost was measured beside its effect.

Quantum agents obey the same bounds. NMR on molecular spins reduced entropy production by feedback, within the information bound [73], and converted information to energy at the floor [74]. Superconducting circuits tracked a demon's memory during feedback [75], verified the generalized integral fluctuation theorem [76], and measured information gained and lost during feedback [77]. Coupling decays, so the receipt must record it at the time of the act. Cold atoms in a three-dimensional optical lattice were sorted by measurement [78]. Linpeng and colleagues measured the energetic cost of quantum measurement with quantum, coherent and thermal light in a circuit quantum electrodynamics setup. Single-photon light cost the least energy per unit of information gained [79]. The measurement that writes a record has a measured price.

Classical engines measured the price at full scale. An optical tweezer engine converted information to work at the theoretical limit [80]. A colloidal engine extracted work that depends on temporal correlations in its input [81]. A DNA hairpin under continuous monitoring yielded more than $k_B T \ln 2$ per cycle when many measurements happened per cycle, consistent with the information content [82]. A levitated microparticle with delayed feedback obeyed a generalized second law across two decades of delay, and feedback failed at large delay [83]. A late act is a weaker act. Feedback trap engines maximized power and velocity [84], exploited noisy measurements optimally [85], and harvested a nonequilibrium bath [86]. A quantum dot linked work fluctuations to dissipation during information-to-work conversion [87]. A trapped ion verified fluctuation theorems that include the demon's own dissipation [88], following theory by Zeng and Wang [89].

Functional-system agents built and run. A simulated nematode with an acceptor of action results learned locomotion and chemotaxis [90]. A physical mobile robot ran the same functional-system controller [91]. Clauses 1 and 3 run in software and on a body. Neither meters joules.

3.3 Opinion

Purposive systems and anticipation. Rosenblueth, Wiener and Bigelow defined purposeful behavior through negative feedback toward a goal [92]. It has no energy term. Anokhin's functional system has afferent synthesis, decision, an acceptor of action results fixed before the act, action, and comparison [93, 94]. This is clauses 1 and 3 in their first systematic form, built on animal experiments. Sudakov stated the result as the system-forming factor [95]. The prediction clause is the organizing principle. Shvyrkov showed from single-neuron recordings that the goal organizes neural activity [96]. Rosen defined an anticipatory system as one that contains a predictive model of itself and its world and acts on the prediction [97]. None of these

sets a joule floor. Sterling defines allostasis as regulation by prediction [98]. Physiology puts the prediction before the act. Virgo, Biehl and McGregor gave conditions under which a system's states can be read as beliefs updated by Bayes' theorem [99]. Biehl and Virgo call a system an agent when it can be read as solving a POMDP (partially observable Markov decision process) [100]. Both are formal definitions. Clause 1 asks for less and fixes more: one template, fixed at t_0, hashed on the receipt.

Autonomy and organization. Maturana and Varela defined the living by self-production [101]. Kauffman put a thermodynamic work cycle inside the definition of an autonomous agent [102]. It has no prediction clause. Di Paolo made adaptivity, the regulation of one's own viability, a condition for sense-making [103]. That is clause 3. Barandiaran and Moreno proposed a minimal criterion for cognitive organization and grounded adaptive regulation in metabolism [104, 105]. Barandiaran, Di Paolo and Rohde defined agency by individuality, interactional asymmetry and normativity [106]. Ruiz-Mirazo, Moreno and Mossio framed autonomy as a self-maintaining metabolic organization [107, 108]. Montévil and Mossio described organisms as closures of constraints on thermodynamic flows [109]. It is the closest of this school to joules. Aguilera and Barandiaran built a minimal model of autonomous agency in stochastic thermodynamics [110]. The autonomy tradition is moving toward the joule. Barandiaran and Almendros applied individuality, normativity and interactional asymmetry to large language models. A model fails the first two and partly fails the third [111]. The definition applies its gate per act: a model call is an act when its receipt carries R^*, m and J. Levin's TAME (Technological Approach to Mind Everywhere) treats agency as graded and testable by experiment at every scale [112]. The definition keeps the gate binary per act and puts the grading in X.

Free energy and the physics of agency. Friston proposes that agents minimize variational free energy [113, 114]. That free energy is informational, not joules, and every bounded system qualifies, so it sets no floor. Rovelli explains agency through the entropy gradient [115]. Agency is thermodynamic at root. Jaeger argues that agency requires self-manufacture and that current artificial intelligence lacks it [116]. That is a demand for a gate before any intelligence claim. Azadi ties agency to computational irreducibility [117]. Wissner-Gross and Freer propose a force toward future path diversity [118]. Its causal temperature is not a bath temperature, so its units are not joules. Kappen's comment critiques it [119]. Fields, Goldstein and Sandved-Smith separate thermodynamic free energy, in joules, from variational free energy in active inference agents, and relate the two [120]. Only the joule quantity enters K1. Bruineberg and colleagues separate the Markov blanket as a tool of inference from the blanket as a physical boundary [17]. Definition 2 takes the boundary from the definition. Bartlett and colleagues make goal-directed information processing, measured by semantic information, the distinctive feature of living systems [121]. That is clauses 2 and 3 as a criterion for life.

---

4. Layer 2. The kinematic bounds of agency

Once an act is eligible, physics bounds how fast agency moves per second and per joule, and what precision and speed cost. The bounds come in four families: speed limits, the TUR, finite-time costs and rate ceilings.

4.1 Proven

Speed limits. Shiraishi, Funo and Saito proved the classical speed limit in Section 2.2 [23]. Moving the agent's state distribution a distance L in time τ costs entropy, and so joules. The bound extends to open quantum systems [122] and to strong coupling with general environments [123]. Vo, Van Vu and Hasegawa derived the speed limit and the TUR from one inequality [124]. Falasco and Esposito bounded the time to complete a transition by the dissipation spent [125]. Ito bounded the information-geometric speed of a distribution by entropy production [126]. Ito and Dechant bounded the rate of change of any observable by Fisher information and dissipation [127]. Nakazato and Ito bounded entropy production by the Wasserstein distance traveled [128]. Dechant, Sasa and Ito split entropy production into the part that moves the system and the part that keeps it running [129]. Van Vu and Saito unified the TUR, minimum dissipation and speed limits through optimal transport [130]. Nagayama, Yoshimura and Ito derived a family of speed limits indexed by generalized means of activity [131]. Yoshimura and Ito proved the TUR and the speed limit in deterministic chemical reaction networks [132]. Aurell, Mejía-Monasterio and Muratore-Ginanneschi solved minimal finite-time dissipation as an optimal transport problem [133]. The kinematic bounds form one family. The cheapest way to move an agent's state is a transport map. Hasegawa derived the speed limit, the TUR and the Heisenberg principle from one bulk-boundary correspondence [134]. Van Vu and Hasegawa bounded irreversible entropy production by a modified Wasserstein distance in classical and quantum Markov systems [135]. Delvenne and Falasco bounded entropy production by the statistics of kinetic observables [136].

Precision costs dissipation. Barato and Seifert stated the TUR, $\mathrm{Var}(Y)/\langle Y\rangle^2 \geq 2/\Sigma$ [25]. A reliable agent pays for its reliability. From $2/\epsilon^2$ at $\epsilon = 0.01$, precision of 1% costs at least 20,000 $k_B T$ [137]. Gingrich, Horowitz, Perunov and England proved the TUR for all steady-state currents [26]. Horowitz and Gingrich proved it for finite observation times [138], which is what agents have. Further results bound the full distribution of fluctuations [139], cover discrete time steps [140], extend the TUR wherever a fluctuation theorem holds [141], and cover Langevin dynamics, underdamped motion and velocity feedback [142, 143, 144]. Correlations between currents tighten the bounds [145]. Potts and Samuelsson showed that any fluctuation relation implies a TUR, including with measurement and feedback [28]. This is the precision bound for agents. A review states the field [146]. Barato and Seifert priced the precision of a clock [147]. Salazar bounded entropy production by information from the detailed fluctuation theorem [148]. Landi and Paternostro reviewed entropy production from classical to quantum [149]. Feedback forms now cover the cases agents meet. Van Vu and Hasegawa bounded fluctuations under arbitrary control protocols by entropy production and a kinetic term [150]. Tanogami, Van Vu and Saito bounded the current of a subsystem by its own entropy production and the information flow between it and its partner [151]. Kumasaki, Tojo, Sagawa and Funo derived a TUR for feedback cooling [152]. Honma and Van Vu derived a finite-time TUR for open quantum systems under continuous monitoring and Markovian feedback, with quantum mutual information beside entropy production [153]. For counting observables, Garrahan bounded fluctuations by dynamical activity and gave the matching bounds for first-passage times [34]. Gingrich and Horowitz derived the conjugate relation for the first passage time to accumulate a net current [154]. Proposition 3 applies these results to confirmed agency.

Finite-time costs. Schmiedl and Seifert found optimal finite-time protocols, with jumps at the ends [155]. The optimal act at finite speed is not the slow act. Sivak and Crooks bounded excess work by the squared thermodynamic length over the duration [30]. Proesmans, Ehrich and Bechhoefer found the minimal cost of erasing a bit in time τ [29, 156]. The Layer 1 floor becomes a Layer 2 curve. Zhen and colleagues stated a universal finite-time bound on reset energy [157]. Speed and efficiency trade against each other by theorem: efficiency at maximum power [158, 159], finite power forbidding Carnot efficiency [160], a three-way trade among power, efficiency and constancy [161], and a universal constraint for low-dissipation engines [162]. Tsirlin and colleagues found minimal dissipation at a fixed rate [163]. Lee, Lee, Kwon and Park tightened the speed limit for highly irreversible processes and obtained a tight finite-time Landauer bound [164]. Van Vu and Saito derived the finite-time quantum Landauer principle and the role of coherence in it [165]. Rolandi and Perarnau-Llobet derived the finite-time Landauer principle beyond weak coupling [166]. Kamijima, Funo and Sagawa derived the finite-time trade-offs between measurement and feedback [167]. Tkachenko showed for physical neural networks that quasi-static inference needs no work, while finite-speed inference costs at least a transport distance (preprint) [168]. Inference has no floor that survives slow operation. The reset does. Proposition 1 prices it.

Holding a goal. Tatikonda and Mitter [169] and Nair and Evans [170] proved the data-rate theorem: holding an unstable linear system needs a channel rate above the sum of the base-2 logarithms of its unstable eigenvalue magnitudes. Nair, Fagnani, Zampieri and Evans reviewed the field [171]. Sandberg, Delvenne, Newton and Mitter found the maximum work a continuously measuring demon extracts in finite time, and showed that any implementation needs an external power source [172]. Horowitz and Sandberg showed that the information terms in second-law inequalities form a hierarchy, each the minimum cost of acquiring that information by a distinct measurement [173]. Proposition 2 turns the bit count into a joule floor per step. It is a joule-priced corollary of the data-rate theorem, not a new law.

Quantum feedback and measurement. The feedback second law holds for quantum systems [3]. Jacobs extended it to weak measurements [174]. Funo, Watanabe and Ueda derived integral fluctuation theorems under quantum measurement and feedback [175]. Guryanova, Friis and Huber proved that ideal projective measurements need infinite resources, and gave the energy cost of a pointer that reproduces the system's statistics [176]. Danageozian, Wilde and Buscemi bounded the measurement heat of error identification by the Groenewold information gain [177]. These price the measurement that writes a record. Proposition 4 keeps the floor of the record itself.

Groups. Wolpert derived a strictly nonzero lower bound on the minimal entropy production rate of interacting subsystems from the constraints on which subsystems each one depends on [178]. Tasnim and Wolpert extended the theory to subsystems that change state together [179]. Wolpert related the entropy production of a whole Bayes net to the precision of currents in its parts [180]. Rolandi, Abiuso and Perarnau-Llobet showed that collective protocols can make dissipated work grow sublinearly with the number of members [181]. Fujimoto and Ito found by a

game-theoretic method that the partial entropy productions of interacting subsystems trade against each other, and that the total is least when the subsystems share the penalty equally [182]. Proposition 7 composes these joules with the group's coupled bits.

Rate ceilings. Margolus and Levitin bounded orthogonal transitions per second by energy [31]. Lloyd applied the limit to a whole computer [183]. Deffner and Campbell reviewed the family of quantum speed limits [184].

Living sensors. Energy dissipation, adaptation speed and accuracy are bound together in bacterial chemotaxis [185]. Learning about the environment requires energy [186], and inference accuracy is tied to dissipation [187]. Sensing precision is limited by receptors, time and energy, with an optimal allocation among them [188]. No resource can be skipped. That fits Mixture of Limits. Erasure inside adaptation has a cost [189]. Information acquired per energy dissipated is the nearest existing quantity to v_J in living sensors [190, 191]. Copying information costs energy [192]. Writing the receipt has a price. Accurate and synchronized biochemical clocks cost free energy [193, 194], and so does reducing noise at high sensitivity [195]. Holding a system away from equilibrium has a minimum power [196]. That is the standing cost of an agent ready to act. Bryant and Machta bounded the energy of sending a bit through a cell's physical channels, in $k_B T$ per bit, as a function of size, distance and latency, and found it many orders of magnitude above unity [197]. Tjalma and colleagues showed that the bits of the past that best predict the future are prohibitively costly for cellular networks [198].

Information rates without joules. Schreiber defined transfer entropy, directed information flow per step [199]. It is per time, not per joule, and it has no eligibility gate. Tishby and Polani charged information per step in a Bellman recursion [200]. Landauer converts those bits to joules. Stratonovich bounded the maximal gain from a given amount of information [201]. That is a ceiling on value per coupled bit.

4.2 Demonstrated

The TUR tested. A two-qubit NMR experiment obeyed generalized TURs and violated the specialized TUR where theory predicts [202]. An optical tweezer information engine violated the original TUR near maximal efficiency and obeyed generalized bounds with mutual information [27]. Agents need the feedback form. A quantum dot Szilard engine confirmed it independently [203]. Atomic-scale conductors examined the TUR with measured current noise, the physics of every chip [204]. Molecular motors were scored by precision per unit dissipation [205]. Observed current fluctuations bound dissipation from below [206]. A receipt of fluctuations is a lower-bound meter for joules.

Speed limits and finite-time costs tested. Single atoms in an optical lattice showed the Mandelstam-Tamm and Margolus-Levitin bounds and the crossover between them [207]. The rate ceiling is measured. A trapped colloid measured time and entropy trade-offs for fast thermal transitions [208]. Brownian particles driven by optimal transport protocols saturated the minimal finite-time dissipation bound [209]. The kinematic optimum is reachable. A feedback trap measured a two-force Brownian machine against linear response bounds [210]. A photonic

experiment tested geometric bounds on entropy production [211]. Dago and Bellon measured erasure in a double-well memory at high speed and traced the overhead above Landauer's bound to its sources [212].

Living sensors measured. Mattingly, Kamino, Machta and Emonet measured the rate at which E. coli acquires information during chemotaxis, and showed that the cells climb gradients close to the limit that rate sets [213]. Information limits behavior, measured.

Heat engines at finite speed. A single colloid ran a micrometer Stirling engine [214]. A trapped bead ran a Brownian Carnot cycle with efficiency at maximum power measured [215]. The energy cost of choosing one of two options was measured [216]. That is the smallest act. A colloidal engine ran on an active bacterial bath [217], and engineered noise gave near-Carnot efficiency at finite power by shortening relaxation [218]. Engineering activity moves the bound. An information engine flipped from refrigerator to heater with noise [219]. The receipt must record the regime, not only means.

Empowerment in simulation. Klyubin, Polani and Nehaniv defined empowerment, the channel capacity from actions to future sensors [9]. It was computed for continuous control [220] and reviewed [221]. It measures how much an agent can move the world, in bits. It is not a rate per joule.

4.3 Opinion

Kinematics is mostly theorem and experiment. The opinions here concern which variable is right. Seifert holds that inference from fluctuations is the route to hidden costs [137]. Rovelli and Wissner-Gross and Freer both propose entropic kinematics [115, 118]. Kolchinsky shows that dissipation alone does not bound how fast replicators grow or decay [222]. Rates are bounded by dissipation together with activity. That is the form of the speed limit, where $\langle A \rangle$ sits beside Σ. Adlam, McQueen and Waegell argue that a purely quantum system cannot be an agent, because building a world model and deliberating need copying, which the no-cloning theorem forbids (preprint) [223]. Clause 3 already asks for a classical record of m.

---

5. Layer 3. Intelligence as the calculus of agency

Intelligence is the derivative of gated agency in joules, read off a completed run. This layer needs three things from the literature: theory that ties learning and decision to joules, measurement of joules per unit of useful work, and positions on what intelligence is.

5.1 Proven

Learning and prediction priced in joules. Goldt and Seifert bounded the information a learning network acquires by the entropy it produces [224]. Learning efficiency is at most one. This is ι for the learning part of an act, with a ceiling. Still showed that optimal memory keeps only predictive information [225]. The marginal joule spent on memory must buy prediction. Boyd, Crutchfield and Gu proved that the agent that extracts the most work from data has the

maximum-likelihood model of that data [226]. Better models yield more work per joule. Boyd, Crutchfield, Gu and Binder stated overfitting and generalization as energetics [227]. Generalization appears as joules. Ehrich, Still and Sivak priced the controller itself, beyond the information it uses [228]. So J must be metered at the wall of the whole system. Ehrich and Sivak gave the ledger of energy and information flows in autonomous machines [229]. Wolpert reviewed computation costs beyond Landauer [230]. Wolpert and colleagues set out the open problems for the energy costs of computation in stochastic thermodynamics [320]. Levy and Baxter showed that neural codes that maximize bits per unit energy differ from codes that maximize bits [231]. The right objective for a brain is information per joule. Balasubramanian, Kimber and Berry set the objective for an exploratory regime as transmission rate per unit power [232]. That is bits per joule. Goldt and Seifert bounded the information a network learns about a rule by the thermodynamic cost of learning, for batch and online learning [233]. Fiderer, Barth, Smith and Briegel defined the work capacity of an environment channel, the maximum rate at which any agent can expect to extract work in a percept-action loop. Work-efficient agents balance prediction against forgetting (preprint) [234]. Elliott, Gu, Garner and Thompson showed that quantum adaptive agents can need far less memory than memory-minimal classical agents [235]. Less memory to reset is a lower floor to pay.

Decisions and bounded optimality. Russell and Subramanian defined bounded optimality: the best program for a given machine and environment [236]. Here the resource is joules, and the optimum is $\arg\min J$ subject to $X \geq X_{\mathrm{req}}$. Russell and Wefald priced each computation by its expected effect on the decision, net of its cost, and stopped deliberation when no computation had positive net value [316]. Horvitz framed the same trade for reasoning under resource constraints [317]. The σ-law selector is that rule with the cost in metered joules. Proposition 9 gives its exact form for a cascade. Genewein and colleagues derived abstraction and hierarchy from utility minus information cost [237]. Mixture of Limits gears are such a hierarchy. Legg and Hutter defined intelligence as complexity-weighted expected reward over all computable environments [238]. It is an integral, but over reward, without energy, and uncomputable. The derivative in joules replaces reward with confirmed coupled bits and the measure with metered joules. Takahashi and Hayashi give the closest existing calculus in joules [239]. They define two ratios under explicit accounting conventions. Thermodynamic epiplexity per joule is $I(W^{\mathrm{post}}; Z \mid W^{\mathrm{pre}})/E_{\mathrm{cons}}$: new bits about an environment variable Z retained in the agent's state, per joule consumed inside a stated boundary. Empowerment per joule is the capacity of the channel from acts to observations per unit expected energetic cost, the capacity-per-unit-cost of Verdú. They prove a closed-cycle Landauer benchmark by joining a thermodynamic learning inequality with data processing, and they show by a reversible copying construction that, with an open boundary and fresh memory brought in free, information gain and in-boundary dissipation decouple. They close with a reporting checklist: boundary, energy balance, baseline, coarse-graining, horizon and reset, throughput, estimator. This track takes four things from that work. Their closed-cycle benchmark is K1 for learning. Their decoupling construction is the reason for act-closed metering in Definition 2 (Lemma 1). Their checklist is adopted in Section 8.12. Their advice to report bits per joule beside bits per second is the variable Prediction 1 optimizes. The definition differs from their metrics in four exact ways. Empowerment per joule is a capacity, a supremum over act distributions; the definition credits the realized flow of the policy that ran (Definition 1), with

capacity as its bound. Their numbers are benchmark-relative efficiency reports, by their own statement; this track ranks systems by J^* behind a correctness gate (Proposition 8). They have no prediction fixed before the act, no consequence clause and no refusal; the definition requires all three before any bit counts. They give no kinematics of time; Prediction 1 derives an optimal act time and a ceiling from the same joule accounting. Ortega, Braun, Dyer, Kim and Tishby set out information-theoretic bounded rationality [240]. Zénon, Solopchuk and Pezzulo cast the cost of cognition as information [241]. Sims applied rate-distortion theory to perception [242]. Their currency is bits. Landauer converts bits to joules at the reset.

5.2 Demonstrated

Brains. Laughlin, de Ruyter van Steveninck and Anderson measured the energy cost per bit in blowfly photoreceptors and interneurons [243]. Higher information rates cost more per bit. That is a measured ι in a living agent and a measured diminishing return, $\alpha_J < 0$ in sensing. Attwell and Laughlin built the energy budget of grey matter by process [244]. Lennie showed that energy limits how many cortical neurons can be active at once [245]. Energy is the constraint, and the rest adapts. Energy per action potential differs widely across neuron types [246]. Same function, different joules. Neural design follows energy efficiency [247]. Levy and Calvert audited the human cortex. Communication costs 35 times as much as computation, and a neuron's computation sits a factor of 10^8 from the best possible bits per joule [248]. That is a distance to the floor. Harris and colleagues found that a visual pathway synapse is sized to maximize information per unit energy, not information [249]. The objective is bits per joule, measured at one synapse. Padamsey, Katsanevaki, Dupuy and Rochefort found that food restriction cut synaptic ATP (adenosine triphosphate) use in mouse visual cortex by 29% and broadened orientation tuning by 32% [250]. Fewer joules bought less precision. Plaçais and Preat showed that starved flies switch off costly aversive long-term memory, and forcing it back reduced survival [251]. Plaçais and colleagues found that raised energy flux in the mushroom body triggers long-term memory [252]. Mery and Kawecki found that forming long-term memory lowered flies' resistance to desiccation [253]. Learning has a measured metabolic price. Hechler, de Lange and Riedl measured lower cortical oxygen use during confident prediction, by up to 12% (preprint) [254]. Prediction first can save joules. Malkin and colleagues gave synapses an energy cost for reliability in trained networks and found signatures of Bayesian inference in the optimum [255]. Li and van Rossum showed that naive synaptic plasticity costs extreme energy and that caching changes before consolidation saves it [256]. Lynn and colleagues inferred broken detailed balance in the human brain from neuroimaging, rising with physical and cognitive exertion [257]. That entropy production is coarse-grained and informational. It is not a joule meter.

Machines. Memory access costs far more energy than arithmetic [258]. Parameter count matters only through the joules it makes you move. Intelligence per watt measures task accuracy per unit power on local accelerators [259]. Its numerator is benchmark accuracy, not gated agency. MLPerf Power standardizes power measurement from microwatts to megawatts, with each result gated by an accuracy target [260]. That gate is the form of J^*. FrugalGPT cascades language models and stops when a scorer accepts the answer [321]. RouteLLM learns to send

each query to a strong or a weak model under a cost target [322]. Their cost is dollars per query and their gate is a quality threshold. Read with metered joules and $C(z)$ as the gate, both are cascades in the sense of Proposition 9. ML.ENERGY measures inference energy automatically [261]. TokenPowerBench benchmarks the power of large language model (LLM) inference [262]. Jin, Wei and Brooks analyze the energy of test-time compute [263]. General-purpose models cost much more energy per task than task-specific ones [264]. The cheapest sufficient model wins in joules. That is Mixture of Limits, measured. Samsi and colleagues measured the energy of large language model inference on GPUs (graphics processing units) [265]. Dillavou and colleagues demonstrated decentralized learning in a physical circuit, with no processor [266]. Stern and colleagues trained self-learning circuits for power-efficient solutions and measured the trade between power and error [267]. Saggio and colleagues demonstrated a quantum speed-up for learning agents in a photonic experiment [268]. Fewer interactions to learn is fewer acts to close.

A kernel-reported acceptor. Section 8.11 reports a preregistered run of an acceptor agent on a laptop, with prediction first, comparison, refusal and binding joule budgets. ι was read off 4,940 receipts. The median confirmed act gave 1.5×10^5 bits per reported joule, about 2×10^{15} times the joules per bit of the floor. Under a binding budget the σ-law selector closed more tasks than the fixed cascade, one Mixture of Limits instance. That comparison was chosen after the run, so it is exploratory, and joules per closed task were equal. The preregistered second run and a second machine repeated the gain at the loosest budget only (Section 8.11.2). Integrated ι ranked a goal-blind agent first on a loose budget, while J^* ranked it last.

Groups simulated. Crosato and colleagues computed the thermodynamic quantities of simulated collective motion across its critical point [269]. Chen and Prokopenko compared thermodynamic efficiency with purely informational utilities as explanations of collective behavior near criticality [270]. ι_G in Proposition 7 is the gated, metered form of such a ratio.

Meters. RAPL (Running Average Power Limit) is the set of on-die energy counters in Intel and AMD processors. Its readings were validated against external measurement [271]. DRAM (dynamic random-access memory) readings were validated separately [272]. Meter quality depends on the processor generation [273]. Apple's powermetrics manual states that its average power values "are estimated and may be inaccurate" and should not be used to compare devices [274]. On Apple silicon, powermetrics readings are reported values. Measured joules for that machine come from an external wall meter.

Compression scored. The Hutter Prize scores lossless compression of the first gigabyte of an English Wikipedia dump, with a prize fund of 500,000 euros and limits on runtime and memory [275]. It bounds time and memory, never joules. A joule cap would make it a calculus-of-agency benchmark for one task class.

Estimates. These sources are valuable, and their joule figures are modeled, extrapolated or computed. They sit beside measured joules, never in their place. Estimates are never `measured_j`. Strubell, Ganesh and McCallum sampled power for short runs and extrapolated to full training of natural language processing models [276]. Patterson and colleagues computed training energy

from reported hardware, runtime and data-center data [277]. Lacoste and colleagues built a calculator from hardware type, runtime and region [278]. The BLOOM footprint mixes metered and modeled inputs [279]. Carbontracker reads on-device counters and predicts full-run totals [280]. Its predictions are estimates. Elsworth and colleagues report a median of 0.24 Wh per Gemini Apps text prompt, counting accelerators, hosts, idle capacity and data-center overhead [281]. It is an operator-reported figure, filed as `reported_j`.

5.3 Opinion

Each position is placed against the definition: intelligence is the derivative of gated agency in joules, read off a completed run.

- Schmidhuber treats intrinsic reward as compression progress, a first derivative of how well an agent compresses its history [282]. It is the closest precedent for intelligence as a derivative read from a run. Its variable is compression, not joules.
- Chollet defines intelligence as skill-acquisition efficiency over priors and experience, with the ARC (Abstraction and Reasoning Corpus) benchmark [283]. It is a ratio. Its denominator is information and data, not joules. The ARC Prize leaderboard plots cost per task against score as a measure of efficiency [284]. The cost is in dollars. A joule meter turns it into J^*.
- Hernández-Orallo builds universal psychometrics across species and machines [285]. It evaluates by tasks. This track evaluates by joules to close tasks. Hernández-Orallo and Dowe's anytime test adapts to the time available [286]. The calculus adapts to the joules available.
- Gershman, Horvitz and Tenenbaum define intelligence as expected utility net of computation cost [287]. This track makes the cost joules.
- Lieder and Griffiths treat cognition as optimal use of limited resources [288]. It is close to Mixture of Limits. The resource is left abstract.
- Balasubramanian reads the brain's design through its energy limits [289].
- Silver, Singh, Precup and Sutton hold that reward is enough [290]. Reward is chosen by a designer. Joules are read off a meter. Vamplew and colleagues reply that scalar reward cannot express multi-objective goals [291]. This track keeps one variable, joules, and moves multiplicity into the gate $X \geq X_{\text{req}}$.
- Hafez and colleagues define agency and intelligence together [292]. It is not joule-native.
- Karagoz makes energy self-sustainment the objective [293]. It has no prediction gate, no kinematics and no joule ranking across systems.
- Chaisson proposes energy rate density, power per unit mass, as a complexity metric across cosmic evolution [294]. It is a rate in watts per kilogram. The calculus divides confirmed bits by joules.

5.4 Where the calculus stands

- **Proven:** learning, memory, prediction and control each have a joule price with a ceiling.
- **Demonstrated:** joules per bit are measured in brains. Joules per task are measured in machines. Meters can be validated.

- **Proven here:** ι is a bounded density on the joule axis under joule-proportional crediting (Proposition 5). Sampled ι converges to it (Proposition 6). Groups compose by Proposition 7. Holding a goal sets an optimal act time and a ceiling on ι (Prediction 1).
- **Demonstrated here:** ι and J^* read off the receipts of a kernel-reported acceptor, with falsifiers fixed in advance (Section 8.11). J^* ranks agents. Integrated ι does not.
- **Added by this track:** intelligence read as dX/dJ off a gated, completed run, with J^* as the ranking. Takahashi and Hayashi come closest, with bits-per-joule ratios under explicit accounting [239]. This track adds the eligibility gate, the realized flow in place of capacity, the gated ranking and its proof (Proposition 8), and the optimal act time (Prediction 1).

---

6. Global findings, by method

The search ran in Russian, Chinese, Japanese, German, French, Spanish, Portuguese, Korean, Italian, Polish, Hindi and Arabic, with English follow-ups. A second pass searched Japanese, Chinese, Russian, German, French, Spanish and Korean again and added Persian, Turkish and Hebrew. Its results are folded into Sections 2 to 5 and 8 by what they prove, measure or define. Results are organized by method and class of result. Language is metadata only. No entry is ranked by where it came from.

6.1 Theorems and reviews: information has a thermodynamic price

- Poplavskii gave an early systematic treatment of the energy cost of acquiring and processing information [295]. Written in Russian, with an English translation. Layer: definition.
- Stratonovich proved value-of-information theorems [201]. English edition of a Russian monograph. Layer: kinematics.
- Ito and Sagawa applied information thermodynamics on networks to E. coli chemotaxis. Transfer entropy bounds robustness, and information-thermodynamic efficiency is high where ordinary thermodynamic efficiency is low [296]. Written in Japanese. Layers: definition and kinematics.
- Sun and Quan reviewed Maxwell's demon and the physical floor on dissipation in information processing [297]. Written in Chinese. Layer: definition.
- Quan, Dong and Sun reviewed mesoscopic thermodynamics: the demon is consistent with the second law once erasure is counted, and power-efficiency constraints were tested [298]. Written in Chinese with an English abstract. Layers: definition and kinematics.
- Parrondo's encyclopedic review gives the current account of information as a thermodynamic resource [299]. Layer: definition.
- Strasberg's doctoral thesis tests common assumptions in information thermodynamics with physical models [300]. Written in English with a German abstract. Layer: definition.
- Further theorems surfaced through searches in Japanese, Russian, Chinese, Korean, Portuguese, Italian and Polish: speed limits and TURs in chemical networks [127, 131, 132], finite-time thermodynamics [159, 162, 163], Langevin TURs [143, 144], information bounds on entropy production [148, 149], and single-shot work costs [38].

6.2 Experiments: the price measured

- Dago's doctoral thesis reports 1-bit erasure and writing near the minimal energy at high speed on an underdamped micro-cantilever [301]. Written in French. Layer: the definition's joule floor reached at kinematic speed.
- Lagoin's doctoral thesis builds macroscopic Maxwell's demons, including a Szilard engine, from a vane in a granular gas [302]. Written in French. Layer: definition. The demon's accounting holds outside the microscopic regime.
- Ciampini and colleagues used extractable work to witness quantum correlations in a photonic experiment [303]. Layer: definition.
- Experiments surfaced through searches in Korean, Portuguese, Hindi, Italian, Spanish, Chinese and Japanese: information engines [27, 80], NMR demons [73, 74], the TUR on qubits [202], active-bath engines [217, 218], geometric bounds [211], symmetry breaking and Carnot cycles [215, 216], trapped-ion demons [66, 88], and feedback conversion and optimal transport [69, 76, 209].

6.3 Theory of functional systems: prediction before the act

The Anokhin school states clauses 1 and 3 in physiological terms. A result template, the acceptor of action results, is formed before the act, compared after it, and the comparison reorganizes behavior. These works were found through Russian-language searches.

- Anokhin's functional system [93, 94].
- Sudakov on the result as the system-forming factor and on probabilistic prediction of behavior [95, 304, 305, 306].
- Shvyrkov on goals organizing neuronal activity and on learning as selection of neuronal systems [96, 307].
- K. V. Anokhin on the brain's cognitive structure as a hypernetwork of functional-system elements [308].
- Saltykov and Grachev on the system-forming factor as anticipation [309].
- Vityaev on purposefulness formalized as rule learning that predicts results [310].

Implemented and run. Cognitive architectures built on functional-system theory form result templates, compare, and learn subgoals, in simulation and on a physical robot [90, 91, 311, 312, 313]. None meters joules. Adding a meter to one of these agents would test the full order of definition, kinematics and calculus on an existing implementation. Differentiable probabilistic logic networks are related calculus tooling, not built on functional systems, with no eligibility gate and no joules [314].

6.4 Autonomy, organization and minimal agency

Found through Spanish, French and German searches. Agency is a self-maintaining organization that regulates its interactions [103, 104, 105, 106, 107, 108]. The organization is a closure of constraints on thermodynamic flows [109]. The autonomy school has stepped into stochastic

thermodynamics [110]. Layer: definition, as the organization that must exist before any act counts.

6.5 Measured joules for machine intelligence

Found through Chinese-language searches on inference energy: automated inference energy measurement [261], per-token power benchmarks [262], and the energy of test-time reasoning [263].

6.6 Philosophy of agency

Rovelli's physics of agency surfaced through German [115]. Jaeger and Azadi complete the set of position papers [116, 117].

6.7 Searches with no primary research

Hindi and Arabic searches returned no primary research in those languages. English follow-ups surfaced further work, listed above by method. Persian, Turkish and Hebrew searches returned theses, reviews and work already cited, with no new primary research.

---

7. Open problems

7.1 What must be proven

P1. The run path is well defined. Proven. Proposition 5 gives $\iota = dX/dJ$ as a density on the joule axis with $0 \leq \iota \leq 1/(k_B T \ln 2)$, under joule-proportional crediting. Proposition 6 gives convergence of sampled ι as the meter's sampling refines, with the meter's relative error as the error term.

P2. Agency-gated Landauer bound. Under closed-cycle accounting, each confirmed coupled bit costs at least $k_B T \ln 2$, so $v_J \leq 1/(k_B T \ln 2)$. In hand: measurement plus erasure work is at least $k_B T I$ [5], and erasure costs at least $k_B T \ln 2$ per bit [6, 7]. Missing: a data-processing step showing that the coupled bits credited in X cannot exceed the bits recorded in the acceptor's memory during the comparison. Clause 3 already requires the confirmation m to come from a physical record. Proposition 1 proves the per-act floor with this step taken as assumption (R2).

P3. Agency speed limit. Proven for acts whose record meets (R1) to (R4). Proposition 1 bounds the mean joules of act k below by $b_k k_B T \ln 2$ plus a finite-time term in τ_k, $\langle A \rangle_k$, the reset error ε and the confirmed bits, through Fano's inequality. Prediction 1 adds the standing joules of holding a goal and gives the optimal act time.

P4. Precision bound for agents. Proven. Proposition 3 bounds the relative variance of confirmed agency by dynamical activity for the count of confirmed acts, and by entropy production with the information term under feedback.

P5. Hindsight theorem. ι is a function of the completed receipt only. No pre-run quantity fixes it. In hand: each m_k compares an outcome to a prediction fixed before the act, so m_k is unknown before act k closes. Missing: a construction of two worlds that agree on everything a pre-run predictor sees and differ in some m_k. With P1, this makes "intelligence is read off the completed run" a theorem.

P6. Ranking survives meter error. Proven. Proposition 8, part 3: if $(1+\epsilon)J^*_{S_1}(q) < (1-\epsilon)J^*_{S_2}(q)$, with ϵ the meter's relative error bound, then $S_1 \succ_q S_2$ under any reading within that error. Missing: validated ϵ for each meter [271].

P7. Mixture of Limits optimality. Partly proven. Proposition 9 gives the exact conditions for independent gears and for two nested gears, and the budget mechanism. The fixed cascade instance tries gears from cheapest up, stops at the first gear whose output passes $C(z)$, and refuses with a receipt if the next gear would break the budget. It achieves the joule minimum up to the cost of failed attempts. In hand: if sufficiency is monotone in gear order and gear costs rise with order, the first sufficient gear is the cheapest sufficient gear. Missing: a regret bound when sufficiency is not monotone, and the conditions under which this rule and the σ-law selector, $a^* = \arg\max[H(a) - \lambda J(a)]$ subject to $J \leq B$, choose the same act.

P8. When agency compounds. Conditions under which $\alpha_J > 0$ over a run, so that earlier receipts lower the joules of later closures. In hand: an earlier OpenIE toy run found $\alpha_J \leq 0$ on near-optimal tasks. Missing: a model of reuse in which stored receipts cut later J by more than the storage and lookup joules. Still gives the memory cost side [225]. Section 8.10 measures the template's share of each act's joules.

7.2 What must be demonstrated

ID	Claim	Evidence required
D1	J^* per task is measured, not estimated, on at least two fabrics	`measured_j` from RAPL on x86 Linux and an external wall meter on Apple silicon
D2	The σ-law selector closes tasks in fewer measured joules than the fixed cascade under binding budgets	Paired runs with the same tasks, gears, predicates, budgets and meters
D3	Refuse with a receipt works	Zero overspends, refusals on infeasible tasks, few refusals on feasible ones
D4	ι is read off receipts and is stable across seeds	Per-run ι_k series with spread within a preset bound
D5	Estimates do not substitute for measurement	Rank agreement between `est_j` and `measured_j`, reported, never merged
D6	The physical floor is respected and the gap is known	Measured joules per confirmed bit divided by $k_B T \ln 2$
D7	Prediction 1: the optimal act time and the ceiling of an agent that holds a goal	A τ sweep at two or more instability rates, with act and standing joules metered separately and

ID	Claim	Evidence required
		together

Section 8.11 reports the first run against D2 to D6, on reported joules. Section 8.11.1 reports the second run on the same machine, with the guard and refusal fixes.

---

8. Experiment plan

Every table, receipt and figure in this plan follows one rule. Estimates are never `measured_j`. A modeled figure is labeled `est_j`. A metered figure is labeled `measured_j` and names its meter.

8.1 Meters

Acronyms: CPU (central processing unit), GPU (graphics processing unit), ANE (Apple Neural Engine), AC (alternating current), NVML (NVIDIA Management Library).

Tier	Meter	What it reads	Label
M0	External wall-plug power meter, logging at 1 Hz or faster	Whole-system AC power	`measured_j`, wall
M1	RAPL through Linux powercap, package, core, uncore and DRAM domains where present	Cumulative on-die energy in microjoules	`measured_j`, rapl
M1g	NVML total energy counter on a discrete GPU	Cumulative GPU board energy	`measured_j`, nvml
M2	Apple powermetrics, CPU and GPU samplers at 100 ms	CPU, GPU and ANE power as reported by the system	`reported_j`, powermetrics

- **RAPL.** Read the counter before and after each act. Handle wraparound. Validate against M0 on a calibration workload [271, 272].
- **powermetrics.** Readings are filed as `reported_j`. Measured joules on Apple silicon come from M0. Within one machine, `reported_j` attributes joules to CPU, GPU and ANE.
- **Idle baseline.** Record 120 s of idle power on every meter before each block. Report gross and idle-subtracted joules side by side. Never report only the subtracted figure.
- **State control.** Fix the CPU frequency governor where the operating system allows. Log ambient temperature and thermal pressure. Close background jobs. Discard one warm-up run per arm.

8.2 Tasks

Each task has a machine-checkable completeness predicate $C(z)$: unit tests for code, exact match for arithmetic, schema validation for extraction, a verified final state for tool use. Two strata are fixed in advance. Near-optimal tasks are those the cheapest gear usually closes. Compounding tasks are sequences where earlier receipts can be reused. X_{req} is set by $C(z)$. Passing is a gate, not a score.

8.3 Arms

All arms share the gears, the predicate, the budget, the meters and a randomized interleaved run order.

1. **σ-law selector:** $a^* = \arg\max[H(a) - \lambda J(a)]$ subject to $J \leq B$, with refusal on a receipt.
2. **Fixed cascade (one Mixture of Limits instance):** the cheapest-sufficient rule, with gears tried in a fixed order from cheapest up. The fair rival. Same information, gears, budget and refusal right. Only the selection rule differs.
3. **Largest gear always:** reference ceiling.
4. **Smallest gear always:** reference floor, with refusal when it fails.

Both selectors are Mixture of Limits instances. The philosophy is to use the cheapest gear that is sufficient, because past the floor additional joules do not purchase verifiable progress. The fixed cascade applies it in a fixed order. The σ-law selector stops when the next act no longer pays for its joules. That is a threshold-convergence rule, so it is a Mixture of Limits instance too. The comparison is between two instances of one philosophy. In code and data files the fixed cascade is the arm `mol` and the σ-law selector is the arm `sigma`.

8.4 Budgets, seeds and statistics

- A pilot runs arm 2 without a budget and records the median measured joules per task, $\tilde{J}_q$.
- Budget levels are $B \in \{0.25, 0.5, 0.75, 1.0\} \times \tilde{J}_q$, plus unconstrained. A level binds only if at least 20% of tasks exceed it in the pilot. Levels that do not bind are reported and excluded from the D2 test.
- At least 10 seeds per arm, task and budget. More if the pilot shows 10 cannot resolve a 10% difference in median J^* at 95% confidence.
- Report mean, median, standard deviation and a 95% bootstrap confidence interval of J^* per task. Compare arms 1 and 2 with a Wilcoxon signed-rank test on per-task median J^*, with effect size.
- The preregistration names the primary endpoints. Secondary comparisons are corrected with the Holm step-down procedure. An analysis chosen after a run is labeled post hoc wherever it appears, in summaries included, and it is never the headline result. It becomes a primary endpoint only in the preregistration of the next run.

8.5 Receipt schema

The receipt is the Notational Intelligence record of the run. One record per act:

```
act_id, run_id, task_id, arm, seed, budget_j, gear
prediction_hash, prediction_ts        # R*, fixed before the act
outcome_hash, outcome_ts              # R
m                                     # 1, mu, or 0
coupled_bits_est                      # realized I(do(a); W'), bias-corrected, with interval
coupled_bits_bound                    # capacity bound, never credited as X
t_start, t_end
```

```
measured_j   {meter, domains, gross_j, idle_w, idle_subtracted_j}
reported_j   {meter, cpu_j, gpu_j, ane_j}    # powermetrics only
wall_j       {meter, gross_j, idle_w, idle_subtracted_j}
est_j        {model, inputs}                 # never merged with measured_j
cumulative_measured_j
refused, refuse_reason
passed_C
record_states, record_bits            # N and H of the act's record M_k
joule_share                           # P_k attribution when acts overlap
template_j   {meter, write_j, hold_j, reset_j}   # joules for R*
trajectory_j {tau, activity, work}    # physical runs only, Section 8.8
```

8.6 Metrics

- J^* per task: measured joules to pass $C(z)$, from M0 where present, otherwise M1.
- Closure rate per arm and budget.
- Overspends: acts that exceeded B. The target is zero.
- Refusal quality: precision and recall of refusals against tasks shown infeasible within B by any arm.
- Intelligence series: $\iota_k = \Delta X_k / \Delta J_k$ per run, and its spread across seeds.
- Acceleration sign: the sign of α_J from regressing ι_k on cumulative J, per stratum.
- Estimate check: Kendall's rank correlation between `est_j` and `measured_j`.
- Distance to floor: measured joules per confirmed coupled bit divided by $k_B T \ln 2$ at the logged temperature.
- Act intelligence: $\iota_k = b_k / J_k$ per act, which equals the density of Proposition 5 on that act.
- Template share: J^{R^*}/J per act and the curve $\iota(w)$ (Section 8.10).

8.7 Falsifiers, fixed before any run

ID	Observation	Claim defeated
F1	The 95% interval for the paired difference in median J^*, σ-law minus fixed cascade, includes or exceeds zero at every binding budget	The σ-law selector beats its fair rival in joules
F2	Any arm records an overspend	That arm's refusal implementation
F3	Refusal recall on infeasible tasks is below 0.9, or more than 10% of refusals fall on tasks another arm closed within B	Refusal correctness
F4	Kendall's correlation between `est_j` and `measured_j` is below 0.8	Use of that estimator for any ranking
F5	The coefficient of variation of mean ι across seeds exceeds 0.25 on a task	Reading ι at that task's granularity
F6	The fitted α_J is not positive with 95% confidence on the compounding stratum	Compounding on those tasks

ID	Observation	Claim defeated
F7	The mean measured joules per confirmed bit over a run fall below $k_B T \ln 2$, or any act of a macroscopic machine does	The accounting itself: a meter or bookkeeping fault, fixed before any other result is reported
F8	M1 and M0 disagree on a calibration workload by more than the error bound in P6	That RAPL domain as a measured source on that machine
F9	A physical act cycle shows work below the Proposition 1 floor at its logged τ and $\langle A \rangle$	The accounting of that run: a missed jump channel, a miscalibrated potential or an unmetered work source
F10	A stationary run shows $\mathrm{Var}(N)/\langle N \rangle^2$ below $1/\langle K \rangle$ beyond its confidence interval	The jump count of that run: some jumps went uncounted
F11	A controller holds the plant with measured joules per step below $k_B T \ln 2 \log_2 \lvert\lambda\rvert$	The meter or the plant model of that run
F12	The maximizer of measured $\iota(\tau)$ lies outside $\tau^* \pm 2\delta\tau^*$ of Prediction 1, or the slope of $\ln \tau^*$ against $\ln s$ lies outside $-1/2$ by more than two standard errors; or a run's ι exceeds the ceiling of Prediction 1	Additivity of act and standing joules on one meter, with ι as their density; for the ceiling, the accounting of that run

8.8 Physical act cycles

These runs test Propositions 1 to 4 at the physical floor. Work is computed from measured trajectories and the calibrated potential, as in [61, 67, 70]. It is filed as `trajectory_j` and never merged with `measured_j`, `reported_j` or `est_j`.

- **Act cycle on a charge register (Propositions 1 and 4).** A single-electron box or quantum dot holds a one-bit record, as in [68, 70, 71]. Each cycle writes the comparison, acts, and resets the record in time τ. A charge detector counts every tunneling jump, so $\langle A \rangle$ is measured. Sweep τ over at least two decades. Result form: work per cycle against $k_B T \ln 2 + k_B T/(2\tau\langle A \rangle)$, the floor of Proposition 1 for $N = 2$ and $b = 1$.
- **Counting confirmed acts (Proposition 3).** On the same register, cycling in steady state, count confirmed acts N and all jumps K over repeated windows of equal length. Result form: $\mathrm{Var}(N)/\langle N \rangle^2$ against $1/\langle K \rangle$, and the spread of T_n against $1/(\langle T_n \rangle \langle k \rangle)$. On a feedback-trap information engine as in [27], run the forward and the backward experiment, compute σ_I from the measured trajectories, and test part 3.
- **The optimal act time (Prediction 1).** This is the lab test.
  - *Setup.* A colloidal bead in water at temperature T sits in a feedback trap [61]. Its position is read every Δt, and a feedback force imposes a virtual potential $U(x, t)$. Work is computed from the trajectory by stochastic energetics, $W = \sum_t \partial_t U(x_t, t)\, \Delta t$ [319]. One coordinate runs the one-bit act cycle in a virtual double well, reset in time τ. The other is held at the top of a virtual inverted well, $U = -\frac{1}{2} k x^2$, whose instability rate Λ is set by k and the bead's mobility.

    - *Blocks.* For each value of Λ, three blocks run interleaved in randomized order. Block A: acts with no goal held, at nine or more log-spaced values of τ over at least two decades; it fits J_0 and a. Block B: the goal held with no acts; it gives the standing power s. Block C: both together, at the same values of τ; it gives $\iota(\tau)$. At least 200 cycles per point.
    - *Sweep of s.* At least four values of Λ spanning at least one decade, so s spans at least one decade.
    - *Fit.* Fit $J(\tau) = J_0 + a/\tau + s\tau$ to block C by weighted least squares. Read the measured τ^* at the maximum of $\iota(\tau)$ by a local quadratic fit in $\ln \tau$. Compute $\sqrt{a/s}$ and its band $\delta\tau^*$ from blocks A and B. Regress $\ln \tau^*$ on $\ln s$ by least squares and report the slope with its standard error.
    - *Falsification.* The claim fails if the slope lies outside $-1/2$ by more than two standard errors, or a measured τ^* lies outside $\sqrt{a/s} \pm 2\delta\tau^*$, or block C's joules per act differ from the sum of blocks A and B beyond their combined 95% interval. The last test is additivity on one meter, the physical content of the prediction.
    - *Power.* The slope's standard error must be below 0.1. Otherwise the run is reported as inconclusive and the sweep is extended. An inconclusive run is not a pass.
    - *Second substrate.* A single-electron box with a charge detector repeats the protocol, with $\langle A \rangle$ counted jump by jump [70]. Agreement on two substrates is the test.
    - All work is computed from trajectories and filed as `trajectory_j`.
- **Holding an unstable plant (Proposition 2).** A digital controller holds a plant with one unstable mode of fixed $|\lambda|$ per step, at f steps per second. Meter the controller with M0 and M1. Log the record's bits per step. Result form: measured joules per step divided by $k_B T \ln 2 \log_2 |\lambda|$, reported as distance to floor. Sweep $|\lambda|$ and f.

8.9 Living agents: measured ι

Result form: ι in a living preparation, as bits per decision divided by joules per decision, both taken in the same preparation and reported with distance to floor. A living ι counts as measured only when both terms are measured. A joule figure computed from conductances or inferred from imaging is filed as `est_j`. Entropy production inferred from brain imaging is coarse-grained and informational, so it is not a joule meter [257].

- **Bacterial chemotaxis.** Information rate from single-cell behavior in a defined gradient, as Mattingly and colleagues measured it [213]. Joules from the oxygen consumption of the same culture, divided by cell number. The comparison floor is the cost of sending a bit through the cell's signaling channel, from Bryant and Machta [197], beside $k_B T \ln 2$.
- **Mouse visual cortex.** Orientation information per trial from two-photon imaging, and synaptic ATP use from whole-cell recordings, as Padamsey and colleagues obtained both [250]. Food restriction moves the joules. The measured response of information to joules is the sign and size of dX/dJ in a cortex.
- **Fly memory.** Energy flux in the mushroom body during long-term memory formation, imaged as Plaçais and colleagues did [252], per learned association, with retention as the confirmation. Survival under starvation and desiccation is the outcome cost [251, 253].

8.10 The joule price of prediction first

Result form. Split each act's metered joules into the template's share and the rest, $J_k = J_k^{R^*} + J_k^{\text{act}}$. $J_k^{R^*}$ covers writing, holding and resetting $R^*(a)$. Resetting a template with $H(R^*)$ bits of entropy costs at least $H(R^*)\, k_B T \ln 2$ [6, 24]. Memory that does not predict costs dissipation [54]. A template raises ι when the act joules it saves exceed its own joules. The measured result is the curve $\iota(w)$ over template width w, the bits of R^*, with its maximizer w^* and the ratio $\Delta J^{\text{act}} / \Delta J^{R^*}$ at each step.

Protocol. Fix tasks, gears, predicates and budgets. Vary w over a preset grid. An act may stop as soon as its outcome is resolved against the template. Log `template_j` and the act's remaining joules for every act, on the same meters.

Evidence in hand: confident prediction lowered human cortical oxygen use by up to 12%, measured by metabolic imaging (preprint) [254]. In cellular networks, the bits of the past that best predict the future are prohibitively costly [198]. In percept-action loops, work-efficient agents balance prediction against forgetting [234].

8.11 First run: a kernel-reported acceptor on a laptop

This run applies Sections 8.1 to 8.7 to one task class on one machine. Code, task sets, receipts and analysis are in the site repository under `experiments/agency-run`. The preregistration and its falsifiers were committed first (e830bff). The pilot and the locked calibration came next (5828420), and then the main run (35c967a).

Method.

- **Agent and task.** A Python process closes Sudoku puzzles. The completeness predicate $C(z)$ passes when the grid is full, every row, column and box holds 1 to 9, and every given is kept. A confirmed task credits $b = n_{\text{empty}} \log_2 9$ bits, the number of fills the act chooses among. That is the capacity bound of Definition 1. The test set holds 30 puzzles, 10 per stratum (easy, medium, hard), generated from a fixed seed. A separate set of 30 calibrates the selectors.
- **Gears.** g0 fills naked singles. g1 adds hidden singles. g2 adds depth-first search.
- **Arms.** The σ-law selector takes $a^* = \arg\max [H(a) - \lambda J(a)]$, with $H(a)$ the calibrated pass probability given the gears already tried, times b, and $\lambda = b/B$. The fixed cascade (one Mixture of Limits instance) takes the cheapest gear whose calibrated pass probability is at least 0.5. Both fix R^* before each act, hash it with SHA-256 (Secure Hash Algorithm, 256-bit) and timestamp it. Both compare after the act and refuse with a receipt when no act fits the budget. The goal-blind baseline draws one gear at random, with no prediction, no comparison, no escalation and no refusal. The budget cuts it off.
- **Budgets and seeds.** Per task, $B \in \{0.5, 1.0, 2.0\} \times J_{g2}(q)$, the pilot median of g2 alone. Levels 0.5 and 1.0 bind: 98% and 40% of pilot runs exceed them. Level 2.0 does not (0.7%). There are 20 seeds per condition, in a seeded interleaved order. In all, 5,400 episodes ran, with 4,940 acts.

- **Energy label.** Every joule is `reported_j`: the kernel's per-process energy estimate for the agent (`ri_energy_nj` from `proc_pid_rusage`), read before and after each act, with no privileges. The process is the agent's wall. `measured_j` is empty, because no wall meter was attached. `est_j` is cycles times 4.20×10^{-10} J, the pilot's median ratio.

Results. Mean ± standard deviation over 20 seeds. A run is one arm at one budget level over all 30 tasks. CV is the coefficient of variation, standard deviation over mean.

Arm	B/J_{g2}	Tasks closed of 30	X (bits)	J (reported, mJ)	ι (bits per reported joule)	CV of ι	Refusals	Cuts	Overspends
σ-law	0.5	0.60 ± 0.50	108 ± 91	20.8 ± 0.3	5,254 ± 4,402	0.84	29.40	0	0
σ-law	1.0	9.10 ± 1.86	1,479 ± 316	43.8 ± 4.0	34,454 ± 10,007	0.29	20.90	0	9
σ-law	2.0	28.40 ± 0.60	4,823 ± 105	66.2 ± 5.8	73,345 ± 6,443	0.09	1.60	0	0
Fixed cascade (one Mixture of Limits instance)	0.5	0.70 ± 0.47	126 ± 85	18.8 ± 0.3	6,801 ± 4,569	0.67	29.30	0	0
Fixed cascade (one Mixture of Limits instance)	1.0	6.45 ± 1.50	1,047 ± 254	44.7 ± 3.6	23,804 ± 7,067	0.30	23.55	0	7
Fixed cascade (one Mixture of Limits instance)	2.0	27.50 ± 0.51	4,668 ± 92	67.0 ± 6.7	70,418 ± 7,766	0.11	2.50	0	0
Goal-blind	0.5	0.25 ± 0.44	45 ± 80	14.1 ± 3.1	2,775 ± 4,999	1.80	0	20.35	20
Goal-blind	1.0	3.85 ± 1.84	623 ± 301	26.6 ± 5.7	24,153 ± 13,364	0.55	0	17.45	0
Goal-blind	2.0	18.25 ± 2.71	3,034 ± 471	34.1 ± 9.7	92,724 ± 15,690	0.17	0	1.35	0

- **Per act.** 1,902 acts were confirmed. The median act ι was 1.53×10^5 bits per reported joule, with interquartile range 8.3×10^4 to 4.7×10^5. Joules per confirmed bit were at least 3.8×10^{14} times $k_B T \ln 2$ at 300 K, with median 2.3×10^{15} times. The floor allows 3.5×10^{20} bits per joule.
- **Joules per closed task.** The median over arms was 0.34 to 0.38 mJ for easy tasks, 1.58 to 1.70 mJ for medium and 4.05 to 4.28 mJ for hard.
- **Selectors, paired by seed (exploratory, post hoc).** This analysis was chosen after the run, so it is exploratory. At $B = 1.0\,J_{g2}$ the σ-law selector closed 2.65 more tasks per run than the fixed cascade (95% bootstrap interval 1.5 to 3.8). Its run ι was higher by 10,650 bits per joule (4,747 to 16,592). At $B = 2.0\,J_{g2}$ it closed 0.9 more (0.55 to 1.25). The σ-law

rule goes straight to the gear that suffices, so the joules of failed light attempts stay in the budget. Joules per closed task were equal: on the 4 tasks both arms closed at $B = 1.0\,J_{g2}$, the paired interval ran from -7.9 to $+15.5$ µJ (R-F1).

- **Package cross-check.** powermetrics, run non-interactively over 30 s windows, read about 12.1 W of CPU power and 13.6 W of GPU power from other work on the machine while the agent slept. The agent added about 0.6 W on its own meter. Background CPU power moved by about 0.5 W between windows, so idle-subtracted package joules came out negative. The per-process meter is the run's meter. D1 and D5 need a wall meter on an otherwise idle machine.

Falsifier outcomes. All were fixed before the run.

ID	Observation	Outcome
R-F1	95% interval of the paired per-task difference in median J^*, σ-law minus fixed cascade, includes zero at every binding level	Fired. At $B = 1.0$, 4 tasks closed by both: interval -7.9 to $+15.5$ µJ. At $B = 0.5$, no task closed by both. On tasks both close, the joules are equal
R-F2	Any episode ends above its budget	Fired. 36 of 5,400. σ-law 9 and fixed cascade 7 at $B = 1.0$, 0.07% to 3.6% over, when the final comparison ran outside the guard. Goal-blind 20 at $B = 0.5$, up to 33% over, on budgets a few meter checks wide
R-F3	Refusal recall below 0.9 on infeasible task levels, or over 10% of refusals on task levels some arm closes in half the seeds	Fired. Recall 1.00 on 41 infeasible task levels. 13.1% of refusals fell on task levels some arm closed
R-F4	Kendall's tau-b between `est_j` and `reported_j` per act below 0.8	Held. 0.96 on 4,000 acts
R-F5	CV of run ι above 0.25 in any condition	Fired at binding levels (0.29 to 1.80). Held at $B = 2.0$ (0.09, 0.11, 0.17)
R-F6	Ranking by J^* differs from ranking by integrated ι	Fired. At $B = 2.0$, J^* ranks σ-law, fixed cascade, goal-blind. Integrated ι ranks goal-blind first. At $B = 1.0$ the last two swap
R-F7	A confirmed act below $k_B T \ln 2$ per bit	Held. The lowest act sat 3.8×10^{14} times above
R-F8	Goal-blind run ι at least the better selector's at a binding level	Held. 2,775 against 6,801 at $B = 0.5$. 24,153 against 34,454 at $B = 1.0$

What the run settles.

- ι is read off receipts, act by act and run by run, as Propositions 5 and 6 require. On a loose budget it is stable across seeds.

- J^* is the ranking, and integrated ι is not. The goal-blind agent had the highest run ι on the loose budget, closing 18 of 30 tasks against 28. Its failures were cheap. J^* is gated by $C(z)$, so it ranks that agent last. This is why Section 2.3 ranks by J^*, with X_{req} as a gate.
- Exploratory, post hoc: under a binding budget the σ-law selector closed more tasks than the fixed cascade, and joules per closed task were equal. The gain lies in which tasks fit the budget, not in a cheaper closure of the same task. Section 8.11.2 states what the later runs replicate.
- Two implementation fixes come before the next run. The comparison and the receipt write move inside the guarded region. A measured closing cost is charged before each act.

8.11.1 Second run: guarded acceptor, 40 seeds

This run is the same task class, the same gears and the same three arms as Section 8.11, on the same laptop. The preregistration for the second run was committed first (610dd4c). The pilot and the locked calibration came next (c88511f), and then the main run (920d7e2). Prediction 1 has no software analogue in this run: the agent has no declared act cost that falls with act time and no standing cost it controls, so the preregistration states that the dose-response design is not run.

Method.

- **Guard.** Every joule of an episode is charged inside the guard: selection, hash, gear steps, comparison, receipt and final read. The opening check uses spent $= 0$. The law arms refuse when the guard fails. The goal-blind arm is cut.
- **Refusal threshold.** Both selectors use the r-quantile of calibration costs for budget-fit tests. r was locked from calibration replay before the main run: $r = 0.1$, with recall 1.0 and precision about 0.976 on the calibration split.
- **Budgets and seeds.** Per task, $B \in \{0.5, 1.0, 2.0\} \times J_{g2}(q)$, the pilot median of g2 alone. Levels 0.5 and 1.0 bind: pilot fractions exceeding them are 0.98 and 0.4. Level 2.0 does not (0.0067). There are 40 seeds per condition. In all, 10,800 episodes ran.
- **Quiet window.** Quiet was not achieved. Probes ran until the limit. The last probe before the start, at 21:46 local time, recorded other-process CPU near 621% of one core, with mediaanalysisd and editor helpers at the top. Baseline package power was about 21.4 W. Package power in the main window was about 22.7 W. The run records `quiet_window = no`.
- **Energy label.** Every joule is `reported_j`: the kernel's per-process energy estimate for the agent (`ri_energy_nj` from `proc_pid_rusage`), with package figures from powermetrics where used for the cross-check. `est_j` is cycles times $1.7826647100706183 \times 10^{-9}$ J, locked in the pilot. `measured_j` is empty: no wall meter was attached.
- **Package cross-check.** In the main window the process meter read 42.35 J. Net package joules were 121.81 J, a ratio of about 2.88 to the process. Baseline package drift was about -0.082 W. Because `quiet_window = no`, the cross-check is declared not achievable. The per-process meter remains the run's meter.

Results. Mean ± standard deviation over 40 seeds. A run is one arm at one budget level over all 30 tasks. CV is the coefficient of variation, standard deviation over mean. J is reported in millijoules for comparison with Section 8.11.

Arm	B/J_{g2}	Tasks closed of 30	X (bits)	J (reported, mJ)	ι (bits per reported joule)	CV of ι	Refusals	Cuts	Overspends
σ-law	0.5	0.62 ± 0.49	113 ± 89	59.5 ± 2.0	1,942 ± 1,525	0.785	29.38	0	0
σ-law	1.0	2.90 ± 1.74	523 ± 314	133.7 ± 13.8	4,150 ± 2,892	0.697	27.10	0	0
σ-law	2.0	22.73 ± 1.30	3,980 ± 206	197.3 ± 18.2	20,359 ± 2,388	0.117	7.28	0	0
Fixed cascade (one Mixture of Limits instance)	0.5	0.57 ± 0.50	104 ± 90	60.1 ± 2.1	1,777 ± 1,548	0.871	29.43	0	0
Fixed cascade (one Mixture of Limits instance)	1.0	2.67 ± 1.25	483 ± 226	140.3 ± 13.5	3,593 ± 1,948	0.542	27.32	0	0
Fixed cascade (one Mixture of Limits instance)	2.0	19.93 ± 1.42	3,509 ± 227	201.2 ± 20.4	17,684 ± 2,909	0.164	10.07	0	0
Goal-blind	0.5	0.12 ± 0.33	23 ± 61	35.4 ± 11.6	513 ± 1,397	2.723	0	22.85	0
Goal-blind	1.0	1.00 ± 0.93	181 ± 169	72.0 ± 19.5	2,661 ± 2,579	0.969	0	20.00	0
Goal-blind	2.0	10.65 ± 2.27	1,859 ± 400	97.3 ± 30.1	20,302 ± 5,766	0.284	0	8.70	0

- **Per act.** 7,960 acts ran. 2,448 were confirmed. Kendall's tau-b between `est_j` and `reported_j` was 0.942 on 4,000 acts. The median act ι was about 3.11×10^4 bits per reported joule. Every confirmed act sat far above $k_B T \ln 2$ per bit: the lowest was about 6.4×10^{14} times that floor.
- **Refusal.** Recall was 1.0 on 49 infeasible pairs. There were 5,223 refusals. The share on tasks closed by some arm was about 0.081.
- **Overspends.** Zero episodes ended above budget. The maximum overspend fraction was -0.03684: every episode finished under budget.
- **Selectors, paired by seed.** This analysis was declared in the preregistration. At $B = 2.0\, J_{g2}$ the σ-law selector closed 2.8 more tasks per run than the fixed cascade (95% bootstrap interval 2.3 to 3.275). Its run ι was higher by 2,674 bits per joule (1,831 to 3,504). On 39 of 40 seeds the σ-law selector closed more. At $B = 0.5$ and $B = 1.0$ the intervals for the closed-task difference include zero. These results fire no falsifier.

Falsifier outcomes. All were fixed before the run. The table states the second-run outcome beside the first-run outcome.

ID	Observation	Run 2	Run 1
R-F1	At every binding level, the 95% interval of the paired per-task difference in median J^*, σ-law minus fixed cascade, includes or exceeds zero	Held. At $B = 1.0$, no task closed by both. At $B = 0.5$, one task closed by both. Shared-task joules still do not show a clear σ-law saving	Fired
R-F2	Any episode ends above its budget	Held. 0 of 10,800. The guard charges comparison, receipt and final read	Fired
R-F3	Refusal recall below 0.9 on infeasible pairs, or over 10% of refusals on pairs some arm closes	Held. Recall 1.0 on 49 infeasible pairs. Share on tasks closed by some arm about 0.081. The r-quantile refusal threshold is the change from Run 1	Fired
R-F4	Kendall's tau-b between `est_j` and `reported_j` per act below 0.8	Held. 0.942 on 4,000 acts	Held
R-F5	CV of run ι above 0.25 in any condition	Fired at binding levels (0.54 to 2.72). Held for both law arms at $B = 2.0$ (0.117, 0.164)	Fired
R-F6	Ranking by J^* differs from ranking by integrated ι	Fired. At $B = 2.0$, J^* ranks σ-law, fixed cascade, goal-blind. Integrated ι ranks σ-law, goal-blind, fixed cascade	Fired
R-F7	A confirmed act below $k_B T \ln 2$ per bit	Held. The lowest act sat about 6.4×10^{14} times above	Held
R-F8	Goal-blind run ι at least the better selector's at a binding level	Held. 513 against 1,942 at $B = 0.5$. 2,661 against 4,150 at $B = 1.0$	Held

What the run settles.

- The guard closes R-F2. Zero overspends.
- The refusal quantile closes R-F3. Recall stays 1.0 and the share of refusals on closable tasks falls under 10%.
- J^* remains the ranking. Integrated ι still disagrees with it on the loose budget (R-F6).
- Run ι is still unstable across seeds at binding budgets (R-F5).
- On shared closed tasks the σ-law selector still does not show fewer joules than the fixed cascade in the sense that would defeat R-F1's observation at every binding level. The paired-by-seed gain at $B = 2.0$ is more closed tasks, not cheaper closures of the same tasks.

8.11.2 Second machine: a replication with a package energy counter

This run repeats the second-run protocol of Section 8.11.1 on a second machine with a different energy counter. It is a replication of the selector comparison, not a measurement of energy. Code, receipts and summaries are in `experiments/agency-run`: `PREREGISTRATION_v2_jetson.md` and its Amendment 1, `results_jetson/summary.md` (part A) and

`results_jetson_cyc/summary.md` (part B). The preregistration addendum was committed first (cb059f0), then the meter backend and harness (1938577), then Amendment 1 (7703822). The part A pilot (6ddd086) and main run (a652503) came next, then the part B pilot (3fde1fb) and main run (1bb2444).

Method.

- **Machine.** A Linux laptop with a 14-core x86 processor: 6 performance cores with 2 threads each and 8 efficiency cores. Turbo was off. Every core was pinned to its base frequency: 2.6 GHz on performance cores and 1.9 GHz on efficiency cores. The agent ran pinned to one performance-core thread, one episode at a time. The median agent-core frequency during blocks was 2.6 GHz. The original settings were restored after the run.
- **Unchanged.** Tasks, predicate, coupled bits, gears g0 to g2, the three arms, the guard, the refusal replay, the budget levels $B \in \{0.5, 1.0, 2.0\} \times J_{g2}(q)$, 40 seeds and 10,800 episodes per part, falsifiers R-F1 to R-F8 and the paired-by-seed analysis are those of Section 8.11.1.
- **Energy counter.** RAPL (Running Average Power Limit) is the processor's on-chip package energy counter. Every interval was charged the package joules minus the idle baseline power of that block times the interval's seconds. RAPL is a model and sensor inside the chip, not a wall meter. It counts every process on the package.
- **Quiet window.** Quiet was not achieved: `quiet_window = no`. A CI (continuous integration) runner built Rust code on the machine throughout. The 40 probes before the start saw other processes using from 1.0 to 20.0 cores and package power from 2.3 to 28.7 W. The per-block gate was met before 180 of 360 blocks in part A and 244 of 360 in part B. 98 and 208 blocks were quiet for their whole length.
- **Part A.** The agent's meter is RAPL read before and after each act. Every figure is `reported_j`.
- **Part B.** Amendment 1 added this part before any pilot, because a harness test showed that acts last about 0.01 to 6 ms while RAPL updates about every 1 ms in steps of about 5 to 60 mJ. The agent's meter is `est_j`: the agent thread's user cycles times c. c was calibrated on RAPL in the part B pilot, net of idle, over 8 rounds of 3 s, on the second attempt: $c = 6.576 \times 10^{-10}$ J per cycle, from 41.11 J of net package energy over 62,516,635,701 cycles. Every part B joule figure is `est_j`. RAPL figures logged beside it stay `reported_j`.
- **Labels.** `measured_j` is empty in both parts. No wall meter or shunt meter was attached.
- **Arm names.** Arm names and their code fields follow Section 8.3. The goal-blind baseline is `blind`.

Part A results: RAPL read per act. Mean ± standard deviation over 40 seeds. J is `reported_j` in joules, as in the summary. Levels 0.5, 1.0 and 2.0 all bind: 0.647, 0.40 and 0.353 of pilot g2 runs exceed them.

Arm	B/J_{g2}	Tasks closed of 30	X (bits)	J (reported, J)	ι (bits per reported joule)	CV of ι	Refusals	Cuts	Overspends
σ-law	0.5	0.00	0	0.0018	0	n/a	30.00	0.00	323
σ-law	1.0	0.00	0	0.0002	0	n/a	30.00	0.00	322
σ-law	2.0	1.00 ± 0.39	181 ± 71	0.0018 ± 0.0336	14,427 ± 18,680	1.295	29.00	0.00	322
Fixed cascade (one Mixture of Limits instance)	0.5	0.00	0	-0.0008	0	n/a	30.00	0.00	326
Fixed cascade (one Mixture of Limits instance)	1.0	0.00	0	-0.0005	0	n/a	30.00	0.00	324
Fixed cascade (one Mixture of Limits instance)	2.0	1.00 ± 0.51	181 ± 91	0.0107 ± 0.0286	15,451 ± 23,767	1.538	29.00	0.00	320
Goal-blind	0.5	0.00	0	0.0026	0	n/a	0.00	30.00	327
Goal-blind	1.0	0.00	0	0.0009	0	n/a	0.00	30.00	323
Goal-blind	2.0	0.28 ± 0.45	50 ± 82	0.0019 ± 0.0180	3,801 ± 13,749	3.618	0.00	28.90	321

- **Counter resolution.** The median episode lasted about 0.04 ms. 10,075 of 10,800 episodes saw no change in the package counter. 10,549 had `reported_j` at or below zero after idle subtraction.
- **Budgets.** Pilot medians of J_{g2} ranged from -0.0110 to 0.0914 J, median 0.0097 J. The guard's open threshold was 0.117 J. Budgets fell below it for 30, 30 and 28 of 30 tasks at the three levels, so the law arms refused nearly every task.
- **Overspends.** Every condition recorded 320 to 327 overspends. The maximum overspend fraction was 10.86. A charge that reads zero for most acts and one whole counter step for a few cannot hold a budget of a few millijoules.
- **Per act.** 137 acts ran. 91 were confirmed. 44 acts read zero. 26 confirmed acts read zero or less. Kendall's tau-b between `est_j` and `reported_j` was 0.368 on 93 acts.
- **Cross-check.** Block package net was 3.445 J against 2.737 J summed over episodes, a ratio of 0.794. Idle package power was 4.01 W before the run and 5.55 W after. The cross-check is declared not achievable.
- **Selectors, paired by seed.** At $B = 2.0\,J_{g2}$ the closed-task difference was 0 (95% bootstrap interval -0.15 to 0.15). The run ι difference was 1,328 bits per reported joule ($-8,971$ to $12,581$). 30 of 40 seeds tied. At 0.5 and 1.0 no arm closed a task.

Part B results: cycle count times a RAPL-calibrated constant. Mean ± standard deviation over 40 seeds. J is `est_j` in joules, as in the summary. Levels 0.5 and 1.0 bind: 0.98 and 0.40 of pilot g2 runs exceed them. Level 2.0 does not (0.0).

Arm	B/J_{g2}	Tasks closed of 30	X (bits)	J (estimated, J)	ι (bits per estimated joule)	CV of ι	Refusals	Cuts	Overspends
σ-law	0.5	1.10 ± 0.98	200 ± 178	0.0345 ± 0.0017	6,027 ± 5,458	0.906	28.90	0.00	0
σ-law	1.0	8.75 ± 3.14	1,574 ± 565	0.0805 ± 0.0098	20,519 ± 8,516	0.415	21.25	0.00	0
σ-law	2.0	28.60 ± 0.98	4,882 ± 154	0.1008 ± 0.0234	50,667 ± 10,088	0.199	1.40	0.00	0
Fixed cascade (one Mixture of Limits instance)	0.5	1.15 ± 0.92	208 ± 167	0.0366 ± 0.0013	5,833 ± 4,757	0.816	28.85	0.00	0
Fixed cascade (one Mixture of Limits instance)	1.0	9.22 ± 3.07	1,660 ± 551	0.0813 ± 0.0106	21,412 ± 8,130	0.380	20.77	0.00	0
Fixed cascade (one Mixture of Limits instance)	2.0	27.77 ± 2.30	4,745 ± 377	0.1099 ± 0.0339	47,108 ± 13,216	0.281	2.23	0.00	0
Goal-blind	0.5	0.20 ± 0.46	36 ± 84	0.0210 ± 0.0055	1,722 ± 3,927	2.280	0.00	21.50	0
Goal-blind	1.0	3.73 ± 2.12	667 ± 381	0.0436 ± 0.0140	16,393 ± 9,246	0.564	0.00	16.68	0
Goal-blind	2.0	15.65 ± 2.50	2,662 ± 433	0.0499 ± 0.0160	57,266 ± 14,596	0.255	0.00	4.17	0

- **Per act.** 7,800 acts ran. 3,847 were confirmed. The median act ι was 69,892 bits per estimated joule, with interquartile range 35,759 to 146,697. Every confirmed act sat far above $k_B T \ln 2$ per bit: the lowest was about 3.6×10^{14} times that floor, the median about 5.0×10^{15} times.
- **Refusal.** Recall was 1.0 on 45 infeasible pairs. There were 4,136 refusals. The share on tasks closed by some arm was about 0.071.
- **Overspends.** Zero episodes ended above budget. The maximum overspend fraction was -0.03977.
- **Shared tasks.** At $B = 1.0$, 9 tasks closed by both selectors. The paired per-task difference in median J^*, σ-law minus fixed cascade, had mean 1.02×10^{-4} J (95% interval -5.23×10^{-4} to 6.03×10^{-4}, Wilcoxon $p = 0.214$). At $B = 0.5$ one task closed by both,

with difference -6.68×10^{-6} J. At $B = 2.0$, 29 tasks closed by both, with mean difference -2.65×10^{-4} J (-6.63×10^{-4} to 5.67×10^{-5}, $p = 0.358$).

- **Selectors, paired by seed.** At $B = 2.0\ J_{g2}$ the σ-law selector closed 0.825 more tasks per run than the fixed cascade (95% bootstrap interval 0.125 to 1.575). Its run ι was higher by 3,559 bits per estimated joule ($-1,068$ to $8,320$). 9 of 40 seeds favored the σ-law selector and 27 tied. At $B = 0.5$ the closed-task difference was -0.05 (-0.45 to 0.35). At $B = 1.0$ it was -0.475 (-1.8 to 0.775).
- **The estimate against the counter.** Kendall's tau-b between each episode's `est_j` and the same episode's RAPL package net was -0.037 on 4,000 episodes. 5,258 of 10,800 episodes saw no change in the package counter. Summed over blocks, `est_j` was 23.23 J against 17.14 J of RAPL package net, a ratio of 1.355. Idle package power was 9.37 W before the run and 3.17 W after. The cross-check is declared not achievable.

Falsifier outcomes. All were fixed before the run. Part B replaces R-F4 with the episode test against RAPL, as Amendment 1 states. In Run 2 and in the two parts here, "Held" means the observation did not occur.

ID	Observation	Part A (RAPL per act)	Part B (cycle model)	Run 2
R-F1	At every binding level, the 95% interval of the paired per-task difference in median J^*, σ-law minus fixed cascade, includes or exceeds zero	Held. One task closed by both, at $B = 2.0$, with difference -0.0052 J. No task at the other levels	Held. At $B = 1.0$ the interval includes zero. At $B = 0.5$ the single shared task gives -6.68×10^{-6} J	Held
R-F2	Any episode ends above its budget	Fired. 320 to 327 per condition	Held. 0 of 10,800	Held
R-F3	Refusal recall below 0.9 on infeasible pairs, or over 10% of refusals on pairs some arm closes	Held. Recall 1.0 on 88 pairs. Share 0.0011	Held. Recall 1.0 on 45 pairs. Share 0.071	Held
R-F4	Part A: Kendall's tau-b between `est_j` and `reported_j` per act below 0.8. Part B: tau-b between episode `est_j` and episode RAPL net below 0.8	Fired. 0.368 on 93 acts	Fired. -0.037 on 4,000 episodes	Held
R-F5	CV of run ι above 0.25 in any condition	Fired (1.295 to 3.618)	Fired. 0.380 to 2.280 at binding levels. At $B = 2.0$: σ-law 0.199, fixed cascade 0.281, goal-blind 0.255	Fired
R-F6	Ranking by J^* differs from ranking by integrated ι	Fired. At $B = 2.0$, J^* ranks σ-law, fixed cascade, goal-blind. Integrated ι ranks fixed cascade first	Fired. At $B = 2.0$, J^* ranks σ-law, fixed cascade, goal-blind. Integrated ι ranks goal-blind, σ-law, fixed cascade	Fired

ID	Observation	Part A (RAPL per act)	Part B (cycle model)	Run 2
R-F7	A confirmed act below $k_B T \ln 2$ per bit	Fired. 26 confirmed acts read zero or less, which places them below the floor. That is counter quantization	Held. The lowest act sat about 3.6×10^{14} times above	Held
R-F8	Goal-blind run ι at least the better selector's at a binding level	Fired. At $B = 0.5$ and 1.0 every arm reads 0	Held. 1,722 against 6,027 at $B = 0.5$. 16,393 against 21,412 at $B = 1.0$	Held

What the replication settles.

- Per-act RAPL cannot meter acts shorter than one millisecond. Most episodes saw no counter update, so the guard refused or overspent from quantization alone. Part A is a meter-resolution result. It says nothing about the selectors.
- With a cycle meter, the selection result has the same sign as Run 2 at $B = 2.0$: the σ-law selector closed more tasks per run. The gain is smaller (0.825 against 2.8), and level 2.0 did not bind on this machine. At the binding levels the intervals include zero.
- The cycle model did not track the package counter episode by episode (tau-b -0.037), and its block total was 1.355 times the counter's net. Part B joules rank acts by cycles. They are not energy readings.
- Shared-task joules show no σ-law saving here either. Where the σ-law selector gains, it gains in tasks closed, not in cheaper closure of the same task.
- No figure in this section is `measured_j`.

What replicates, and what does not. The run 1 selector result was exploratory. Two later runs tested it under preregistration.

Claim from run 1 (exploratory)	Run 2 (preregistered)	Second machine, part B (preregistered)
More tasks closed at $B = 1.0\,J_{g2}$: +2.65 (1.5 to 3.8)	Not replicated. The interval includes zero	Not replicated. -0.475 (-1.8 to 0.775)
More tasks closed at $B = 2.0\,J_{g2}$: +0.9 (0.55 to 1.25)	Replicated. +2.8 (2.3 to 3.275)	Replicated. +0.825 (0.125 to 1.575)
Joules per closed task	Equal. No shared-task saving	Equal. Shared-task intervals include zero

The replicated result is narrow. At the loosest budget level the σ-law selector closes more tasks than the fixed cascade, on two machines. No run shows fewer joules for the same closed task, and the binding-budget gain of run 1 did not replicate. Part A is a meter-resolution result and tests neither claim.

Measurement protocol for the next run. Both parts fail at the meter, so the protocol changes before the selectors are tested again.

1. **Idle machine.** Stop nothing that belongs to someone else. Wait for a quiet window, or schedule one: other processes at or below 0.5 cores for the whole run, with `quiet_window = yes` recorded. If quiet is not reached, the run does not start.
2. **Wall or shunt meter.** Attach a wall power meter, or a shunt meter on the supply rail, sampled faster than one act. Only its readings fill `measured_j`.
3. **Resolution check first.** Before the pilot, record the meter's update interval and step size against the median act length. If the meter cannot resolve one act, meter blocks of acts and attribute by count, and say so in the preregistration.
4. **Calibrate the estimate on the meter.** Fit `est_j` on the wall or shunt meter at block level, and require R-F4 to hold before `est_j` drives the guard.
5. **Pin and record.** Pin frequency, record idle baselines before and after, report drift, and keep the package counter as a labeled cross-check.
6. **Task closure is primary.** The next run's preregistration makes task closure the primary endpoint: the paired-by-seed difference in tasks closed, σ-law minus fixed cascade, at each budget level, with the Holm step-down correction across levels. Shared-task joules (R-F1) are secondary.

8.12 Reporting

Publish all receipts, raw meter traces, idle baselines and analysis code with the result. Report measured, reported and estimated joules in separate columns. Never sum across tiers. State which falsifiers fired, with the same prominence as those that passed. Every run also answers the checklist of Takahashi and Hayashi [239]: the accounting boundary; the energy balance terms (stored energy and exported work, or a statement that they are negligible); the baseline and whether joules are total or incremental; the coarse-graining of outcomes; the horizon and reset protocol; wall-clock time and mean power beside bits per joule; and the estimator of coupled bits with its bias correction.

8.13 Where the products sit

- **Notational Intelligence** is the receipt format in Section 8.5.
- **Mixture of Limits** is the philosophy both selectors instantiate. Arm 2 is its fixed cascade instance.
- **The Reality Code Engine** reads $\iota = dX/dJ$ in hindsight from its hash-chained log, with X the coding tasks through the gate and J their labeled joules (The Reality Code Engine, Section 9).

Hardware and product placement moved to the Metabolic Intelligence white paper. The meters of this plan are M0 and M1 in Section 8.1.

References

Entries marked “preprint” are cited by their arXiv posting. Entries published in a language other than English say so.

1. Pearl, J. (1995). Causal diagrams for empirical research. *Biometrika* 82:669-688. https://doi.org/10.1093/biomet/82.4.669
2. Everitt, T., Hutter, M., Kumar, R. and Krakovna, V. (2019). Reward tampering problems and solutions in reinforcement learning: a causal influence diagram perspective. arXiv:1908.04734 (preprint).
3. Sagawa, T. and Ueda, M. (2008). Second law of thermodynamics with discrete quantum feedback control. *Physical Review Letters* 100:080403. https://doi.org/10.1103/PhysRevLett.100.080403
4. Sagawa, T. and Ueda, M. (2010). Generalized Jarzynski equality under nonequilibrium feedback control. *Physical Review Letters* 104:090602. https://doi.org/10.1103/PhysRevLett.104.090602
5. Sagawa, T. and Ueda, M. (2009). Minimal energy cost for thermodynamic information processing: measurement and information erasure. *Physical Review Letters* 102:250602. https://doi.org/10.1103/PhysRevLett.102.250602
6. Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development* 5:183-191. https://doi.org/10.1147/rd.53.0183
7. Bennett, C. H. (1982). The thermodynamics of computation: a review. *International Journal of Theoretical Physics* 21:905-940. https://doi.org/10.1007/BF02084158
8. Ay, N. and Polani, D. (2008). Information flows in causal networks. *Advances in Complex Systems* 11:17-41. https://doi.org/10.1142/S0219525908001465
9. Klyubin, A. S., Polani, D. and Nehaniv, C. L. (2005). Empowerment: a universal agent-centric measure of control. *2005 IEEE Congress on Evolutionary Computation* 1:128-135. https://doi.org/10.1109/CEC.2005.1554676
10. Kenton, Z., Kumar, R., Farquhar, S., Richens, J. et al. (2023). Discovering agents. *Artificial Intelligence* 322:103963. https://doi.org/10.1016/j.artint.2023.103963
11. James, R. G., Barnett, N. and Crutchfield, J. P. (2016). Information flows? A critique of transfer entropies. *Physical Review Letters* 116:238701. https://doi.org/10.1103/PhysRevLett.116.238701
12. Lizier, J. T. and Prokopenko, M. (2010). Differentiating information transfer and causal effect. *The European Physical Journal B* 73:605-615. https://doi.org/10.1140/epjb/e2010-00034-5
13. Janzing, D., Balduzzi, D., Grosse-Wentrup, M. and Schölkopf, B. (2013). Quantifying causal influences. *The Annals of Statistics* 41. https://doi.org/10.1214/13-AOS1145
14. Albantakis, L., Marshall, W., Hoel, E. and Tononi, G. (2019). What caused what? A quantitative account of actual causation using dynamical causal networks. *Entropy* 21:459. https://doi.org/10.3390/e21050459

15. Aguilera, M., Millidge, B., Tschantz, A. and Buckley, C. L. (2022). How particular is the physics of the free energy principle? *Physics of Life Reviews* 40:24-50. https://doi.org/10.1016/j.plrev.2021.11.001
16. Biehl, M., Pollock, F. A. and Kanai, R. (2021). A technical critique of some parts of the free energy principle. *Entropy* 23:293. https://doi.org/10.3390/e23030293
17. Bruineberg, J., Dołęga, K., Dewhurst, J. and Baltieri, M. (2022). The emperor's new Markov blankets. *Behavioral and Brain Sciences* 45:e183. https://doi.org/10.1017/S0140525X21002351
18. Orseau, L., McGregor McGill, S. and Legg, S. (2018). Agents and devices: a relative definition of agency. arXiv:1805.12387 (preprint).
19. Albantakis, L., Barbosa, L., Findlay, G., Grasso, M. et al. (2023). Integrated information theory (IIT) 4.0: formulating the properties of phenomenal existence in physical terms. *PLOS Computational Biology* 19:e1011465. https://doi.org/10.1371/journal.pcbi.1011465
20. Krakauer, D., Bertschinger, N., Olbrich, E., Flack, J. C. and Ay, N. (2020). The information theory of individuality. *Theory in Biosciences* 139:209-223. https://doi.org/10.1007/s12064-020-00313-7
21. Hoel, E. P., Albantakis, L. and Tononi, G. (2013). Quantifying causal emergence shows that macro can beat micro. *Proceedings of the National Academy of Sciences* 110:19790-19795. https://doi.org/10.1073/pnas.1314922110
22. Rosas, F. E., Mediano, P. A. M., Biehl, M., Chandaria, S. and Polani, D. (2020). Causal blankets: theory and algorithmic framework. In *Active Inference: IWAI 2020*, Communications in Computer and Information Science, Springer, pp. 187-198. https://doi.org/10.1007/978-3-030-64919-7_19
23. Shiraishi, N., Funo, K. and Saito, K. (2018). Speed limit for classical stochastic processes. *Physical Review Letters* 121:070601. https://doi.org/10.1103/PhysRevLett.121.070601
24. Esposito, M. and Van den Broeck, C. (2011). Second law and Landauer principle far from equilibrium. *EPL* 95:40004. https://doi.org/10.1209/0295-5075/95/40004
25. Barato, A. C. and Seifert, U. (2015). Thermodynamic uncertainty relation for biomolecular processes. *Physical Review Letters* 114:158101. https://doi.org/10.1103/PhysRevLett.114.158101
26. Gingrich, T. R., Horowitz, J. M., Perunov, N. and England, J. L. (2016). Dissipation bounds all steady-state current fluctuations. *Physical Review Letters* 116:120601. https://doi.org/10.1103/PhysRevLett.116.120601
27. Paneru, G., Dutta, S., Tlusty, T. and Pak, H. K. (2020). Reaching and violating thermodynamic uncertainty bounds in information engines. *Physical Review E* 102:032126. https://doi.org/10.1103/PhysRevE.102.032126
28. Potts, P. P. and Samuelsson, P. (2019). Thermodynamic uncertainty relations including measurement and feedback. *Physical Review E* 100:052137. https://doi.org/10.1103/PhysRevE.100.052137
29. Proesmans, K., Ehrich, J. and Bechhoefer, J. (2020). Finite-time Landauer principle. *Physical Review Letters* 125:100602. https://doi.org/10.1103/PhysRevLett.125.100602
30. Sivak, D. A. and Crooks, G. E. (2012). Thermodynamic metrics and optimal paths. *Physical Review Letters* 108:190602. https://doi.org/10.1103/PhysRevLett.108.190602

31. Margolus, N. and Levitin, L. B. (1998). The maximum speed of dynamical evolution. *Physica D* 120:188-195. https://doi.org/10.1016/S0167-2789(98)00054-2
32. Touchette, H. and Lloyd, S. (2000). Information-theoretic limits of control. *Physical Review Letters* 84:1156-1159. https://doi.org/10.1103/PhysRevLett.84.1156
33. Touchette, H. and Lloyd, S. (2004). Information-theoretic approach to the study of control systems. *Physica A* 331:140-172. https://doi.org/10.1016/j.physa.2003.09.007
34. Garrahan, J. P. (2017). Simple bounds on fluctuations and uncertainty relations for first-passage times of counting observables. *Physical Review E* 95:032134. https://doi.org/10.1103/PhysRevE.95.032134
35. Bennett, C. H. (1973). Logical reversibility of computation. *IBM Journal of Research and Development* 17:525-532. https://doi.org/10.1147/rd.176.0525
36. Sagawa, T. and Ueda, M. (2012). Fluctuation theorem with information exchange: role of correlations in stochastic thermodynamics. *Physical Review Letters* 109:180602. https://doi.org/10.1103/PhysRevLett.109.180602
37. Hartle, H., Wolpert, D., Stier, A., Kempes, C. P. and Manzano, G. (2024). Work extraction with feedback control using limited resources. arXiv:2407.05507 (preprint).
38. Horodecki, M. and Oppenheim, J. (2013). Fundamental limitations for quantum and nanoscale thermodynamics. *Nature Communications* 4:2059. https://doi.org/10.1038/ncomms3059
39. Parrondo, J. M. R., Horowitz, J. M. and Sagawa, T. (2015). Thermodynamics of information. *Nature Physics* 11:131-139. https://doi.org/10.1038/nphys3230
40. Seifert, U. (2012). Stochastic thermodynamics, fluctuation theorems and molecular machines. *Reports on Progress in Physics* 75:126001. https://doi.org/10.1088/0034-4885/75/12/126001
41. Conant, R. C. and Ashby, W. R. (1970). Every good regulator of a system must be a model of that system. *International Journal of Systems Science* 1:89-97. https://doi.org/10.1080/00207727008920220
42. Richens, J. and Everitt, T. (2024). Robust agents learn causal world models. *International Conference on Learning Representations (ICLR 2024)*, oral, honorable mention. arXiv:2402.10877.
43. Richens, J., Abel, D., Bellot, A. and Everitt, T. (2025). General agents contain world models. arXiv:2506.01622 (preprint).
44. Horowitz, J. M. and Esposito, M. (2014). Thermodynamics with continuous information flow. *Physical Review X* 4:031015. https://doi.org/10.1103/PhysRevX.4.031015
45. Ito, S. and Sagawa, T. (2013). Information thermodynamics on causal networks. *Physical Review Letters* 111:180603. https://doi.org/10.1103/PhysRevLett.111.180603
46. Hartich, D., Barato, A. C. and Seifert, U. (2014). Stochastic thermodynamics of bipartite systems: transfer entropy inequalities and a Maxwell's demon interpretation. *Journal of Statistical Mechanics* P02016. https://doi.org/10.1088/1742-5468/2014/02/P02016
47. Allahverdyan, A. E., Janzing, D. and Mahler, G. (2009). Thermodynamic efficiency of information and heat flow. *Journal of Statistical Mechanics* P09011. https://doi.org/10.1088/1742-5468/2009/09/P09011

48. Esposito, M. and Schaller, G. (2012). Stochastic thermodynamics for “Maxwell demon” feedbacks. *EPL* 99:30003. https://doi.org/10.1209/0295-5075/99/30003
49. Mandal, D. and Jarzynski, C. (2012). Work and information processing in a solvable model of Maxwell’s demon. *PNAS* 109:11641-11645. https://doi.org/10.1073/pnas.1204263109
50. Boyd, A. B., Mandal, D. and Crutchfield, J. P. (2016). Identifying functional thermodynamics in autonomous Maxwellian ratchets. *New Journal of Physics* 18:023049. https://doi.org/10.1088/1367-2630/18/2/023049
51. Boyd, A. B., Mandal, D. and Crutchfield, J. P. (2018). Thermodynamics of modularity: structural costs beyond the Landauer bound. *Physical Review X* 8:031036. https://doi.org/10.1103/PhysRevX.8.031036
52. Wolpert, D. H. and Kolchinsky, A. (2020). Thermodynamics of computing with circuits. *New Journal of Physics* 22:063047. https://doi.org/10.1088/1367-2630/ab82b8
53. Kolchinsky, A. and Wolpert, D. H. (2017). Dependence of dissipation on the initial distribution over states. *Journal of Statistical Mechanics* 083202. https://doi.org/10.1088/1742-5468/aa7ee1
54. Still, S., Sivak, D. A., Bell, A. J. and Crooks, G. E. (2012). Thermodynamics of prediction. *Physical Review Letters* 109:120604. https://doi.org/10.1103/PhysRevLett.109.120604
55. Kolchinsky, A. and Wolpert, D. H. (2018). Semantic information, autonomous agency and non-equilibrium statistical physics. *Interface Focus* 8:20180041. https://doi.org/10.1098/rsfs.2018.0041
56. Ortega, P. A. and Braun, D. A. (2013). Thermodynamics as a theory of decision-making with information-processing costs. *Proceedings of the Royal Society A* 469:20120683. https://doi.org/10.1098/rspa.2012.0683
57. Kolchinsky, A. and Wolpert, D. H. (2020). Thermodynamic costs of Turing machines. *Physical Review Research* 2:033312. https://doi.org/10.1103/PhysRevResearch.2.033312
58. Kolchinsky, A. and Wolpert, D. H. (2021). Work, entropy production, and thermodynamics of information under protocol constraints. *Physical Review X* 11:041024. https://doi.org/10.1103/PhysRevX.11.041024
59. Manzano, G., Kardeş, G., Roldán, É. and Wolpert, D. H. (2024). Thermodynamics of computations with absolute irreversibility, unidirectional transitions, and stochastic computation times. *Physical Review X* 14:021026. https://doi.org/10.1103/PhysRevX.14.021026
60. Bérut, A., Arakelyan, A., Petrosyan, A., Ciliberto, S., Dillenschneider, R. and Lutz, E. (2012). Experimental verification of Landauer’s principle linking information and thermodynamics. *Nature* 483:187-189. https://doi.org/10.1038/nature10872
61. Jun, Y., Gavrilov, M. and Bechhoefer, J. (2014). High-precision test of Landauer’s principle in a feedback trap. *Physical Review Letters* 113:190601. https://doi.org/10.1103/PhysRevLett.113.190601
62. Orlov, A. O., Lent, C. S., Thorpe, C. C., Boechler, G. P. and Snider, G. L. (2012). Experimental test of Landauer’s principle at the sub-k_BT level. *Japanese Journal of Applied Physics* 51:06FE10. https://doi.org/10.1143/JJAP.51.06FE10

63. Hong, J., Lambson, B., Dhuey, S. and Bokor, J. (2016). Experimental test of Landauer's principle in single-bit operations on nanomagnetic memory bits. *Science Advances* 2:e1501492. https://doi.org/10.1126/sciadv.1501492
64. Gavrilov, M. and Bechhoefer, J. (2016). Erasure without work in an asymmetric double-well potential. *Physical Review Letters* 117:200601. https://doi.org/10.1103/PhysRevLett.117.200601
65. Gavrilov, M., Chétrite, R. and Bechhoefer, J. (2017). Direct measurement of weakly nonequilibrium system entropy is consistent with Gibbs-Shannon form. *PNAS* 114:11097-11102. https://doi.org/10.1073/pnas.1708689114
66. Yan, L. L., Xiong, T. P., Rehan, K., Zhou, F., Liang, D. F. et al. (2018). Single-atom demonstration of the quantum Landauer principle. *Physical Review Letters* 120:210601. https://doi.org/10.1103/PhysRevLett.120.210601
67. Dago, S., Pereda, J., Barros, N., Ciliberto, S. and Bellon, L. (2021). Information and thermodynamics: fast and precise approach to Landauer's bound in an underdamped micromechanical oscillator. *Physical Review Letters* 126:170601. https://doi.org/10.1103/PhysRevLett.126.170601
68. Scandi, M., Barker, D., Lehmann, S., Dick, K. A., Maisi, V. F. and Perarnau-Llobet, M. (2022). Minimally dissipative information erasure in a quantum dot via thermodynamic length. *Physical Review Letters* 129:270601. https://doi.org/10.1103/PhysRevLett.129.270601
69. Toyabe, S., Sagawa, T., Ueda, M., Muneyuki, E. and Sano, M. (2010). Experimental demonstration of information-to-energy conversion and validation of the generalized Jarzynski equality. *Nature Physics* 6:988-992. https://doi.org/10.1038/nphys1821
70. Koski, J. V., Maisi, V. F., Pekola, J. P. and Averin, D. V. (2014). Experimental realization of a Szilard engine with a single electron. *PNAS* 111:13786-13789. https://doi.org/10.1073/pnas.1406966111
71. Koski, J. V., Maisi, V. F., Sagawa, T. and Pekola, J. P. (2014). Experimental observation of the role of mutual information in the nonequilibrium dynamics of a Maxwell demon. *Physical Review Letters* 113:030601. https://doi.org/10.1103/PhysRevLett.113.030601
72. Koski, J. V., Kutvonen, A., Khaymovich, I. M., Ala-Nissila, T. and Pekola, J. P. (2015). On-chip Maxwell's demon as an information-powered refrigerator. *Physical Review Letters* 115:260602. https://doi.org/10.1103/PhysRevLett.115.260602
73. Camati, P. A., Peterson, J. P. S., Batalhão, T. B., Micadei, K., Souza, A. M. et al. (2016). Experimental rectification of entropy production by Maxwell's demon in a quantum system. *Physical Review Letters* 117:240502. https://doi.org/10.1103/PhysRevLett.117.240502
74. Peterson, J. P. S., Sarthour, R. S., Souza, A. M., Oliveira, I. S., Goold, J. et al. (2016). Experimental demonstration of information to energy conversion in a quantum system at the Landauer limit. *Proceedings of the Royal Society A* 472:20150813. https://doi.org/10.1098/rspa.2015.0813
75. Cottet, N., Jezouin, S., Bretheau, L., Campagne-Ibarcq, P., Ficheux, Q. et al. (2017). Observing a quantum Maxwell demon at work. *PNAS* 114:7561-7564. https://doi.org/10.1073/pnas.1704827114

76. Masuyama, Y., Funo, K., Murashita, Y., Noguchi, A., Kono, S. et al. (2018). Information-to-work conversion by Maxwell's demon in a superconducting circuit quantum electrodynamical system. *Nature Communications* 9:1291. https://doi.org/10.1038/s41467-018-03686-y
77. Naghiloo, M., Alonso, J. J., Romito, A., Lutz, E. and Murch, K. W. (2018). Information gain and loss for a quantum Maxwell's demon. *Physical Review Letters* 121:030604. https://doi.org/10.1103/PhysRevLett.121.030604
78. Kumar, A., Wu, T.-Y., Giraldo, F. and Weiss, D. S. (2018). Sorting ultracold atoms in a three-dimensional optical lattice in a realization of Maxwell's demon. *Nature* 561:83-87. https://doi.org/10.1038/s41586-018-0458-7
79. Linpeng, X., Bresque, L., Maffei, M., Jordan, A. N. et al. (2022). Energetic cost of measurements using quantum, coherent, and thermal light. *Physical Review Letters* 128:220506. https://doi.org/10.1103/PhysRevLett.128.220506
80. Paneru, G., Lee, D. Y., Tlusty, T. and Pak, H. K. (2018). Lossless Brownian information engine. *Physical Review Letters* 120:020601. https://doi.org/10.1103/PhysRevLett.120.020601
81. Admon, T., Rahav, S. and Roichman, Y. (2018). Experimental realization of an information machine with tunable temporal correlations. *Physical Review Letters* 121:180601. https://doi.org/10.1103/PhysRevLett.121.180601
82. Ribezzi-Crivellari, M. and Ritort, F. (2019). Large work extraction and the Landauer limit in a continuous Maxwell demon. *Nature Physics* 15:660-664. https://doi.org/10.1038/s41567-019-0481-0
83. Debiossac, M., Grass, D., Alonso, J. J., Lutz, E. and Kiesel, N. (2020). Thermodynamics of continuous non-Markovian feedback control. *Nature Communications* 11:1360. https://doi.org/10.1038/s41467-020-15148-5
84. Saha, T. K., Lucero, J. N. E., Ehrich, J., Sivak, D. A. and Bechhoefer, J. (2021). Maximizing power and velocity of an information engine. *PNAS* 118:e2023356118. https://doi.org/10.1073/pnas.2023356118
85. Saha, T. K., Lucero, J. N. E., Ehrich, J., Sivak, D. A. and Bechhoefer, J. (2022). Bayesian information engine that optimally exploits noisy measurements. *Physical Review Letters* 129:130601. https://doi.org/10.1103/PhysRevLett.129.130601
86. Saha, T. K., Ehrich, J., Gavrilov, M., Still, S., Sivak, D. A. and Bechhoefer, J. (2023). Information engine in a nonequilibrium bath. *Physical Review Letters* 131:057101. https://doi.org/10.1103/PhysRevLett.131.057101
87. Barker, D., Scandi, M., Lehmann, S., Thelander, C., Dick, K. A., Perarnau-Llobet, M. and Maisi, V. F. (2022). Experimental verification of the work fluctuation-dissipation relation for information-to-work conversion. *Physical Review Letters* 128:040602. https://doi.org/10.1103/PhysRevLett.128.040602
88. Yan, L.-L., Bu, J.-T., Zeng, Q., Zhang, K., Cui, K.-F. et al. (2024). Experimental verification of demon-involved fluctuation theorems. *Physical Review Letters* 133:090402. https://doi.org/10.1103/PhysRevLett.133.090402
89. Zeng, Q. and Wang, J. (2021). New fluctuation theorems on Maxwell's demon. *Science Advances* 7:eabf1807. https://doi.org/10.1126/sciadv.abf1807

90. Demin, A. V. and Vityaev, E. E. (2014). Learning in a virtual model of the C. elegans nematode for locomotion and chemotaxis. *Biologically Inspired Cognitive Architectures* 7:9-14. https://doi.org/10.1016/j.bica.2013.11.005
91. Putintsev, N. I., Isupov, O. V. and Vityaev, E. E. (2015). Adaptive control system for a mobile agent in a physical environment based on functional systems theory. *Russian Journal of Genetics: Applied Research* 5:601-608. https://doi.org/10.1134/S2079059715060131
92. Rosenblueth, A., Wiener, N. and Bigelow, J. (1943). Behavior, purpose and teleology. *Philosophy of Science* 10:18-24. https://doi.org/10.1086/286788
93. Anokhin, P. K. (1968). The functional system as a unit of organism integrative activity. In *Systems Theory and Biology*, Springer, pp. 376-403. https://doi.org/10.1007/978-3-642-88343-9_15
94. Anokhin, P. K. (1974). The biological roots of the conditioned reflex. In *Biology and Neurophysiology of the Conditioned Reflex and Its Role in Adaptive Behavior*, Pergamon, pp. 1-24. https://doi.org/10.1016/B978-0-08-021516-7.50008-2
95. Sudakov, K. V. (1997). The theory of functional systems: general postulates and principles of dynamic organization. *Integrative Physiological and Behavioral Science* 32:392-414. https://doi.org/10.1007/BF02688634
96. Shvyrkov, V. B. (1980). Goal as a system-forming factor in behavior and learning. In *Neural Mechanisms of Goal-directed Behavior and Learning*, Academic Press, pp. 199-219. https://doi.org/10.1016/B978-0-12-688980-2.50018-7
97. Rosen, R. (2012). *Anticipatory Systems: Philosophical, Mathematical, and Methodological Foundations*, 2nd ed. Springer. https://doi.org/10.1007/978-1-4614-1269-4
98. Sterling, P. (2012). Allostasis: a model of predictive regulation. *Physiology & Behavior* 106:5-15. https://doi.org/10.1016/j.physbeh.2011.06.004
99. Virgo, N., Biehl, M. and McGregor, S. (2021). Interpreting dynamical systems as Bayesian reasoners. In *Machine Learning and Principles and Practice of Knowledge Discovery in Databases* (ECML PKDD 2021 workshops), Communications in Computer and Information Science, Springer, pp. 726-762. https://doi.org/10.1007/978-3-030-93736-2_52
100. Biehl, M. and Virgo, N. (2023). Interpreting systems as solving POMDPs: a step towards a formal understanding of agency. In *Active Inference: IWAI 2022*, Communications in Computer and Information Science 1721, Springer, pp. 16-31. https://doi.org/10.1007/978-3-031-28719-0_2
101. Maturana, H. R. and Varela, F. J. (1980). *Autopoiesis and Cognition: The Realization of the Living*. Reidel. https://doi.org/10.1007/978-94-009-8947-4
102. Kauffman, S. (2003). Molecular autonomous agents. *Philosophical Transactions of the Royal Society A* 361:1089-1099. https://doi.org/10.1098/rsta.2003.1186
103. Di Paolo, E. A. (2005). Autopoiesis, adaptivity, teleology, agency. *Phenomenology and the Cognitive Sciences* 4:429-452. https://doi.org/10.1007/s11097-005-9002-y
104. Barandiaran, X. and Moreno, A. (2006). On what makes certain dynamical systems cognitive: a minimally cognitive organization program. *Adaptive Behavior* 14:171-185. https://doi.org/10.1177/105971230601400208
105. Barandiaran, X. and Moreno, A. (2008). Adaptivity: from metabolism to behavior. *Adaptive Behavior* 16:325-344. https://doi.org/10.1177/1059712308093868

106. Barandiaran, X. E., Di Paolo, E. and Rohde, M. (2009). Defining agency: individuality, normativity, asymmetry, and spatio-temporality in action. *Adaptive Behavior* 17:367-386. https://doi.org/10.1177/1059712309343819
107. Ruiz-Mirazo, K. and Moreno, A. (2011). Autonomy in evolution: from minimal to complex life. *Synthese* 185:21-52. https://doi.org/10.1007/s11229-011-9874-z
108. Moreno, A. and Mossio, M. (2015). *Biological Autonomy: A Philosophical and Theoretical Enquiry*. Springer. https://doi.org/10.1007/978-94-017-9837-2
109. Montévil, M. and Mossio, M. (2015). Biological organisation as closure of constraints. *Journal of Theoretical Biology* 372:179-191. https://doi.org/10.1016/j.jtbi.2015.02.029
110. Aguilera, M. and Barandiaran, X. E. (2024). Thermina: a minimal model of autonomous agency from the lens of stochastic thermodynamics. *ALIFE 2024: Proceedings of the 2024 Artificial Life Conference*. https://doi.org/10.1162/isal_a_00826
111. Barandiaran, X. E. and Almendros, L. S. (2024). Transforming agency: on the mode of existence of large language models. arXiv:2407.10735 (preprint).
112. Levin, M. (2022). Technological approach to mind everywhere: an experimentally-grounded framework for understanding diverse bodies and minds. *Frontiers in Systems Neuroscience* 16:768201. https://doi.org/10.3389/fnsys.2022.768201
113. Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience* 11:127-138. https://doi.org/10.1038/nrn2787
114. Friston, K. (2013). Life as we know it. *Journal of the Royal Society Interface* 10:20130475. https://doi.org/10.1098/rsif.2013.0475
115. Rovelli, C. (2020). Agency in physics. arXiv:2007.05300 (preprint).
116. Jaeger, J. (2023). Artificial intelligence is algorithmic mimicry: why artificial “agents” are not (and won’t be) proper agents. arXiv:2307.07515 (preprint).
117. Azadi, P. (2025). Computational irreducibility as the foundation of agency. arXiv:2505.04646 (preprint).
118. Wissner-Gross, A. D. and Freer, C. E. (2013). Causal entropic forces. *Physical Review Letters* 110:168702. https://doi.org/10.1103/PhysRevLett.110.168702
119. Kappen, H. J. (2013). Comment: causal entropic forces. arXiv:1312.4185 (preprint).
120. Fields, C., Goldstein, A. and Sandved-Smith, L. (2024). Making the thermodynamic cost of active inference explicit. *Entropy* 26:622. https://doi.org/10.3390/e26080622
121. Bartlett, S., Eckford, A. W., Egbert, M., Lingam, M. et al. (2025). Physics of life: exploring information as a distinctive feature of living systems. *PRX Life* 3:037003. https://doi.org/10.1103/rsx4-8x5f
122. Funo, K., Shiraishi, N. and Saito, K. (2019). Speed limit for open quantum systems. *New Journal of Physics* 21:013006. https://doi.org/10.1088/1367-2630/aaf9f5
123. Shiraishi, N. and Saito, K. (2021). Speed limit for open systems coupled to general environments. *Physical Review Research* 3:023074. https://doi.org/10.1103/PhysRevResearch.3.023074
124. Vo, V. T., Van Vu, T. and Hasegawa, Y. (2020). Unified approach to classical speed limit and thermodynamic uncertainty relation. *Physical Review E* 102:062132. https://doi.org/10.1103/PhysRevE.102.062132

125. Falasco, G. and Esposito, M. (2020). Dissipation-time uncertainty relation. *Physical Review Letters* 125:120604. https://doi.org/10.1103/PhysRevLett.125.120604
126. Ito, S. (2018). Stochastic thermodynamic interpretation of information geometry. *Physical Review Letters* 121:030605. https://doi.org/10.1103/PhysRevLett.121.030605
127. Ito, S. and Dechant, A. (2020). Stochastic time evolution, information geometry, and the Cramér-Rao bound. *Physical Review X* 10:021056. https://doi.org/10.1103/PhysRevX.10.021056
128. Nakazato, M. and Ito, S. (2021). Geometrical aspects of entropy production in stochastic thermodynamics based on Wasserstein distance. *Physical Review Research* 3:043093. https://doi.org/10.1103/PhysRevResearch.3.043093
129. Dechant, A., Sasa, S.-i. and Ito, S. (2022). Geometric decomposition of entropy production in out-of-equilibrium systems. *Physical Review Research* 4:L012034. https://doi.org/10.1103/PhysRevResearch.4.L012034
130. Van Vu, T. and Saito, K. (2023). Thermodynamic unification of optimal transport: thermodynamic uncertainty relation, minimum dissipation, and thermodynamic speed limits. *Physical Review X* 13:011013. https://doi.org/10.1103/PhysRevX.13.011013
131. Nagayama, R., Yoshimura, K. and Ito, S. (2025). Infinite variety of thermodynamic speed limits with general activities. *Physical Review Research* 7:013307. https://doi.org/10.1103/PhysRevResearch.7.013307
132. Yoshimura, K. and Ito, S. (2021). Thermodynamic uncertainty relation and thermodynamic speed limit in deterministic chemical reaction networks. *Physical Review Letters* 127:160601. https://doi.org/10.1103/PhysRevLett.127.160601
133. Aurell, E., Mejía-Monasterio, C. and Muratore-Ginanneschi, P. (2011). Optimal protocols and optimal transport in stochastic thermodynamics. *Physical Review Letters* 106:250601. https://doi.org/10.1103/PhysRevLett.106.250601
134. Hasegawa, Y. (2023). Unifying speed limit, thermodynamic uncertainty relation and Heisenberg principle via bulk-boundary correspondence. *Nature Communications* 14:2828. https://doi.org/10.1038/s41467-023-38074-8
135. Van Vu, T. and Hasegawa, Y. (2021). Geometrical bounds of the irreversibility in Markovian systems. *Physical Review Letters* 126:010601. https://doi.org/10.1103/PhysRevLett.126.010601
136. Delvenne, J.-C. and Falasco, G. (2024). Thermokinetic relations. *Physical Review E* 109:014109. https://doi.org/10.1103/PhysRevE.109.014109
137. Seifert, U. (2017). Stochastic thermodynamics: from principles to the cost of precision. arXiv:1707.03759 (lecture notes).
138. Horowitz, J. M. and Gingrich, T. R. (2017). Proof of the finite-time thermodynamic uncertainty relation for steady-state currents. *Physical Review E* 96:020103. https://doi.org/10.1103/PhysRevE.96.020103
139. Pietzonka, P., Barato, A. C. and Seifert, U. (2016). Universal bounds on current fluctuations. *Physical Review E* 93:052145. https://doi.org/10.1103/PhysRevE.93.052145
140. Proesmans, K. and Van den Broeck, C. (2017). Discrete-time thermodynamic uncertainty relation. *EPL* 119:20001. https://doi.org/10.1209/0295-5075/119/20001

141. Hasegawa, Y. and Van Vu, T. (2019). Fluctuation theorem uncertainty relation. *Physical Review Letters* 123:110602. https://doi.org/10.1103/PhysRevLett.123.110602
142. Dechant, A. and Sasa, S.-i. (2018). Entropic bounds on currents in Langevin systems. *Physical Review E* 97:062101. https://doi.org/10.1103/PhysRevE.97.062101
143. Lee, J. S., Park, J.-M. and Park, H. (2019). Thermodynamic uncertainty relation for underdamped Langevin systems driven by a velocity-dependent force. *Physical Review E* 100:062132. https://doi.org/10.1103/PhysRevE.100.062132
144. Lee, J. S., Park, J.-M. and Park, H. (2021). Universal form of thermodynamic uncertainty relation for Langevin dynamics. *Physical Review E* 104:L052102. https://doi.org/10.1103/PhysRevE.104.L052102
145. Dechant, A. and Sasa, S.-i. (2021). Improving thermodynamic bounds using correlations. *Physical Review X* 11:041061. https://doi.org/10.1103/PhysRevX.11.041061
146. Horowitz, J. M. and Gingrich, T. R. (2020). Thermodynamic uncertainty relations constrain non-equilibrium fluctuations. *Nature Physics* 16:15-20. https://doi.org/10.1038/s41567-019-0702-6
147. Barato, A. C. and Seifert, U. (2016). Cost and precision of Brownian clocks. *Physical Review X* 6:041053. https://doi.org/10.1103/PhysRevX.6.041053
148. Salazar, D. S. P. (2021). Information bound for entropy production from the detailed fluctuation theorem. *Physical Review E* 103:022122. https://doi.org/10.1103/PhysRevE.103.022122
149. Landi, G. T. and Paternostro, M. (2021). Irreversible entropy production: from classical to quantum. *Reviews of Modern Physics* 93:035008. https://doi.org/10.1103/RevModPhys.93.035008
150. Van Vu, T. and Hasegawa, Y. (2020). Thermodynamic uncertainty relations under arbitrary control protocols. *Physical Review Research* 2:013060. https://doi.org/10.1103/PhysRevResearch.2.013060
151. Tanogami, T., Van Vu, T. and Saito, K. (2023). Universal bounds on the performance of information-thermodynamic engine. *Physical Review Research* 5:043280. https://doi.org/10.1103/PhysRevResearch.5.043280
152. Kumasaki, K., Tojo, K., Sagawa, T. and Funo, K. (2026). Thermodynamic uncertainty relation for feedback cooling. *Physical Review E* 113:024134. https://doi.org/10.1103/4pq6-7djm
153. Honma, R. and Van Vu, T. (2026). Thermodynamic uncertainty relation with quantum feedback. *Physical Review Letters*. https://doi.org/10.1103/6d8l-94pv. Preprint: arXiv:2602.22651.
154. Gingrich, T. R. and Horowitz, J. M. (2017). Fundamental bounds on first passage time fluctuations for currents. *Physical Review Letters* 119:170601. https://doi.org/10.1103/PhysRevLett.119.170601
155. Schmiedl, T. and Seifert, U. (2007). Optimal finite-time processes in stochastic thermodynamics. *Physical Review Letters* 98:108301. https://doi.org/10.1103/PhysRevLett.98.108301
156. Proesmans, K., Ehrich, J. and Bechhoefer, J. (2020). Optimal finite-time bit erasure under full control. *Physical Review E* 102:032105. https://doi.org/10.1103/PhysRevE.102.032105

157. Zhen, Y.-Z., Egloff, D., Modi, K. and Dahlsten, O. (2021). Universal bound on energy cost of bit reset in finite time. *Physical Review Letters* 127:190602. https://doi.org/10.1103/PhysRevLett.127.190602
158. Esposito, M., Kawai, R., Lindenberg, K. and Van den Broeck, C. (2010). Efficiency at maximum power of low-dissipation Carnot engines. *Physical Review Letters* 105:150603. https://doi.org/10.1103/PhysRevLett.105.150603
159. Tu, Z. C. (2008). Efficiency at maximum power of Feynman's ratchet as a heat engine. *Journal of Physics A* 41:312003. https://doi.org/10.1088/1751-8113/41/31/312003
160. Shiraishi, N., Saito, K. and Tasaki, H. (2016). Universal trade-off relation between power and efficiency for heat engines. *Physical Review Letters* 117:190601. https://doi.org/10.1103/PhysRevLett.117.190601
161. Pietzonka, P. and Seifert, U. (2018). Universal trade-off between power, efficiency, and constancy in steady-state heat engines. *Physical Review Letters* 120:190602. https://doi.org/10.1103/PhysRevLett.120.190602
162. Ma, Y.-H., Xu, D., Dong, H. and Sun, C.-P. (2018). Universal constraint for efficiency and power of a low-dissipation heat engine. *Physical Review E* 98:042112. https://doi.org/10.1103/PhysRevE.98.042112
163. Tsirlin, A. M., Mironova, V. A., Amelkin, S. A. and Kazakov, V. (1998). Finite-time thermodynamics: conditions of minimal dissipation for thermodynamic processes with given rate. *Physical Review E* 58:215-223. https://doi.org/10.1103/PhysRevE.58.215
164. Lee, J. S., Lee, S., Kwon, H. and Park, H. (2022). Speed limit for a highly irreversible process and tight finite-time Landauer's bound. *Physical Review Letters* 129:120603. https://doi.org/10.1103/PhysRevLett.129.120603
165. Van Vu, T. and Saito, K. (2022). Finite-time quantum Landauer principle and quantum coherence. *Physical Review Letters* 128:010602. https://doi.org/10.1103/PhysRevLett.128.010602
166. Rolandi, A. and Perarnau-Llobet, M. (2023). Finite-time Landauer principle beyond weak coupling. *Quantum* 7:1161. https://doi.org/10.22331/q-2023-11-03-1161
167. Kamijima, T., Funo, K. and Sagawa, T. (2024). Finite-time thermodynamic bounds and tradeoff relations for information processing. arXiv:2409.08606 (preprint).
168. Tkachenko, A. V. (2025). Thermodynamic cost of inference and learning in physical neural networks. arXiv:2503.09980 (preprint). Title of the current arXiv version.
169. Tatikonda, S. and Mitter, S. (2004). Control under communication constraints. *IEEE Transactions on Automatic Control* 49:1056-1068. https://doi.org/10.1109/TAC.2004.831187
170. Nair, G. N. and Evans, R. J. (2004). Stabilizability of stochastic linear systems with finite feedback data rates. *SIAM Journal on Control and Optimization* 43:413-436. https://doi.org/10.1137/S0363012902402116
171. Nair, G. N., Fagnani, F., Zampieri, S. and Evans, R. J. (2007). Feedback control under data rate constraints: an overview. *Proceedings of the IEEE* 95:108-137. https://doi.org/10.1109/JPROC.2006.887294

172. Sandberg, H., Delvenne, J.-C., Newton, N. J. and Mitter, S. K. (2014). Maximum work extraction and implementation costs for nonequilibrium Maxwell's demons. *Physical Review E* 90:042119. https://doi.org/10.1103/PhysRevE.90.042119
173. Horowitz, J. M. and Sandberg, H. (2014). Second-law-like inequalities with information and their interpretations. *New Journal of Physics* 16:125007. https://doi.org/10.1088/1367-2630/16/12/125007
174. Jacobs, K. (2009). Second law of thermodynamics and quantum feedback control: Maxwell's demon with weak measurements. *Physical Review A* 80:012322. https://doi.org/10.1103/PhysRevA.80.012322
175. Funo, K., Watanabe, Y. and Ueda, M. (2013). Integral quantum fluctuation theorems under measurement and feedback control. *Physical Review E* 88:052121. https://doi.org/10.1103/PhysRevE.88.052121
176. Guryanova, Y., Friis, N. and Huber, M. (2020). Ideal projective measurements have infinite resource costs. *Quantum* 4:222. https://doi.org/10.22331/q-2020-01-13-222
177. Danageozian, A., Wilde, M. M. and Buscemi, F. (2022). Thermodynamic constraints on quantum information gain and error correction: a triple trade-off. *PRX Quantum* 3:020318. https://doi.org/10.1103/PRXQuantum.3.020318
178. Wolpert, D. H. (2020). Minimal entropy production rate of interacting systems. *New Journal of Physics* 22:113013. https://doi.org/10.1088/1367-2630/abc5c6
179. Tasnim, F. and Wolpert, D. H. (2023). Stochastic thermodynamics of multiple co-evolving systems: beyond multipartite processes. *Entropy* 25:1078. https://doi.org/10.3390/e25071078
180. Wolpert, D. H. (2020). Uncertainty relations and fluctuation theorems for Bayes nets. *Physical Review Letters* 125:200602. https://doi.org/10.1103/PhysRevLett.125.200602
181. Rolandi, A., Abiuso, P. and Perarnau-Llobet, M. (2023). Collective advantages in finite-time thermodynamics. *Physical Review Letters* 131:210401. https://doi.org/10.1103/PhysRevLett.131.210401
182. Fujimoto, Y. and Ito, S. (2024). Game-theoretical approach to minimum entropy productions in information thermodynamics. *Physical Review Research* 6:013023. https://doi.org/10.1103/PhysRevResearch.6.013023
183. Lloyd, S. (2000). Ultimate physical limits to computation. *Nature* 406:1047-1054. https://doi.org/10.1038/35023282
184. Deffner, S. and Campbell, S. (2017). Quantum speed limits: from Heisenberg's uncertainty principle to optimal quantum control. *Journal of Physics A* 50:453001. https://doi.org/10.1088/1751-8121/aa86c6
185. Lan, G., Sartori, P., Neumann, S., Sourjik, V. and Tu, Y. (2012). The energy-speed-accuracy trade-off in sensory adaptation. *Nature Physics* 8:422-428. https://doi.org/10.1038/nphys2276
186. Mehta, P. and Schwab, D. J. (2012). Energetic costs of cellular computation. *PNAS* 109:17978-17982. https://doi.org/10.1073/pnas.1207814109
187. Lang, A. H., Fisher, C. K., Mora, T. and Mehta, P. (2014). Thermodynamics of statistical inference by cells. *Physical Review Letters* 113:148103. https://doi.org/10.1103/PhysRevLett.113.148103

188. Govern, C. C. and ten Wolde, P. R. (2014). Optimal resource allocation in cellular sensing systems. *PNAS* 111:17486-17491. https://doi.org/10.1073/pnas.1411524111
189. Sartori, P., Granger, L., Lee, C. F. and Horowitz, J. M. (2014). Thermodynamic costs of information processing in sensory adaptation. *PLoS Computational Biology* 10:e1003974. https://doi.org/10.1371/journal.pcbi.1003974
190. Barato, A. C., Hartich, D. and Seifert, U. (2013). Information-theoretic versus thermodynamic entropy production in autonomous sensory networks. *Physical Review E* 87:042104. https://doi.org/10.1103/PhysRevE.87.042104
191. Barato, A. C., Hartich, D. and Seifert, U. (2014). Efficiency of cellular information processing. *New Journal of Physics* 16:103024. https://doi.org/10.1088/1367-2630/16/10/103024
192. Ouldridge, T. E., Govern, C. C. and ten Wolde, P. R. (2017). Thermodynamics of computational copying in biochemical systems. *Physical Review X* 7:021004. https://doi.org/10.1103/PhysRevX.7.021004
193. Cao, Y., Wang, H., Ouyang, Q. and Tu, Y. (2015). The free-energy cost of accurate biochemical oscillations. *Nature Physics* 11:772-778. https://doi.org/10.1038/nphys3412
194. Zhang, D., Cao, Y., Ouyang, Q. and Tu, Y. (2019). The energy cost and optimal design for synchronization of coupled molecular oscillators. *Nature Physics* 16:95-100. https://doi.org/10.1038/s41567-019-0701-7
195. Sartori, P. and Tu, Y. (2015). Free energy cost of reducing noise while maintaining a high sensitivity. *Physical Review Letters* 115:118102. https://doi.org/10.1103/PhysRevLett.115.118102
196. Horowitz, J. M., Zhou, K. and England, J. L. (2017). Minimum energetic cost to maintain a target nonequilibrium state. *Physical Review E* 95:042102. https://doi.org/10.1103/PhysRevE.95.042102
197. Bryant, S. J. and Machta, B. B. (2023). Physical constraints in intracellular signaling: the cost of sending a bit. *Physical Review Letters* 131:068401. https://doi.org/10.1103/PhysRevLett.131.068401
198. Tjalma, A. J., Galstyan, V., Goedhart, J., Slim, L. et al. (2023). Trade-offs between cost and information in cellular prediction. *Proceedings of the National Academy of Sciences* 120:e2303078120. https://doi.org/10.1073/pnas.2303078120
199. Schreiber, T. (2000). Measuring information transfer. *Physical Review Letters* 85:461-464. https://doi.org/10.1103/PhysRevLett.85.461
200. Tishby, N. and Polani, D. (2011). Information theory of decisions and actions. In *Perception-Action Cycle*, Springer, pp. 601-636. https://doi.org/10.1007/978-1-4419-1452-1_19
201. Stratonovich, R. L. (2020). *Theory of Information and its Value* (English edition of the 1975 Russian monograph). Springer. https://doi.org/10.1007/978-3-030-22833-0
202. Pal, S., Saryal, S., Segal, D., Mahesh, T. S. and Agarwalla, B. K. (2020). Experimental study of the thermodynamic uncertainty relation. *Physical Review Research* 2:022044. https://doi.org/10.1103/PhysRevResearch.2.022044
203. Barker, D., Lehmann, S., Dick, K. A., Samuelsson, P. et al. (2025). Information thermodynamics in a quantum dot Szilard engine: experimentally investigating fluctuation theorems and thermodynamic uncertainty relations. arXiv:2511.08541 (preprint).

204. Friedman, H. M., Agarwalla, B. K., Shein-Lumbroso, O., Tal, O. and Segal, D. (2020). Thermodynamic uncertainty relation in atomic-scale quantum conductors. *Physical Review B* 101:195423. https://doi.org/10.1103/PhysRevB.101.195423
205. Hwang, W. and Hyeon, C. (2018). Energetic costs, precision, and transport efficiency of molecular motors. *Journal of Physical Chemistry Letters* 9:513-520. https://doi.org/10.1021/acs.jpclett.7b03197
206. Li, J., Horowitz, J. M., Gingrich, T. R. and Fakhri, N. (2019). Quantifying dissipation using fluctuating currents. *Nature Communications* 10:1666. https://doi.org/10.1038/s41467-019-09631-x
207. Ness, G., Lam, M. R., Alt, W., Meschede, D., Sagi, Y. and Alberti, A. (2021). Observing crossover between quantum speed limits. *Science Advances* 7:eabj9119. https://doi.org/10.1126/sciadv.abj9119
208. Pires, L. B., Goerlich, R., Luna da Fonseca, A., Debiossac, M., Hervieux, P.-A. et al. (2023). Optimal time-entropy bounds and speed limits for Brownian thermal shortcuts. *Physical Review Letters* 131:097101. https://doi.org/10.1103/PhysRevLett.131.097101
209. Oikawa, S., Nakayama, Y., Ito, S., Sagawa, T. and Toyabe, S. (2025). Experimentally achieving minimal dissipation via thermodynamically optimal transport. *Nature Communications* 16:10424. https://doi.org/10.1038/s41467-025-66519-9
210. Proesmans, K., Dreher, Y., Gavrilov, M., Bechhoefer, J. and Van den Broeck, C. (2016). Brownian duet: a novel tale of thermodynamic efficiency. *Physical Review X* 6:041010. https://doi.org/10.1103/PhysRevX.6.041010
211. Mancino, L., Cavina, V., De Pasquale, A., Sbroscia, M., Booth, R. I. et al. (2018). Geometrical bounds on irreversibility in open quantum systems. *Physical Review Letters* 121:160602. https://doi.org/10.1103/PhysRevLett.121.160602
212. Dago, S. and Bellon, L. (2022). Dynamics of information erasure and extension of Landauer's bound to fast processes. *Physical Review Letters* 128:070604. https://doi.org/10.1103/PhysRevLett.128.070604
213. Mattingly, H. H., Kamino, K., Machta, B. B. and Emonet, T. (2021). *Escherichia coli* chemotaxis is information limited. *Nature Physics* 17:1426-1431. https://doi.org/10.1038/s41567-021-01380-3
214. Blickle, V. and Bechinger, C. (2012). Realization of a micrometre-sized stochastic heat engine. *Nature Physics* 8:143-146. https://doi.org/10.1038/nphys2163
215. Martínez, I. A., Roldán, É., Dinis, L., Petrov, D., Parrondo, J. M. R. and Rica, R. A. (2016). Brownian Carnot engine. *Nature Physics* 12:67-70. https://doi.org/10.1038/nphys3518
216. Roldán, É., Martínez, I. A., Parrondo, J. M. R. and Petrov, D. (2014). Universal features in the energetics of symmetry breaking. *Nature Physics* 10:457-461. https://doi.org/10.1038/nphys2940
217. Krishnamurthy, S., Ghosh, S., Chatterji, D., Ganapathy, R. and Sood, A. K. (2016). A micrometre-sized heat engine operating between bacterial reservoirs. *Nature Physics* 12:1134-1138. https://doi.org/10.1038/nphys3870
218. Krishnamurthy, S., Ganapathy, R. and Sood, A. K. (2023). Overcoming power-efficiency tradeoff in a micro heat engine by engineered system-bath interactions. *Nature Communications* 14:6842. https://doi.org/10.1038/s41467-023-42350-y

219. Paneru, G., Dutta, S., Sagawa, T., Tlusty, T. and Pak, H. K. (2020). Efficiency fluctuations and noise induced refrigerator-to-heater transition in information engines. *Nature Communications* 11:1012. https://doi.org/10.1038/s41467-020-14823-x
220. Jung, T., Polani, D. and Stone, P. (2011). Empowerment for continuous agent-environment systems. *Adaptive Behavior* 19:16-39. https://doi.org/10.1177/1059712310392389
221. Salge, C., Glackin, C. and Polani, D. (2013). Empowerment: an introduction. arXiv:1310.1863 (preprint).
222. Kolchinsky, A. (2024). Thermodynamic dissipation does not bound replicator growth and decay rates. arXiv:2404.01130 (preprint).
223. Adlam, E. C., McQueen, K. J. and Waegell, M. (2025). Agency cannot be a purely quantum phenomenon. arXiv:2510.13247 (preprint).
224. Goldt, S. and Seifert, U. (2017). Stochastic thermodynamics of learning. *Physical Review Letters* 118:010601. https://doi.org/10.1103/PhysRevLett.118.010601
225. Still, S. (2020). Thermodynamic cost and benefit of memory. *Physical Review Letters* 124:050601. https://doi.org/10.1103/PhysRevLett.124.050601
226. Boyd, A. B., Crutchfield, J. P. and Gu, M. (2022). Thermodynamic machine learning through maximum work production. *New Journal of Physics* 24:083040. https://doi.org/10.1088/1367-2630/ac4309
227. Boyd, A. B., Crutchfield, J. P., Gu, M. and Binder, F. C. (2025). Thermodynamic overfitting and generalization: energetics of predictive intelligence. *New Journal of Physics* 27:063901. https://doi.org/10.1088/1367-2630/addf71
228. Ehrich, J., Still, S. and Sivak, D. A. (2023). Energetic cost of feedback control. *Physical Review Research* 5:023080. https://doi.org/10.1103/PhysRevResearch.5.023080
229. Ehrich, J. and Sivak, D. A. (2023). Energy and information flows in autonomous systems. *Frontiers in Physics* 11:1108357. https://doi.org/10.3389/fphy.2023.1108357
230. Wolpert, D. H. (2019). The stochastic thermodynamics of computation. *Journal of Physics A* 52:193001. https://doi.org/10.1088/1751-8121/ab0850
231. Levy, W. B. and Baxter, R. A. (1996). Energy efficient neural codes. *Neural Computation* 8:531-543. https://doi.org/10.1162/neco.1996.8.3.531
232. Balasubramanian, V., Kimber, D. and Berry, M. J., II (2001). Metabolically efficient information processing. *Neural Computation* 13:799-815. https://doi.org/10.1162/089976601300014358
233. Goldt, S. and Seifert, U. (2017). Thermodynamic efficiency of learning a rule in neural networks. *New Journal of Physics* 19:113001. https://doi.org/10.1088/1367-2630/aa89ff
234. Fiderer, L. J., Barth, P. C., Smith, I. D. and Briegel, H. J. (2025). Information thermodynamics of agents: the work capacity of channels with memory. arXiv:2504.06209 (preprint).
235. Elliott, T. J., Gu, M., Garner, A. J. P. and Thompson, J. (2022). Quantum adaptive agents with efficient long-term memories. *Physical Review X* 12:011007. https://doi.org/10.1103/PhysRevX.12.011007
236. Russell, S. J. and Subramanian, D. (1995). Provably bounded-optimal agents. *Journal of Artificial Intelligence Research* 2:575-609. https://doi.org/10.1613/jair.133

237. Genewein, T., Leibfried, F., Grau-Moya, J. and Braun, D. A. (2015). Bounded rationality, abstraction, and hierarchical decision-making: an information-theoretic optimality principle. *Frontiers in Robotics and AI* 2:27. https://doi.org/10.3389/frobt.2015.00027
238. Legg, S. and Hutter, M. (2007). Universal intelligence: a definition of machine intelligence. *Minds and Machines* 17:391-444. https://doi.org/10.1007/s11023-007-9079-x
239. Takahashi, K. and Hayashi, Y. (2026). Thermodynamic limits of physical intelligence. In *Artificial General Intelligence: 19th International Conference, AGI 2026, Proceedings, Part II*, Lecture Notes in Computer Science. Springer. https://doi.org/10.1007/978-3-032-33195-3_24. arXiv:2602.05463.
240. Ortega, P. A., Braun, D. A., Dyer, J., Kim, K.-E. and Tishby, N. (2015). Information-theoretic bounded rationality. arXiv:1512.06789 (preprint).
241. Zénon, A., Solopchuk, O. and Pezzulo, G. (2019). An information-theoretic perspective on the costs of cognition. *Neuropsychologia* 123:5-18. https://doi.org/10.1016/j.neuropsychologia.2018.09.013
242. Sims, C. R. (2016). Rate-distortion theory and human perception. *Cognition* 152:181-198. https://doi.org/10.1016/j.cognition.2016.03.020
243. Laughlin, S. B., de Ruyter van Steveninck, R. R. and Anderson, J. C. (1998). The metabolic cost of neural information. *Nature Neuroscience* 1:36-41. https://doi.org/10.1038/236
244. Attwell, D. and Laughlin, S. B. (2001). An energy budget for signaling in the grey matter of the brain. *Journal of Cerebral Blood Flow and Metabolism* 21:1133-1145. https://doi.org/10.1097/00004647-200110000-00001
245. Lennie, P. (2003). The cost of cortical computation. *Current Biology* 13:493-497. https://doi.org/10.1016/S0960-9822(03)00135-0
246. Sengupta, B., Stemmler, M., Laughlin, S. B. and Niven, J. E. (2010). Action potential energy efficiency varies among neuron types in vertebrates and invertebrates. *PLoS Computational Biology* 6:e1000840. https://doi.org/10.1371/journal.pcbi.1000840
247. Sengupta, B., Stemmler, M. B. and Friston, K. J. (2013). Information and efficiency in the nervous system: a synthesis. *PLoS Computational Biology* 9:e1003157. https://doi.org/10.1371/journal.pcbi.1003157
248. Levy, W. B. and Calvert, V. G. (2021). Communication consumes 35 times more energy than computation in the human cortex, but both costs are needed to predict synapse number. *Proceedings of the National Academy of Sciences* 118:e2008173118. https://doi.org/10.1073/pnas.2008173118
249. Harris, J. J., Jolivet, R., Engl, E. and Attwell, D. (2015). Energy-efficient information transfer by visual pathway synapses. *Current Biology* 25:3151-3160. https://doi.org/10.1016/j.cub.2015.10.063
250. Padamsey, Z., Katsanevaki, D., Dupuy, N. and Rochefort, N. L. (2022). Neocortex saves energy by reducing coding precision during food scarcity. *Neuron* 110:280-296.e10. https://doi.org/10.1016/j.neuron.2021.10.024
251. Plaçais, P.-Y. and Preat, T. (2013). To favor survival under food shortage, the brain disables costly memory. *Science* 339:440-442. https://doi.org/10.1126/science.1226018

252. Plaçais, P.-Y., de Tredern, É., Scheunemann, L., Trannoy, S. et al. (2017). Upregulated energy metabolism in the *Drosophila* mushroom body is the trigger for long-term memory. *Nature Communications* 8:15510. https://doi.org/10.1038/ncomms15510
253. Mery, F. and Kawecki, T. J. (2005). A cost of long-term memory in *Drosophila*. *Science* 308:1148. https://doi.org/10.1126/science.1111331
254. Hechler, A., de Lange, F. P. and Riedl, V. (2023). The energy metabolic footprint of predictive processing in the human brain. bioRxiv preprint, version 2 posted 2024. https://doi.org/10.1101/2023.12.08.570804
255. Malkin, J., O'Donnell, C., Houghton, C. J. and Aitchison, L. (2024). Signatures of Bayesian inference emerge from energy-efficient synapses. *eLife* 12:RP92595. https://doi.org/10.7554/eLife.92595
256. Li, H. L. and van Rossum, M. C. W. (2020). Energy efficient synaptic plasticity. *eLife* 9:e50804. https://doi.org/10.7554/eLife.50804
257. Lynn, C. W., Cornblath, E. J., Papadopoulos, L., Bertolero, M. A. and Bassett, D. S. (2021). Broken detailed balance and entropy production in the human brain. *Proceedings of the National Academy of Sciences* 118:e2109889118. https://doi.org/10.1073/pnas.2109889118
258. Horowitz, M. (2014). Computing's energy problem (and what we can do about it). *2014 IEEE International Solid-State Circuits Conference*, pp. 10-14. https://doi.org/10.1109/ISSCC.2014.6757323
259. Saad-Falcon, J., Narayan, A., Akengin, H. O., Griffin, J. W. et al. (2025). Intelligence per watt: measuring intelligence efficiency of local AI. arXiv:2511.07885 (preprint).
260. Tschand, A., Rajan, A. T. R., Idgunji, S., Ghosh, A. et al. (2025). MLPerf Power: benchmarking the energy efficiency of machine learning systems from µWatts to MWatts for sustainable AI. *2025 IEEE International Symposium on High Performance Computer Architecture (HPCA)*, 1201-1216. https://doi.org/10.1109/HPCA61900.2025.00092. arXiv:2410.12032.
261. Chung, J.-W., Ma, J. J., Wu, R., Liu, J. et al. (2025). The ML.ENERGY benchmark: toward automated inference energy measurement and optimization. arXiv:2505.06371 (preprint).
262. Niu, C., Zhang, W., Li, J., Zhao, Y. et al. (2025). TokenPowerBench: benchmarking the power consumption of LLM inference. arXiv:2512.03024 (preprint).
263. Jin, Y., Wei, G.-Y. and Brooks, D. (2025). The energy cost of reasoning: analyzing energy usage in LLMs with test-time compute. arXiv:2505.14733 (preprint).
264. Luccioni, S., Jernite, Y. and Strubell, E. (2024). Power hungry processing: watts driving the cost of AI deployment? *ACM FAccT 2024*, pp. 85-99. https://doi.org/10.1145/3630106.3658542
265. Samsi, S., Zhao, D., McDonald, J., Li, B. et al. (2023). From words to watts: benchmarking the energy costs of large language model inference. *2023 IEEE High Performance Extreme Computing Conference (HPEC)*, pp. 1-9. https://doi.org/10.1109/HPEC58863.2023.10363447
266. Dillavou, S., Stern, M., Liu, A. J. and Durian, D. J. (2022). Demonstration of decentralized physics-driven learning. *Physical Review Applied* 18:014040. https://doi.org/10.1103/PhysRevApplied.18.014040

267. Stern, M., Dillavou, S., Jayaraman, D., Durian, D. J. and Liu, A. J. (2024). Training self-learning circuits for power-efficient solutions. *APL Machine Learning* 2:016114. https://doi.org/10.1063/5.0181382
268. Saggio, V., Asenbeck, B. E., Hamann, A., Strömberg, T. et al. (2021). Experimental quantum speed-up in reinforcement learning agents. *Nature* 591:229-233. https://doi.org/10.1038/s41586-021-03242-7
269. Crosato, E., Spinney, R. E., Nigmatullin, R., Lizier, J. T. and Prokopenko, M. (2018). Thermodynamics and computation during collective motion near criticality. *Physical Review E* 97:012120. https://doi.org/10.1103/PhysRevE.97.012120
270. Chen, Q. and Prokopenko, M. (2025). Why collective behaviours self-organize to criticality: a primer on information-theoretic and thermodynamic utility measures. *Royal Society Open Science* 12:241655. https://doi.org/10.1098/rsos.241655
271. Khan, K. N., Hirki, M., Niemi, T., Nurminen, J. K. and Ou, Z. (2018). RAPL in action: experiences in using RAPL for power measurements. *ACM Transactions on Modeling and Performance Evaluation of Computing Systems* 3:1-26. https://doi.org/10.1145/3177754
272. Desrochers, S., Paradis, C. and Weaver, V. M. (2016). A validation of DRAM RAPL power measurements. *Proceedings of the Second International Symposium on Memory Systems*, pp. 455-470. https://doi.org/10.1145/2989081.2989088
273. Hackenberg, D., Schöne, R., Ilsche, T., Molka, D., Schuchart, J. and Geyer, R. (2015). An energy efficiency feature survey of the Intel Haswell processor. *2015 IEEE IPDPS Workshops*, pp. 896-904. https://doi.org/10.1109/IPDPSW.2015.70
274. Apple. powermetrics(1) manual page. macOS developer tools documentation.
275. Hutter Prize for Lossless Compression of Human Knowledge. http://prize.hutter1.net/
276. Strubell, E., Ganesh, A. and McCallum, A. (2019). Energy and policy considerations for deep learning in NLP. arXiv:1906.02243 (preprint).
277. Patterson, D., Gonzalez, J., Le, Q., Liang, C. et al. (2021). Carbon emissions and large neural network training. arXiv:2104.10350 (preprint).
278. Lacoste, A., Luccioni, A., Schmidt, V. and Dandres, T. (2019). Quantifying the carbon emissions of machine learning. arXiv:1910.09700 (preprint).
279. Luccioni, A. S., Viguier, S. and Ligozat, A.-L. (2022). Estimating the carbon footprint of BLOOM, a 176B parameter language model. arXiv:2211.02001 (preprint).
280. Anthony, L. F. W., Kanding, B. and Selvan, R. (2020). Carbontracker: tracking and predicting the carbon footprint of training deep learning models. arXiv:2007.03051 (preprint).
281. Elsworth, C., Huang, K., Patterson, D., Schneider, I. et al. (2025). Measuring the environmental impact of delivering AI at Google scale. arXiv:2508.15734 (preprint).
282. Schmidhuber, J. (2010). Formal theory of creativity, fun, and intrinsic motivation (1990-2010). *IEEE Transactions on Autonomous Mental Development* 2:230-247. https://doi.org/10.1109/TAMD.2010.2056368
283. Chollet, F. (2019). On the measure of intelligence. arXiv:1911.01547 (preprint).
284. ARC Prize Foundation. ARC-AGI leaderboard. https://arcprize.org/leaderboard (accessed 8 October 2026).

285. Hernández-Orallo, J. (2017). *The Measure of All Minds: Evaluating Natural and Artificial Intelligence*. Cambridge University Press. https://doi.org/10.1017/9781316594179
286. Hernández-Orallo, J. and Dowe, D. L. (2010). Measuring universal intelligence: towards an anytime intelligence test. *Artificial Intelligence* 174:1508-1539. https://doi.org/10.1016/j.artint.2010.09.006
287. Gershman, S. J., Horvitz, E. J. and Tenenbaum, J. B. (2015). Computational rationality: a converging paradigm for intelligence in brains, minds, and machines. *Science* 349:273-278. https://doi.org/10.1126/science.aac6076
288. Lieder, F. and Griffiths, T. L. (2020). Resource-rational analysis: understanding human cognition as the optimal use of limited computational resources. *Behavioral and Brain Sciences* 43:e1. https://doi.org/10.1017/S0140525X1900061X
289. Balasubramanian, V. (2021). Brain power. *PNAS* 118:e2107022118. https://doi.org/10.1073/pnas.2107022118
290. Silver, D., Singh, S., Precup, D. and Sutton, R. S. (2021). Reward is enough. *Artificial Intelligence* 299:103535. https://doi.org/10.1016/j.artint.2021.103535
291. Vamplew, P., Smith, B. J., Källström, J., Ramos, G., Rădulescu, R. et al. (2022). Scalar reward is not enough: a response to Silver, Singh, Precup and Sutton (2021). *Autonomous Agents and Multi-Agent Systems* 36:41. https://doi.org/10.1007/s10458-022-09575-5
292. Hafez, W., Wei, C., Pena, R., Nazeri, A. et al. (2026). A mathematical theory of agency and intelligence. arXiv:2602.22519 (preprint).
293. Karagoz, A. (2025). Energentic intelligence: from self-sustaining systems to enduring artificial life. arXiv:2506.04916 (preprint).
294. Chaisson, E. J. (2011). Energy rate density as a complexity metric and evolutionary driver. *Complexity* 16:27-40. https://doi.org/10.1002/cplx.20323
295. Poplavskii, R. P. (1975). Thermodynamic models of information processes. *Uspekhi Fizicheskikh Nauk* 115:465 (in Russian), https://doi.org/10.3367/UFNr.0115.197503d.0465. English translation: *Soviet Physics Uspekhi* 18:222-241, https://doi.org/10.1070/PU1975v018n03ABEH001955
296. Ito, S. and Sagawa, T. (2017). Information thermodynamics on networks and its application to biological information processing. *Butsuri* (日本物理学会誌) 72(9):658 (in Japanese). https://doi.org/10.11316/butsuri.72.9_658
297. Sun, C.-P. and Quan, H.-T. (2013). Maxwell's demon and the physical limits on information processing. *Wuli* (物理, Physics) 42(11) (in Chinese). doi:10.7693/wl20131101
298. Quan, H.-T., Dong, H. and Sun, C.-P. (2023). Theory and experiments of mesoscopic statistical thermodynamics. *Acta Physica Sinica* 72:230501 (in Chinese, with English abstract). https://doi.org/10.7498/aps.72.20231608
299. Parrondo, J. M. R. (2023). Thermodynamics of information. arXiv:2306.12447 (encyclopedia chapter preprint).
300. Strasberg, P. Thermodynamics and information processing at the nanoscale. Doctoral thesis, Technische Universität Berlin.
301. Dago, S. (2022). Thermodynamique stochastique: pilotage de micro-oscillateurs et applications à l'étude et l'optimisation du traitement de l'information. Doctoral thesis, École normale supérieure de Lyon (in French). https://theses.fr/2022LYSEN019

302. Lagoin, M. (2023). Thermodynamique dans les systèmes stationnaires hors équilibre macroscopiques: démons de Maxwell et machines thermiques aléatoires. Doctoral thesis, École normale supérieure de Lyon (in French).
303. Ciampini, M. A., Mancino, L., Orieux, A., Vigliar, C. et al. (2017). Experimental extractable work-based multipartite separability criteria. *npj Quantum Information* 3:10. https://doi.org/10.1038/s41534-017-0011-9
304. Sudakov, K. V. (2004). Functional systems theory and the probabilistic prediction of behavior. *Neuroscience and Behavioral Physiology* 34:505-507. https://doi.org/10.1023/B:NEAB.0000022638.96382.dd
305. Sudakov, K. V. (2015). Theory of functional systems: a keystone of integrative biology. In M. Nadin (ed.), *Anticipation: Learning from the Past*, Springer, pp. 153-173. https://doi.org/10.1007/978-3-319-19446-2_9
306. Sudakov, K. V. (2022). Cognitive activity from the perspective of functional systems theory. In *Russian Cognitive Neuroscience*, Brill, pp. 87-117. https://doi.org/10.1163/9789004505667_005
307. Shvyrkov, V. B. (1985). Toward a psychophysiological theory of behavior. *Advances in Psychology*, pp. 47-71. https://doi.org/10.1016/S0166-4115(08)61596-4
308. Anokhin, K. V. (2021). The cognitome: seeking the fundamental neuroscience of a theory of consciousness. *Neuroscience and Behavioral Physiology* 51:915-937. https://doi.org/10.1007/s11055-021-01149-4
309. Saltykov, A. and Grachev, S. (2015). Anticipation and the concept of system-forming factor in the theory of functional systems. In M. Nadin (ed.), *Anticipation: Learning from the Past*, Springer, pp. 507-520. https://doi.org/10.1007/978-3-319-19446-2_30
310. Vityaev, E. E. (2015). Purposefulness as a principle of brain activity. In M. Nadin (ed.), *Anticipation: Learning from the Past*, Springer, pp. 231-254. https://doi.org/10.1007/978-3-319-19446-2_13
311. Vityaev, E. E. and Demin, A. V. (2011). Recursive subgoals discovery based on the functional systems theory. *Frontiers in Artificial Intelligence and Applications* (BICA 2011), IOS Press. https://doi.org/10.3233/978-1-60750-959-2-425
312. Vityaev, E. E. and Demin, A. V. (2018). Cognitive architecture based on the functional systems theory. *Procedia Computer Science* 145:623-628. https://doi.org/10.1016/j.procs.2018.11.072
313. Vityaev, E., Kolonin, A., Kurpatov, A., Molchanov, A. et al. (2022). Brain principles programming. arXiv:2202.12710 (preprint).
314. Potapov, A., Belikov, A., Bogdanov, V., Scherbatiy, A. et al. (2019). Differentiable probabilistic logic networks. arXiv:1907.04592 (preprint).
315. Paninski, L. (2003). Estimation of entropy and mutual information. *Neural Computation* 15:1191-1253. https://doi.org/10.1162/089976603321780272
316. Russell, S. and Wefald, E. (1991). Principles of metareasoning. *Artificial Intelligence* 49:361-395. https://doi.org/10.1016/0004-3702(91)90015-C
317. Horvitz, E. J. (1987). Reasoning about beliefs and actions under computational resource constraints. *Proceedings of the Third Conference on Uncertainty in Artificial Intelligence (UAI 1987)*, 429-447. arXiv:1304.2759.

318. Simon, H. A. and Kadane, J. B. (1975). Optimal problem-solving search: all-or-none solutions. *Artificial Intelligence* 6:235-247. https://doi.org/10.1016/0004-3702(75)90002-8
319. Sekimoto, K. (1998). Langevin equation and thermodynamics. *Progress of Theoretical Physics Supplement* 130:17-27. https://doi.org/10.1143/PTPS.130.17
320. Wolpert, D. H., Korbel, J., Lynn, C. W., Tasnim, F., Grochow, J. A., Kardeş, G. et al. (2024). Is stochastic thermodynamics the key to understanding the energy costs of computation? *PNAS* 121:e2321112121. https://doi.org/10.1073/pnas.2321112121
321. Chen, L., Zaharia, M. and Zou, J. (2023). FrugalGPT: how to use large language models while reducing cost and improving performance. arXiv:2305.05176 (preprint).
322. Ong, I., Almahairi, A., Wu, V., Chiang, W.-L. et al. (2024). RouteLLM: learning to route LLMs with preference data. arXiv:2406.18665 (preprint).